

Appendix A: Accepted Timber Supply Analysis Information Package



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March 3, 2011

Mr. Mike Davis, RPF
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Dear Mike Davis:

Thank-you for submitting the Tree Farm Licence (TFL) 6 Timber Supply Analysis Information Package (Version 2, February 2011) prepared for TFL 6 Management Plan 10. I have extensively reviewed the document with specialists at the Ministry of Forests, Mines and Lands (MFML) and the Ministry of Natural Resource Operations (MNRO). As the MFML Timber Supply Forester responsible for reviewing this information package, I accept the document for use in the timber supply analysis for TFL 6.



I wish to point out that this letter does not mean that the MFML endorses every aspect of this information package. During the allowable annual cut (AAC) determination information session, MFML and MNRO staff will advise the chief forester regarding the technical validity of the analysis and the implications of its assumptions and results. The chief forester will consider this advice when he determines the AAC for TFL 6.

Yours truly,

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Tree Farm Licence 6

Timber Supply Analysis Information Package

In Preparation of

MANAGEMENT PLAN 10

**Submitted to the Ministry of Forests, Mines and Lands
Forest Analysis & Inventory Branch**

**Version 2
February 2011**



**Mike Davis, R.P.F.
Planning Forester
Western Forest Products Inc.**

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1 INTRODUCTION

This Information Package (IP) provides a summary of data, assumptions, and modelling procedures to be used in the Timber Supply Analysis (TSA) for Western Forest Products' (WFP) Tree Farm Licence (TFL) 6 Management Plan (MP) #10.

The timber supply analysis will be conducted using Remsoft's Spatial Planning System (www.remsoft.com). Woodstock is the aspatial component of the suite and addresses the majority of the model objectives and constraints. Woodstock performs a similar function as the Ministry of Forests and Range's FSSIM model whereby management zones and constraints are defined, yield curves incorporated and applied to an aggregated area file. The primary difference between Woodstock and FSSIM is that Woodstock is capable of using either optimization or sequential simulation in developing a harvest forecast.

Stanley, the spatial component of the suite, applies the Woodstock harvest forecast to specific polygons on the land base. Stanley aggregates individual polygons into suitable harvest units (blocks) based on specified minimum, maximum and target block sizes. The model also enforces green-up and adjacency requirements as it schedules the harvest spatially.

For this analysis, optimization will be used in Woodstock to develop the base case harvest schedule. The optimization will be subject to a number of harvest constraints including the requirement to produce a long-term sustainable harvest forecast. Stanley will not be used for the analysis.

WFP will complete the timber supply analysis to estimate timber harvest over a 250-year planning horizon (in five-year planning periods) based on the current harvestable land base, existing old forest timber volumes, and regenerating forest growth rates. The harvest forecast will project the timber supply impacts of current environmental protection and management practices including operational requirements of the Forest and Range Practices Act (FRPA), approved Forest Stewardship Plan (FSP), orders, other regulations and guidelines with significance to timber supply. Sensitivity analyses will be used to investigate the expected impacts of different management scenarios, and to examine the relative importance of variations in assumptions. These may include the removal of area from the timber harvesting land base (THLB), imposing forest-cover harvest constraints, or changes in growth & yield (G&Y) estimates.

The timber supply forecast will attempt to achieve the long-term harvest potential, and minimize the rate of change during the transition from the current level of harvest to the mid- and long-term sustainable levels.



Figure 1 - TFL 6

2 PROCESS

This Information Package is being submitted for acceptance by the Timber Supply Forester at Forest Analysis and Inventory Branch of the Ministry of Forests, Mines and Lands (MFML – formerly Ministry of Forests and Range (MoFR)). This version addresses comments and questions raised during the review process associated with the draft Information package (Version 1 – October 2010).

Version 1 of this Information Package was submitted for review to the MFML Timber Supply Forester at Forest Analysis and Inventory Branch and Ministry of Natural Resource Operations (MNRO – formerly MoFR) staff in the North Island – Central Coast Resource District (NICCRD – formerly NICC Forest District). It was also made available for review by stakeholders, the general public and potentially affected First Nations.

This revised and accepted Information Package will guide the timber supply analysis. Once completed, the timber supply analysis will be made available for review as a component of a draft Management Plan #10 for TFL 6. The final accepted timber supply analysis will be considered by the Chief Forester in determining the new Allowable Annual Cut (AAC) for TFL 6.

3 TIMBER SUPPLY FORECASTS/OPTIONS/SENSITIVITY ANALYSES

This section describes the landbase and management scenarios to be included in the timber supply analysis. The details, assumptions, and sensitivities of each are also described.

3.1 Current Management Option

The current management option (or the “Base Case”) represents the present operational requirements and management practices on the TFL. The forecast of current management incorporates existing land use designations, including Resource Management Zones¹, and current regulations and guidelines including the Forest and Range Practices Act and the approved Forest Stewardship Plan. This option is used as the basis for analysing various timber supply projections.

Current management on TFL 6 includes:

- The operable landbase of forested area accessible using conventional and non-conventional (helicopter) harvesting methods.
- Restricted performance in the non-conventional landbase (see Section 10.3.3).
- Exclusion of uneconomic forest stands.
- Harvesting of both mature and immature stands.
- Biodiversity and Landscape Units – Established Old Growth Management Areas (OGMAs) in the San Josef and Marble landscape units are removed from the THLB. Also removed are draft OGMAs in the Holberg, Keogh, Mahatta, and Neroutsos landscape units. Mature seral targets are incorporated for the two Special Management Zones within TFL 6.
- Established Ungulate Winter Ranges (UWRs) and Wildlife Habitat Areas are removed from the THLB.
- Riparian management based on the FSP results/strategies and a review of stand-level retention within harvested cutblocks between 1997 and 2008.
- Wildlife Tree and other stand-level retention within the THLB are accounted for by a percentage area reduction.
- Visual quality objectives (VQOs) modelled based on proposed VQOs resulting from the recently accepted (May 2010) Visual Landscape Inventory for TFL 6 with upper range disturbance assumed. The North Island – Central Coast Resource District (NICCRD) used these proposed VQOs in undertaking the required steps for establishing VQOs as legal objectives through the Government Actions Regulation (GAR). The established VQOs differ slightly from those that will be used in the analysis. The differences will be discussed in the timber supply analysis.
- Green-up heights for cutblock adjacency are assigned based on Resource Management Zones established in the Vancouver Island Higher Level Plan. Special and General zones have a 3m green-up requirement while Enhanced zones have a 1.3m green-up height.
- Silviculture to meet free growing requirements is carried out on all regenerated stands. The majority of harvested areas are planted.
- Known tree improvement gains will be applied to existing stands ≤ 10 years old and future regenerated stands.

¹ Vancouver Island land Use Plan (VILUP) Resource Management Zones and Resource Management Zone objectives approved by Government in December 2000.

- Minimum harvest age varies by harvesting system and the minimum harvestable volume is 350m³ per hectare. Both minimum age and minimum volume requirements must be met before a stand can be harvested.
- A relatively small area of deciduous leading stands is excluded from the THLB while minor deciduous volumes in mixed stands contribute to the analysis.

3.2 Sensitivity Analyses

Sensitivity analyses will be conducted for the current management option to examine the potential impact of uncertainty in several key attributes. These include the removal of operable areas from the THLB, imposing forest-cover harvest constraints, or changes in growth & yield estimates.

Table 1 - Sensitivity Analyses

Issue Tested	Proposed Sensitivity Analysis
Landbase available for harvesting	▪ exclude all class V (unstable) terrain plus helicopter operable landbase
Growth and yield	▪ adjust natural stand volumes +/-10% ▪ adjust managed stand volumes +/-10% ▪ apply SIBEC estimates of site index
Forest management / Silviculture	▪ exclude future fertilization ▪ less Hw planting
Operability	▪ remove heli volume constraint ▪ exclude helicopter operable landbase
Visual management	▪ assume mid-range disturbance limit
Wildlife habitat / Biodiversity	▪ retain old forests to full target to maintain marbled murrelet habitat in EFZ #8 (Mahatta – Neroutsos) – VILUP objective #16
Minimum harvest ages	▪ add 10 years to the minimum harvest ages

3.3 Alternate Harvest Flow

The harvest level in the current management option will adjust over 10-year intervals (to coincide with the term of AAC determinations) in the first part of the run towards the estimated Long Term Harvest Level (LTHL). The results of the base case will determine potential alternate harvest flows. One option will be to investigate the mid-term timber supply impacts of continuing the initial harvest for 2 or more ten-year intervals.

During preparation of the timber supply analysis, the need for further sensitivity analyses or harvest flows may become apparent. If warranted, additional analyses will be included in the final timber supply analysis for consideration by the Chief Forester.

4 HARVEST MODEL

The TFL 6 timber supply analysis including harvest level and forest inventory projections will be developed using Remsoft's Spatial Planning System (www.remsoft.com).

Woodstock is a pseudo-spatial timber supply model that projects harvesting activities across a land base over a specific period of time. These models are referred to as pseudo-spatial because the data used to create the model has spatial components to it; however, the harvest schedules produced by these models are not spatially explicit. It is possible to bring spatial context into this type of model by applying constraints to spatial attributes of the land base such as landscape units or watersheds; however harvest schedules produced using these types of models report the timing of the harvest of different types of stands as opposed to specific polygons harvested in each period. For these reasons it is not possible to explicitly model spatial management objectives such as cutblock size, adjacency and green-up requirements, or patch size targets using this type of model.

Woodstock uses optimization to establish a harvest schedule that incorporates objectives such as visual quality, biodiversity, wildlife habitat with the objective of timber harvest. In Woodstock, harvest volume is maximized subject to the maintenance of other values on the land base.

5 FOREST COVER INVENTORY

5.1 Vegetation Resources Inventory

A Vegetation Resource Inventory (VRI) project for TFL 6 was initiated in 1999. Phase I (forest cover polygon boundaries delineated and attributes estimated using aerial photography) was completed in 2000 using 1:15,000 scale colour imagery taken in 1995. Phase II (ground sampling) and the Net Volume Adjustment Factor (NVAF) sampling was carried out in 2001. The last component, the statistical adjustment, was completed and reported on by Timberline Natural Resource Group (now Teco Natural Resource Group) in 2009. The results have been reviewed by the MFML, Forest Analysis & Inventory Branch and approved for use in the TFL 6 MP #10 timber supply analysis.

The VRI project was funded by Forest Renewal BC (FRBC) and Forest Investment Account (FIA).

5.2 Age Class Distribution

Table 2 and Figure 2 indicate the current age class distribution of the forested land base of TFL 6.

Table 2 - Age Class Distribution

Age Class	Age (years)	Productive Forest		Current THLB		% Productive in THLB	
		Area (ha) ¹	Volume ('000 m ³) ²	Area (ha) ¹	Volume ('000 m ³) ²	Area	Volume
1	1-20	24,924	107	22,535	93	90.4%	86.5%
2	21-40	30,008	3,992	25,705	3,346	85.7%	83.8%
3	41-60	18,746	6,967	14,521	5,420	77.5%	77.8%
4	61-80	11,101	5,682	8,362	4,459	75.3%	78.5%
5	81-100	5,259	3,574	4,027	2,750	76.6%	76.9%
6	101-120	4,601	3,362	3,465	2,537	75.3%	75.5%
7	121-140	4,647	3,401	3,789	2,779	81.5%	81.7%
8	141-250	3,255	2,248	1,663	1,261	51.1%	56.1%
9	>250	44,518	35,526	23,744	19,948	53.3%	56.1%
Total		147,059	64,861	107,811	42,592	73.3%	65.7%

¹ Areas are updated to end of 2008.

² Volumes are projected to 2011 in order for volume estimates to coincide with the middle year of the first 5-year period, therefore average volume, for each 5-year period

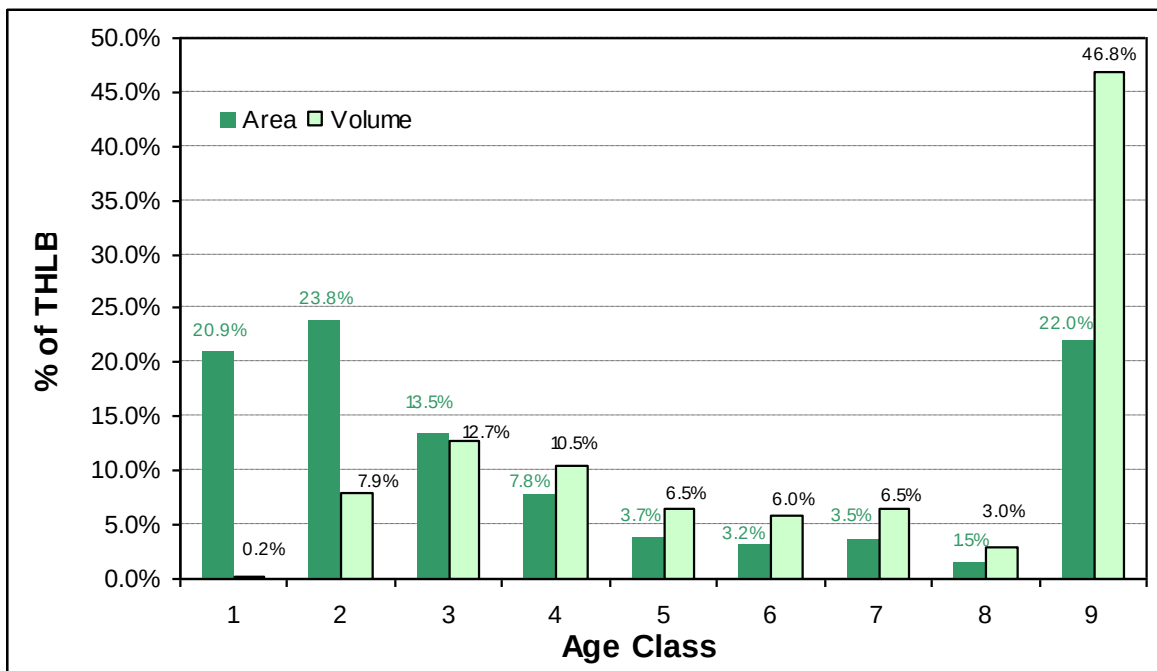


Figure 2 - THLB Age Class Distribution

6 DESCRIPTION OF LANDBASE

This section describes the TFL 6 landbase and the methods used to determine the portion of the landbase that contributes to timber harvesting (THLB). Some portions of the productive landbase, while not contributing to harvest, are crucial in meeting the demands for non-timber resource sustainability. Areas within all tables in this section may not sum due to rounding to the nearest hectare.

6.1 AAC and Landbase Changes and Allocations

At the time of the TFL 6 MP #9 AAC Determination in 2001 the TFL area was estimated as 198,113 ha and the AAC was determined as 1,460,000 m³, of which 13,242 m³ was attributed to BCTS and the remaining 1,446,758 m³ was allocated to WFP.

In 2003, the provincial government enacted the *Forest Revitalization Act*. This legislation reallocated 20% of the AAC of major licensees such as WFP to others such as BC Timber Sales, First Nations and small tenures such as Community Forests and Woodlots. For TFL 6 the effect was the reallocation of 86,000 m³ of AAC from WFP to others: 69,422 m³ to BCTS (for a new total of 82,664 m³); 11,578 m³ to First Nations; and 5,000 m³ to a Community Forest. WFP's AAC was reduced by 81,000 m³ as of the end of 2004 and by a further 5,000 m³ as of the end of 2005. Areas have been deleted from TFL 6 for the total BCTS volume and the Community Forest volume. Refer to Table 3 for a summary of changes in area and AAC since the MP #9 analysis.

Table 3 - Changes in TFL 6 Area and AAC Allocations since MP #9 AAC Determination

Description	Total Area (ha)	AAC (m ³ /year)
TFL 6 MP #9 AAC Determination	198,113	1,460,000
Minor Area Deletions	-112	0
Deletion of Private Land (2007)	-14,027 ¹	-116,800
Deletion of BCTS area (2009)	-11,339	-82,664
Deletion of Tri-Port Community Forest (2010)	-1,072	-5,000
Boundary Adjustments	-122	0
Net TFL 6 (February 2011)	171,441	1,255,536

6.1.1 Minor Area Deletions

The total TFL 6 area was reduced by 112 ha due to four small area deletions. More specifically:

6.1.1.1 Port Alice Industrial Park

In July 2004, 5 ha of private land were deleted from TFL 6 via Instrument No. 92 for development of a light industrial park located in the vicinity of the community of Port Alice.

¹ Area determined by deducting minor private land deletions included in the 112 ha of minor area deletions (for Port Alice Industrial Park (5 ha) and Marble River Park/Misty Lake ER (55 ha)) from total Crown Grants listed in March 1, 2000 Licence document.

6.1.1.2 Addition to Kingcome Timber Supply Area

Also in July 2004, the TFL 6 boundary was adjusted via Instrument No. 90 in the vicinity of Mahatta River such that approximately 22 ha of crown land was deleted from TFL 6 and added to the Kingcome Timber Supply Area (TSA #33).

6.1.1.3 Cluxewe Saltmarsh Reserve

In March 2002, 30 ha of crown land was deleted from TFL 6 via Instrument No. 84 to expand the Cluxewe Saltmarsh Reserve located near the mouth of the Cluxewe River.

6.1.1.4 Marble River Park and Misty Lake Ecological Reserve

In May 2000, WFP and the provincial government entered into a land sale agreement. As a result, in June 2001 Instrument No. 88 was executed - deleting 41 ha of private land from TFL 6 in the vicinity of the Marble River Park and Varney Bay and deleting 14 ha of private land in the vicinity of Misty Lake.

6.1.2 Areas removed from TFL 6

Since the MP #9 AAC determination, the total area within TFL 6 has been reduced due to removals:

6.1.2.1 Private land

An area of 14,027 ha was removed in January 2007 (Instrument No. 94). The TFL 6 AAC was reduced by 116,800 m³ to 1,343,200 m³.

6.1.2.2 BCTS Area

In July 2009, Instrument No. 97 deleted 11,339 ha from TFL 6 that was subsequently added to the newly created Pacific Timber Supply Area. This area was attributed 82,664 m³ of the TFL 6 AAC.

6.1.2.3 Tri-Port Community Forest

In a January 2010 amendment to TFL 6 (Minister Order #3(4) 7-3 under the *Forest Revitalization Act*), 1,072 ha were removed from TFL 6 for creating the Tri-Port Community Forest (Port McNeill, Port Hardy, Port Alice). The TFL 6 AAC attributed to this area is 5,000 m³.

6.1.3 Boundary Adjustments

The height-of-land tenure boundaries used to create the data set to be used in this timber supply analysis differ slightly from the boundaries used in MP #9 with the net effect being a 122 ha reduction in the size of TFL 6.

6.1.4 Remaining First Nations AAC Allocation

Management areas for a further 11,578 m³ of *Forest Revitalization* AAC allocation to First Nations remain to be identified within the TFL.

6.2 Timber Harvesting Land Base Determination

Areas and timber volume estimates for the current THLB and the total long-term land base in TFL 6 are presented in Table 4 - values are reported for both Schedule A (Timber Licences within the TFL) and Schedule B (Crown land) land classes. Areas and volumes have been compiled from a stand database constructed for the preparation of this information package.

For MP #9 the total reductions were 48,366 ha, which was 24% of the total area. In addition to the area reductions, a 4% yield reduction was made in MP #9 to account for Wildlife Tree Patches. In effect this reduced the THLB by a further 5,990 ha for a total area reduction of 54,356 ha or 27.5% of the total area. For MP #10, the total reductions are 63,631 ha or 37% of the total area of TFL 6. OGMAs and different riparian management assumptions are the main reasons for the larger landbase reductions.

The following sections show total area classified in each category noted in Table 4 and serve to summarise the area deducted from the land base in the order the categories appear in Table 4 (i.e. overlapping constraints are addressed in a hierarchy).

Table 4 - Timber Harvesting Land Base for TFL 6

Classification	Productive Area (ha)	Net Area (ha)			Volume ('000 m ³)		
		Sched A	Sched B	Total	Sched A	Sched B	Total
Total Landbase	-	26,311	145,130	171,441	-	-	-
Less Non-forest	-	1,020	12,463	13,483	0	0	0
Less Existing Roads	-	643	3,292	3,935	0	0	0
Total Forested	-	24,648	129,375	154,023	-	-	-
Less Non-productive	-	980	5,985	6,964	0	0	0
Total Productive	147,059	23,669	123,390	147,059	11,236	53,625	64,861
Less Inoperable (including uneconomic)	12,438	1,798	10,641	12,438	1,139	6,326	7,464
Total Operable	134,621	21,871	112,750	134,621	10,097	47,299	57,397
<i>Reductions:</i>							
Riparian Management	11,736	1,716	8,917	10,632	1,050	4,362	5,412
Ungulate Winter Ranges	1,821	288	1,026	1,313	246	847	1,092
Old Growth Management Areas (established)	9,040	988	2,762	3,750	775	2,105	2,880
Old Growth Management Areas (draft)	8,135	576	2,894	3,469	450	2,064	2,514
Wildlife Habitat Areas	659	0	3	3	0	2	2
Recreation Sites and Trails	136	20	30	50	13	22	35
Deciduous-leading	3,460	117	1,658	1,774	34	455	489
Cultural Heritage Resources	16,734	27	106	132	12	52	65
Total Operable Reductions	-	3,730	17,394	21,124	2,580	9,908	12,488
Reduced Landbase	-	18,141	95,356	113,497	7,517	37,391	44,909
Less allowance for stand-level retention	134,621	888	4,798	5,686	372	1,944	2,316
Current THLB	-	17,253	90,558	107,811	7,145	35,447	42,592
Less future roads	1,806	256	1,236	1,491	-	-	-
Long-term Landbase	132,815	16,998	89,322	106,319	7,145	35,447	42,592

6.3 Non-Forest

The non-forest portion of TFL 6 includes area where merchantable tree species are largely absent. Most of this area is in brush and wet areas (Table 5).

Table 5 - Non-forest area in TFL 6

Description	Gross Non-forest Area (ha)	Total Area Reduction (ha)
Alpine	421	421
Rock and slides	775	775
Swamp, Marsh, Creek, River, Lake	4,589	4,589
Dump, Camps and Sort	120	120
Hydro and Telephone RoW	186	186
Brush	3,588	3,588
Other	3,804	3,804
TOTAL	13,483	13,483

6.4 Existing Roads

Existing roads are excluded from the timber harvesting land base. This reduction is due to the combination of classified and unclassified roads. Classified roads are those that are mapped as forest cover polygons distinctly separate from adjacent polygons; a total of 138 ha of classified roads (all highway) were removed. Unclassified roads have been mapped as lineal features. For the purposes of determining the total area of unclassified roads, operational staff were consulted: mainlines are assumed to occupy a 12 metre unproductive width, branch roads are assumed to occupy a 10 metre unproductive width, and spur/stub roads are assumed to occupy an 8 metre unproductive width. As all trails and the majority of the landings are rehabilitated and restocked following logging, the associated area reduction is thought insignificant. Table 6 summarizes the areas of existing roads in the TFL.

Table 6 - Existing roads

Road Classification	Length (km)	Width (m)	Gross Area (ha)	Area Reduction (ha)
Highway	54.6	30	164	144
Mainline	660.9	12	793	766
Branch	2,874.7	10	2,875	2,837
Spur/Stub	238.0	8	190	188
Total	3,828.2	-	4,022	3,935

The area removed by roads is relatively low compared to other coastal management units. This is mainly due to the geography of TFL 6 which is dominated by gentle rolling terrain. This terrain affords longer average yarding distances resulting in higher development per kilometre of road constructed than most other coastal units.

6.5 Non-Productive Forests

TFL 6 includes 7,538 ha of non-productive forest (Table 7). These areas are mostly mature stands. These non-productive forests can contribute to landscape level biodiversity. While not incorporated into the biodiversity calculations these components provide a margin of safety around the biodiversity requirements.

Table 7 - Non-productive area

Description	Gross Non-productive Area (ha)	Total Area Reduction (ha)
Non-productive (including low productivity ecosites (CP, MH, MH1, S7, S8) – PG 5)	7,538	6,964

6.6 Operability

Operability classes have been developed for the TFL 6 landbase that reflect the harvesting system, timber quality, terrain stability and economic accessibility. A terrain stability inventory was completed for Block 1 to MoFR standards in 1995 and for Block 2 in 1992, since updated to MoFR standards in 1998.

The reclassification of the TFL's operability was completed in early 1999 based on a methodology reflecting the elements noted above and approved by the MoFR Port McNeill Forest District in September 1998. The classification resulted in two categories of inoperable areas being defined. The first category strictly relates to the area being physically inaccessible and thus, not available for timber harvesting (I). The second category uses an economic criterion to determine operability (Ocm/Ohm). Physical inoperability relates to the presence of a physical barrier or terrain constraint leaving accessibility virtually impossible. Classifying areas as operable with an economic constraint relates to the inability to harvest stands in a cost-effective manner given the low market value of the timber at the time of the classification. Two classes are recognized in this analysis: (1) Ocm – economically marginal areas accessible by conventional harvesting systems; and (2) Ohm – economically marginal heli-logging areas.

Of the net inoperable land base, 839 ha are currently classified as Ocm/Ohm and 11,600 ha are currently classified as I. The area classified as inoperable and therefore deducted from the productive forest land base is 12,438 ha. Table 8 summarizes the areas removed from the THLB.

Table 8 – Area and Volume by Operability Type

Description	Productive Area (ha)	Total Area Reduction (ha)	Productive Forest Volume ('000 m3)
Conventional (Oce)	129,221	-	53,166
Helicopter (Ohe)	5,400	-	4,231
Operable subtotal	134,621	-	57,397
Inoperable (I)	11,600	11,600	6,962
Marginally economic conventional harvesting (Ocm)	424	424	244
Marginally economic heli harvesting (Ohm)	415	415	259
Inoperable subtotal	12,438	12,438	7,464
Total	147,059	12,438	64,861

Other than unstable areas identified as inoperable no additional netdowns are made for the terrain stability inventory. A review of the terrain stability classes within the THLB produced by the assumptions in this document indicates that 7.6% of the THLB is mapped as Class IV (potentially unstable) and 2.3% as Class V (unstable). Spatial logging history records indicate that a total of 5488 ha of Class IV and 1725 ha of Class V terrain has been logged in the current TFL 6 - equal to 6.1% and 1.9% of the total area logged respectively.

6.7 Riparian Management Areas

Detailed riparian features mapping is ongoing for TFL 6 through cutblock development. Operational stream inventories associated with development planning have been conducted since 1988 (with the introduction of the *Coastal Fisheries Forestry Guidelines*) and various reconnaissance fish and fish habitat inventory projects have been completed over time. Therefore, a combination of 1:5,000 scale (operational) and 1:20,000 scale (strategic) stream data was available. There are likely additional streams on the landbase that have not yet been mapped. These would tend to be smaller streams (S4 or S6, but could include S3 and S5 streams).

This detailed information provided the basis for estimating riparian classes and hence reserve areas for watercourses. The approach employed in preparing the data for the timber supply analysis was to utilise the available stream classification in the GIS to apply Riparian Management Areas (RMAs) to known streams based on the sum of FRPA Riparian Reserve Zone (RRZ) widths and assumed levels of retention within Riparian Management Zones (RMZs). For S1-B, S2 and S3 streams it is assumed that 50% of the RMZ is retained. For S4 and S5 streams RMZ retention is assumed to be 25% and for S6 streams the assumption is that 10% of the RMZ is retained. For lakes and wetlands the RMZ retention levels are as listed in Table 9. These RMZ values are based on a review of cutblocks harvested from 1997 to 2008 and the retention of riparian areas along streams both within and adjacent to the cutblocks. The RMA retention levels for S4, S5 and S6 streams (25%, 25%, and 10% respectively) were marginally increased

from what was indicated by the review of harvested cutblocks as an attempt to account for unmapped streams.

A 40 m “reserve zone” (netdown) will be applied to the ocean shoreline within TFL 6 to account for management of visual quality, operability issues and the presence of numerous eagle nests within this shoreline area.

Table 9 - Riparian Management Areas

Riparian Feature Class	Size Class	Feature Size	Reserve Zone (m)	Management Zone		Area Reduction (ha)	
				Width (m)	Netdown (%)	Reserve Zone	Management Zone
<i>Ocean</i>	-	km					
Shoreline	-	362.2	40	0	0	1,015	0
<i>Streams</i>	<i>Width (m)</i>	<i>Area (ha)</i>					
S1-A	>=100	-	0	100	60	-	-
S1-B	>20.0 - 99.9	310.3	50	20	50	1,033	207
S2	>5.0 - 20.0	352.6	30	20	50	1,629	543
<i>Streams</i>	<i>Width (m)</i>	km					
S1-A	>=100	-	0	100	60	-	-
S1-B	>20.0 - 99.9	-	50	20	50	-	-
S2	>5.0 - 20.0	143.2	30	20	50	Included with S2 above	
S3	>1.5 - 5.0	683.2	20	20	50	2,313	1,156
S4	<1.5	603.7	0	30	25	0	752
S5	>3.0	443.8	0	30	25	0	438
S6	<3.0	3,802.7	0	20	10	0	1,132
<i>Lakes</i>	<i>Area (ha)</i>	<i>Area (ha)</i>					
L1-A	>=1000	1,602.7	0	20*	100	0	72
L1-B	>5.0 - 999.9	884.8	10	20*	50	57	57
L2 (dry zones)	1.0 - 5.0	-	10	20	25	-	-
L3 (wet zones)	1.0 - 5.0	99.7	0	30	25	0	11
L4 (dry zones)	0.5 - 1.0	-	0	30	25	-	-
<i>Wetlands</i>	<i>Area (ha)</i>	<i>Area (ha)</i>					
W1	>5.0	318.9	10	40	25	38	38
W2 (dry zones)	1.0 - 5.0	-	10	20	25	-	-
W3 (wet zones)	1.0 - 5.0	610.8	0	30	25	0	88
W4 (dry zones)	0.5 - 1.0	-	0	30	25	-	-
W5	>5.0	366.4	10	40	25	27	27
Total						6,112	4,520

* WFP RMZ for TSA purposes only, not FPPR RMZ

Operationally riparian management areas are established using slope distance; these zones are modelled using horizontal distance. Therefore the area removed from the THLB for riparian management in the GIS data used in the timber supply analysis is slightly greater than the area that would be removed operationally using the same RMA widths.

6.8 Ungulate Winter Ranges

Revised Ungulate Winter Ranges (UWRs) for Columbian black-tailed deer and Roosevelt elk were approved in November 2004 (U-1-010) with an amendment to the order made in May 2007. The 16 UWRs have a total area of 1,873 ha. Due to conflicting tenure information, 5.3 ha of UWR NAH-08 from the Kingcome TSA (U-1-011) falls within the boundaries of TFL 6 being used for the timber supply analysis. These UWRs are included in the data set for the analysis and will be excluded from the THLB.

Table 10 - Ungulate Winter Ranges area

Description	Productive UWR Area (ha)	Total Area Reduction (ha)
Ungulate Winter Ranges	1,821	1,313

6.9 Old Growth Management Areas

Landscape Units (LUs) and Biodiversity Emphasis Options (BEOs) were designated through the *Order Establishing Provincial Non-Spatial Old Growth Objectives* effective June 30, 2004 (NSOG order). This order is in effect until Old Growth Management Areas (OGMAs) are spatially determined through Landscape Unit planning. For TFL 6 OGMAs have been established within the San Josef (January 2005) and the Marble (July 2010) landscape units. Within the Marble Enhanced Forestry Zone (EFZ), VILUP Objective #10 was applied to reduce the total old growth retention target by one-third within the CWHvm1 in the Marble LU subject to identifying suitable second growth for recruitment.

Draft OGMAs in the Holberg, Neroutsos, Keogh, and Mahatta (WFP tenures only (TFL 6 and unregulated Timber Licences)) landscape units have been identified to meet the NSOG order. These draft OGMAs will be used in the timber supply analysis but must complete a public and First Nations' review process before coming legal.

The draft OGMAs for landscape units with a Low BEO identify enough area to meet the old seral target drawn down to 1/3 for the first rotation (80 years) for all biogeoclimatic (BEC) variants. In some cases enough area has been identified to meet the full target. Targets for the end of the second rotation (160 years) and third rotation (240 years) where necessary will be addressed as forest cover constraints (see Section 10.2.3 for details).

Table 11 - TFL 6 OGMA Status and areas

Landscape Unit	BEO	OGMA Status (July 2010)	OGMA Area (ha)	
			Productive	Area Reduction
Marble	Intermediate	Established	3,849	1,688
San Josef	Intermediate	Established	5,191	2,061
Established OGMAs (subtotal)			9,040	3,750
Holberg	Low	Draft	1,925	984
Keogh	Low	Draft	851	569
Mahatta	Low	Draft	2,244	586
Neroutsos	Low	Draft	3,115	1,330
Draft OGMAs (subtotal)			8,135	3,469
OGMAs Total			17,175	7,219

6.10 Wildlife Habitat Areas

At the time the timber supply analysis data set was put together a total of seven Wildlife Habitat Areas (WHAs) had been approved within the boundaries of TFL 6 (WHAs 1-089, 1-106, 1-107, 1-199, 1-200, 1-201 and 1-202). One WHA was established for Northern Goshawk and six WHAs were established for Marbled Murrelets. The WHAs have a total area of 659 ha with minimal impact to the THLB due to, for the most part, being spatially coincidental with OGMAs.

Table 12 - Wildlife Habitat Areas

Description	Productive Wildlife Habitat Area (ha)	Total Area Reduction (ha)
Wildlife Habitat Area - Marbled Murrelet	504	2
Wildlife Habitat Area – Northern Goshawk	154	1
TOTAL	659	3

It should be noted that for the purposes of the Identified Wildlife Management Strategy (IWMS) policy regarding timber supply impact, the THLB impact of these WHAs is determined using MP#9 data and are different than the impacts indicated above in Table 12. Furthermore, the analysis will not make an allowance for full implementation of the IWMS timber supply impact policy (“1% budget”) as this would be speculation as to where the impact will be allocated within the district (IWMS impacts are tracked at the district level).

The Notice issued under Section 7(2) of the Forest Planning and Practices Regulation for “Indicators of the Amount, Distribution and Attributes of Wildlife Habitat Required for the Survival of Species at Risk in the North Island – Central Coast Forest District” requires licensees to maintain an equal amount of suitable habitat for marbled murrelet as existed in the non-THLB land base at the time of the notice (TSR2). For TFL 6 TSR2 is MP #9. An analysis was done of the marbled murrelet habitat inventory

(conducted in 2005 and 2006) against the TSR2 (MP #9) THLB and non-contributing landbase for the current boundaries of TFL 6. This analysis indicated that as of the date of the habitat inventory there was 2,886 ha of habitat in the TSR2 non-contributing landbase (Amount #1 for the Section 7 notice).

The legal OGMAs (Marble and San Josef LUs) captured roughly 697 ha of TSR2 THLB identified as marbled murrelet habitat. These are the only two landscape units within TFL 6 with a “THLB allocation” for OGMAs; therefore, the habitat captured in the legal OGMAs addresses Amount #3 for the Section 7 notice. An earlier analysis indicated that TFL 6 contained approximately 29.6% of the TSR2 THLB for the NICCFD. Applying this factor to the amount of remaining mature THLB budget for marbled murrelet for the district (1,322 ha) results in 391 ha of “THLB allocation” (TSR2) for marbled murrelet within TFL 6 (Amount #4 from the Section 7 notice). Therefore, the maximum amount of habitat assumed to be required to be maintained in TFL 6 is 3,974 ha (2,886 + 697 + 391). Analyzing the MP 10 data indicates that there are at least 4,631 ha of marbled murrelet habitat in the MP 10 non-contributing landbase. Since the model is not allowed to harvest the non-contributing landbase at least this amount of habitat will be maintained. Therefore, no constraint needs to be added to the model to ensure the Section 7 notice for marbled murrelets is met.

6.11 Recreation Features Inventory

Extensive areas of moderately significant recreation features have been identified by the recreation inventory completed in 1990. The majority of these features are associated with riparian features or scenic areas - riparian (see Section 6.7) and visual (see Section 10.2.1) management assumptions are assumed to address those features. More significant features requiring special forest management are identified as Recreation Sites, Trails and Interpretive Forest Sites. These features are netted out by applying a 10 m buffer to the features (see Table 13).

Table 13 – Recreation Sites and Trails

Description	Productive Recreation Area (ha)	Total Area Reduction (ha)
Beaver Lake	1	1
Grant Bay	64	17
Hecht Beach	0	0
Mahatta River	3	1
Marble River	26	13
O'Connell Lake	2	1
Spencer Cove	3	2
Recreation Sites Total	99	35
Beaver Lake Trails	2	0
Brown's Bridge Trail	1	0
Cluxewe Beach Trail	0	0
Grant Bay Trail	0	0
Goodspeed River Riparian Trail	1	0
Hecht Beach Trail	1	0
Lady Ellen Trails	2	0
Spruce Bay Old Growth Trail	34	14
Recreation Trails Total	37	15
Recreation TOTAL	136	50

6.12 Deciduous Stands

Table 14 shows the area of stands defined as deciduous leading in the inventory. This represents about 2.3% of the productive forest area. Harvesting may occur in these areas, but in recent years has been limited. For this analysis these areas are excluded from the THLB, whereas any deciduous volume component in conifer-leading stands will contribute to timber supply.

Table 14 - Deciduous-leading Forest

Description	Productive Deciduous Area (ha)	Total Area Reduction (ha)
Deciduous-leading Stands	3,460	1,774

6.13 Cultural Heritage Resources

Archaeological sites consist of the physical remains of past human activity. Protection and management of impacts to archaeological sites is governed by the *Heritage Conservation Act*. The term cultural heritage resources applies to a variety of heritage resources that is defined in the *Forest Act* as “an object, a site or the location of a traditional societal practice that is of historical, cultural or archaeological significance to British Columbia, a community or an aboriginal people.” Under FRPA objectives set by

government for cultural heritage resources are to conserve, or, if necessary, protect cultural heritage resources that are:

- (a) the focus of a traditional use by an aboriginal people that is of continuing importance to that people, and
- (b) not regulated under the *Heritage Conservation Act*.

An archaeological overview assessment (AOA) for the former Port McNeill Forest District was completed in 1995 by I.R. Wilson Consultants Ltd. The purpose of the AOA is to identify and assess archaeological resource potential. Specifically the AOA provides a basis for predictions regarding archaeological site variability, density and distribution and provide a framework within which to judge the significance of sites.

A Traditional Use Study (TUS) for the Quatsino First Nation's traditional territory was undertaken in 1996 and maintained to present day. The Galgalis Traditional Use Study was compiled for the Kwakiutl Territorial Fisheries Commission in 1998. This study is a collection of input from a number of the Kwakwaka'wakw First Nations on Northern Vancouver Island, the south central coast of BC and the islands in between. Last year, WFP was able to secure FIA funding to work with the Kwakiutl First Nation in taking the initial steps that will lead to the eventual update and maintenance of the information.

First Nations who have completed traditional use studies hold the detailed information regarding traditional use sites and values identified within their asserted traditional territories. TUS information is not normally shared with forest licensees but where this information exists is considered by decision-makers when making statutory decisions.

Approximately 230 proposed cutblocks (totalling roughly 6,300 ha) in TFL 6 have been intensively surveyed for CMTs. This stand-level information has been entered into WFP's GIS database. Of these, 74 proposed cutblocks had CMTs found, and of those 74 cutblocks, 70 were within 1 kilometre of the ocean shoreline or the Marble River.

To address the THLB impacts of protecting cultural heritage resources a 1% incremental netdown will be applied to the portion of TFL 6 that is both within an EFZ and within 1 km of the ocean; elsewhere it is assumed that management of cultural heritage resources is addressed by other netdowns (mainly stand-level retention). The 1% incremental netdown is an estimate of the incremental area retained specifically to conserve CMTs and other cultural heritage resources that is not addressed by other netdowns including stand-level retention discussed in the next section.

Table 15 – Cultural Heritage Resources

Description	Productive CHR Area (ha)	Total Area Reduction (ha)
Cultural Heritage Resources	16,734	132

6.14 Area Reductions to Reflect Stand-level Retention in Cutblocks

Stand-level retention is modelled based on two components: Wildlife Tree Retention Areas (WTRA – formerly referred to as Wildlife Tree Patches (WTPs)) and areas retained to meet retention silviculture system requirements. WTRA are associated with all cutblocks whereas the retention silviculture system is employed in varying amounts according to the Western Forest Strategy. The stand-level retention netdowns are also assumed to partially or fully address management of cultural heritage resources and karst features (see Sections 6.13 and 6.16 respectively).

6.14.1 Wildlife Tree Retention Areas

Where feasible and wildlife objectives can be met, WTRA are located in constrained areas such as riparian reserves, inoperable stands or unstable slopes. In order to capture those WTRA located in otherwise harvestable areas a THLB area reduction is applied.

The January 2005 order establishing OGMA within the San Josef LU also established wildlife tree retention objectives by BEC subzone for that LU: 6% within CWHvh; 10% within CWHvm; 1% within MHmm. For all other LUs it is assumed that the WTRA requirement is 7% for all BEC subzones. The overall area-weighted WTRA requirement is 7.7%.

6.14.2 Western Forest Strategy

Western's Forest Strategy (WFS) includes a program for conserving biodiversity on company tenures. The approach is to vary the use of retention systems and the amount of stand level retention by Resource Management Zones of the Vancouver Island Land Use Plan and by ecosection and is being phased in over the next few years. For this analysis it will be assumed the WFP strategy is fully implemented from the beginning of the analysis period.

In Enhanced Forestry Zones (EFZs) the retention system will be used for 30% of the harvested area in the Nahwitti Lowland Ecosection and 50% of the harvested area in the Northern Island Mountains Ecosection found within TFL 6. The minimum long-term stand-level retention target is 10%.

In General Management Zones (GMZs) the retention system will be used for 40% of the harvested area in the Nahwitti Lowland Ecosection and 60% of the harvested area in the Northern Island Mountains Ecosection found within TFL 6. The minimum long-term stand-level retention target is 15%.

In Special Management Zones (SMZs) the VILUP Higher Level Plan Order specifies: “applying a variety of silvicultural systems, patch sizes and patch shapes across the zone, subject to a maximum cutblock size of 5 ha if clearcut, clearcut with reserves or seed tree silvicultural systems are applied, and 40 ha if shelterwood, selection or retention silvicultural systems are applied.” A minimum of 20% long-term stand-level retention is recommended for SMZs in the Western Forest Strategy based on both social and biological criteria.

This retention is long-term - it must remain for at least one rotation. Applying the above retention system requirements to the Ecosection/VILUP Zone combinations present within TFL 6 results in at least 40% of the total harvest area being in retention system cutblocks (with the remaining being clearcut or clearcut-with-reserves) and an area-weighted average overall minimum stand-level retention requirement of 10.4%. Assuming the first 7.7% of the stand-level retention target is for WTRA (see Section 6.14.1) and applies to all cutblocks, the minimum incremental impact of the WFP Forest Strategy is estimated to be an area-weighted average of 2.7% (10.4% minus 7.7%).

6.14.3 THLB Netdown

In MP#9, a uniform 4% volume reduction was applied to account for WTP retention and an analysis of the gross area of cutblocks harvested from 1997 to 2008 (12 years) indicated that was an appropriate netdown. With the assumption for this timber supply analysis being that stand-level retention will be higher due to implementation of the Western Forest Strategy, an area-weighted average incremental netdown of 5% for stand-level retention will be used. Rather than applying this netdown uniformly across the landbase, it is distributed by RMZ type, landscape unit and BEC subzone (see Table 16) to reflect the varying levels of WTRA and stand-level retention expected across these and as detailed above

Table 16 - THLB % Netdowns for Stand-level Retention

RMZ Type	Landscape Unit	BEC Subzone	Productive Area (ha)	THLB % reduction for WTRA	THLB % reduction for WFS	Total THLB % reduction	Area reduction (ha)
Enhanced	San Josef	CWHvh	3,233	2.9%	0.6%	3.5%	91
		CWHvm	25,996	4.8%	0.0%	4.8%	1,016
		MHmm	0	0.5%	1.3%	1.8%	0
	Others	All	85,061	3.4%	0.9%	4.3%	2,887
General	All	All	17,342	3.4%	2.1%	5.5%	643
Special	San Josef	CWHvh	5,852	2.9%	6.8%	9.7%	372
		CWHvm	9,575	4.8%	4.8%	9.6%	677
		MHmm	0	0.5%	9.2%	9.7%	0
Total / Area-weighted average			147,059	3.7%	1.3%	5.0%	5,687

6.15 Future Roads

A projected road system was developed as part of the physical operability classification for TFL 6. This road system was digitized into the GIS in conjunction with the operability classification, which allowed for the same approach used with existing roads to predict area summaries (see Section 6.4). All future roads are assumed to be branch roads as there are no significant areas not yet developed within TFL 6 requiring mainline access. The area available for timber production will be reduced when the model harvests these polygons.

Table 17 indicates the area of future roads in the TFL that have yet to be developed.

Table 17 - Future roads

Description	Length (km)	Width (m)	Gross Road Area (ha)	Total Area Reduction (ha)
Future roads	1,947	10	1,947	1,491

6.16 Caves and Karst

Portions of TFL 6 are underlain with limestone bedrock that forms part of the Quatsino formation. As a result of the dissolving action of water on limestone (a soluble bedrock) karst topography (characterized by fluted and pitted rock surfaces, vertical shafts, sinkholes, sinking streams, springs, subsurface drainage systems, and caves) often forms. On March 23, 2007, the District Manager of the NICCFD established a GAR order indentifying karst caves, important features and elements within very high and high vulnerability karst and significant surface karst features as resource features requiring management under FRPA.

The impact of protecting karst features on timber supply is uncertain. To date, little area has been reserved during operational planning to protect karst features. Estimates of impacts will improve as operational planning proceeds in karst areas. For this analysis, no netdowns for karst management will be made as it is assumed that any reserves required are accounted for by the stand-level retention allowances (see Section 0).

7 INVENTORY AGGREGATION

This section describes the delineation of the TFL landbase and definition of stand types needed to complete the timber supply analysis. The TFL area is categorized in a hierarchy of different management zones to allow for a variety of forest cover constraints (e.g. biodiversity). Stand types are grouped in analysis units (AU) for projecting timber yields based on similar leading species, history and productivity. Areas within all tables in this section may not sum due to rounding to the nearest hectare.

7.1 Management Zones

Unique forest cover objectives will be modelled through the different management zones. Landscape Units, Special Management Zones (SMZs), General Management Zones (GMZs) and Enhanced Forestry Zones (EFZs) are delineated in the data (Table 18 and Table 19) and will be used to apply forest cover constraints (see Section 10.2 for details).

To reduce the number of unique resource management zones in the data, some minor revisions were made:

1. The area within TFL 6 around the village of Port Alice that is not within a SMZ, GMZ or EFZ was assigned to the Mahatta-Neroutsos EFZ. This affected 312 ha of productive forest and 189 ha of THLB.
2. 11 ha of productive forest (10 ha of THLB) originally identified within the Nahwitti-Tsulquate GMZ were assigned to the Holberg EFZ. The boundary between these two EFZ's is intended to be the TFL 6 boundary.
3. Due to a different height-of-land interpretation, 110 ha of productive forest (46 ha of THLB) originally identified within the Klaskish GMZ were assigned to the Mahatta-Neroutsos EFZ.
4. 31 ha of productive forest (6 ha of THLB) originally identified within the Marble River Protected Area were assigned to the Marble GMZ. This revision is the result of different boundaries for the Marble River Park between the provincial park data and the RMZ data.

Table 18 - Area by Resource Management Zone

Mgmt Zone	Mgmt Unit	Seral ¹ Stage	Productive Forest (ha)	THLB Area (ha)	Management Considerations (from Vancouver Island Summary Land Use Plan)
EFZ 5	Holberg	Early	10,490	9,275	Enhanced Forestry Zone suited for enhanced timber harvesting and production, while maintaining fish values and watershed integrity.
		Mid	6,388	5,151	
		Mature	2,241	1,638	
		Old	6,170	3,713	
		Total	25,289	19,778	

¹ Early seral is <40 years old; Mid seral is 40-80 years old in CWH zone and 40-120 years old in MH zone; Mature seral is 81-250 years old in CWH zone and 121-250 years old in MH zone; Old seral is >250 years old.

Mgmt Zone	Mgmt Unit	Seral ¹ Stage	Productive Forest (ha)	THLB Area (ha)	Management Considerations (from Vancouver Island Summary Land Use Plan)
EFZ 6	Keogh - Cluxewe	Early	6,240	5,572	Enhanced Forestry Zone suited for enhanced silviculture, with limited opportunity for enhanced timber harvesting; integration of visual values along coastline and highway corridor, as well as recreational opportunities along Keogh River.
		Mid	5,409	4,418	
		Mature	1,538	1,110	
		Old	1,921	1,292	
		Total	15,108	12,392	
SMZ 4	Koprino	Early	2,021	1,748	Special Management Zone should be focal area (within the landscape unit) for the retention of old forest and associated wildlife habitat, as well as for mature and old forest connectivity.
		Mid	241	167	
		Mature	364	272	
		Old	3,016	1,366	
		Total	5,642	3,553	
EFZ 8	Mahatta - Neroutsos	Early	15,740	13,888	Enhanced Forestry Zone suited for enhanced timber harvesting and silviculture; wildlife values in Mahatta system and marbled Murrelet values in noted drainages require specific integration through maintenance of old seral forest; objectives for other resources are to be integrated at the basic stewardship level.
		Mid	8,990	6,366	
		Mature	6,125	4,312	
		Old	14,323	7,809	
		Total	45,178	32,375	
GMZ 7	Marble	Early	5,025	4,139	General Management Zone particularly suited for enhanced silviculture in second growth stands; high fisheries values, wildlife values/capability, as well as ecosystem representation and connectivity functions result in intermediate biodiversity significance; integration of recreational values associated with lakes.
		Mid	5,179	4,113	
		Mature	2,411	1,554	
		Old	4,606	1,243	
		Total	17,222	11,048	
EFZ 4	San Josef - Koprino	Early	13,160	11,704	Enhanced Forestry Zone suited for enhanced timber harvesting and production, while maintaining fish values and watershed integrity.
		Mid	2,661	2,049	
		Mature	3,873	3,137	
		Old	9,140	5,496	
		Total	28,835	22,386	
SMZ 2	West Coast Nahwitti Lowlands	Early	2,240	1,901	Special Management Zone with main focus on special management for significant scenic and recreational values which are concentrated along narrow coastal strip; additional consideration should be on maintenance of the high riparian fish and coastal wildlife values.
		Mid	994	631	
		Mature	1,209	922	
		Old	5,342	2,824	
		Total	9,785	6,279	
Grand Total			147,059	107,811	

Table 19 - Area by landscape unit and BEC variant

Landscape Unit	BEC	Seral Stage	Productive Forest (ha)	Non Contributing Area		THLB Area	
				ha	%	ha	%
Holberg	CWHvh1	Early	2,011	224	11%	1,787	89%
		Mid	936	216	23%	720	77%
		Mature	134	27	20%	107	80%
		Old	1,847	698	38%	1,149	62%
	CWHvh1 Total		4,928	1,165	24%	3,763	76%
	CWHvm1	Early	8,118	970	12%	7,148	88%
		Mid	5,352	1,035	19%	4,317	81%
		Mature	2,107	539	26%	1,568	74%
		Old	3,792	1,552	41%	2,241	59%
	CWHvm1 Total		19,370	4,096	21%	15,274	79%
	CWHvm2	Early	645	39	6%	606	94%
		Mid	156	11	7%	145	93%
		Mature	36	2	5%	34	95%
Old		698	268	38%	429	62%	
CWHvm2 Total		1,535	320	21%	1,214	79%	
Holberg Total			25,832	5,581	22%	20,251	78%
Keogh	CWHvm1	Early	5,040	559	11%	4,481	89%
		Mid	4,613	850	18%	3,763	82%
		Mature	1,532	474	31%	1,058	69%
		Old	1,336	485	36%	851	64%
	CWHvm1 Total		12,521	2,368	19%	10,153	81%
	CWHvm2	Early	806	96	12%	709	88%
		Mid	480	97	20%	383	80%
		Mature	0	0	N/A	0	N/A
		Old	222	90	41%	132	59%
	CWvm2 Total		1,508	284	19%	1,224	81%
Keogh Total			14,029	2,651	19%	11,378	81%
Mahatta	CWHvm1	Early	7,796	1,030	13%	6,767	87%
		Mid	5,076	1,363	27%	3,713	73%
		Mature	1,851	509	27%	1,343	73%
		Old	3,897	1,866	48%	2,031	52%
	CWHvm1 Total		18,620	4,767	26%	13,853	74%
	CWHvm2	Early	1,298	114	9%	1,184	91%
		Mid	169	13	8%	156	92%
		Mature	218	154	71%	64	29%
		Old	1,906	971	51%	935	49%
	CWHvm2 Total		3,591	1,251	35%	2,340	65%
	MHmm1	Early	0	0	N/A	0	N/A
		Mid	0	0	N/A	0	N/A
		Mature	0	0	N/A	0	N/A
Old		3	3	100%	0	0%	
MHmm1 Total		3	3	100%	0	0%	
Mahatta Total			22,214	6,021	27%	16,193	73%

Landscape Unit	BEC	Seral Stage	Productive Forest (ha)	Non Contributing Area		THLB Area	
				ha	%	ha	%
Marble	CWHvm1	Early	4,749	841	18%	3,908	82%
		Mid	5,221	1,085	21%	4,136	21%
		Mature	2,094	592	28%	1,502	72%
		Old	2,638	1,822	69%	816	31%
	CWHvm1 Total		14,702	4,341	30%	10,361	70%
	CWHvm2	Early	407	41	10%	366	90%
		Mid	281	26	9%	255	91%
		Mature	312	256	82%	56	18%
		Old	2,290	1,589	69%	702	31%
	CWHvm2 Total		3,290	1,912	58%	1,378	42%
Marble Total			17,992	6,253	35%	11,739	65%
Neroutsos	CWHvm1	Early	5,238	605	12%	4,632	88%
		Mid	3,402	1,162	34%	2,240	66%
		Mature	3,121	714	23%	2,407	77%
		Old	5,169	2,159	42%	3,010	58%
	CWHvm1 Total		16,929	4,640	27%	12,289	73%
	CWHvm2	Early	1,365	98	7%	1,267	93%
		Mid	290	51	18%	239	82%
		Mature	488	310	64%	178	36%
		Old	3,264	1,490	46%	1,774	54%
	CWHvm2 Total		5,407	1,949	36%	3,458	64%
Neroutsos Total			22,336	6,590	30%	15,746	70%
San Josef	CWHvh1	Early	2,631	364	14%	2,268	86%
		Mid	165	73	44%	92	56%
		Mature	820	208	25%	612	75%
		Old	5,469	2,467	45%	3,002	55%
	CWHvh1 Total		9,085	3,112	34%	5,973	66%
	CWHvm1	Early	14,311	1,663	12%	12,648	88%
		Mid	3,712	980	26%	2,732	74%
		Mature	4,979	1,013	20%	3,966	80%
		Old	10,894	4,703	43%	6,191	57%
	CWHvm1 Total		33,895	8,359	25%	25,537	75%
CWHvm2	Early	501	45	9%	456	91%	
	Mid	11	6	53%	5	47%	
	Mature	69	19	27%	50	73%	
	Old	1,095	613	56%	482	44%	
CWHvm2 Total		1,675	682	41%	993	59%	
San Josef Total			44,655	12,152	27%	32,503	73%
GRAND TOTAL			147,059	39,248	27%	107,811	73%

7.2 Analysis Units

The forested area in the THLB is aggregated into groups of similar stands to produce growth and yield information needed to model timber supply. For existing stands, analysis units (AUs) are based on ecosystem groupings within site productivity classes, age class, leading species, and silviculture treatments. For future stands the leading species criteria is replaced with establishment method (planting or natural regeneration). These criteria are described in more detail in the following sections.

7.2.1 Ecosystem assignment

Ecosystems were assigned using the TFL 6 Ecosite classification system completed in 1985 by Terence Lewis, Ph.D., P.Ag., P.Geo.. This classification system differs from the later published MoFR's BEC system (1994) in that it also incorporates terrain types. In 2007, Lewis updated the classification system based on learnings and research results from the previous 20 years. The revised classification system aligns more closely to the MoFR's BEC system. The 1985 Lewis classification system is being used to define analysis units in this analysis due to site productivity estimates being based on that system (see section 8.1 for details). Each polygon in the TFL was assigned to one analysis unit level ecosite. To limit the number of unique combinations some ecosites were grouped based on similar site productivity and similar future management assumptions (Table 20).

Table 20 - Analysis Units Ecosystems

Ecosite Group	Area (ha)	
	Productive Forest	THLB
S3	6,955	3,384
S1HA	72,028	59,142
S4, S5	816	276
S12, S13	16,553	9,375
S12F	3,023	2,619
M1, M3, M5	14,537	9,631
S1CH	19,873	15,620
S2, S6, S10	9,698	6,109
S2F	1,031	615
M2, M4	2,545	1,039
Total	147,059	107,811

7.2.2 Age class

Stands were assigned to five different age classes based on management era. Ages are based on known or estimated ages as of the year 2011. 2011 was chosen as this is the mid-year of the first five-year period (2009-2013) of the analysis so this represents the average age, and therefore average estimated volume, for the first period of the analysis.

7.2.2.1 Unmanaged Stands

7.2.2.1.1 Mature

Mature stands are defined as stands greater than 140 years of age. Volume in these stands is estimated using MFML's *Variable Density Yield Projection* (VDYP) version 6.6d.

7.2.2.1.2 Immature

These are stands that are between 51 years old and 140 years old. The assumption is that these stands are the results of natural regeneration following harvesting or natural disturbances. Volume in these stands is estimated using VDYP.

7.2.2.2 Managed Stands

These stands have been established since active forest management began within the TFL. Most of these stands are the result of planting but there are naturally regenerated stands present in these age ranges. Volume in these stands is estimated using MFML's *Table Interpolation Program for Stand Yields* (TIPSY) version 4.1.

7.2.2.2.1 Stands aged 11 – 50 years

These stands have been established since the inception of active forest management but with insignificant genetic gain values. Natural regeneration is more prevalent in this age range (especially the older ages in this range) than in the 1-10 year old range.

7.2.2.2.2 Stands aged 1 – 10 years

These most recently established stands have genetic gain values and, on average, higher establishment densities than the older managed stands. The higher establishment densities are intended to reflect the higher proportion of planting compared to the older managed stands.

7.2.2.2.3 Future stands

These stands have genetic gain values equal to the 1 – 10 year old stands and will assume to be influenced by higher levels of stand-level retention from the previous harvest due to the use of the retention silviculture system.

7.2.3 Leading Species

For existing stands, the leading species is used to differentiate analysis units in order to better reflect the differences in expected growth rates by species. To reduce the possible number of unique combinations, some resulting analysis units are combined where this differentiation resulted in little area (generally less than 5 ha) within an analysis unit.

7.2.4 Silviculture Treatments

For existing stands, fertilization and juvenile spacing treatments are used to differentiate analysis units in order to better reflect the differences in expected growth rates in stands following these treatments. For analysis unit allocation, fertilization takes precedence over spacing if a stand has been both fertilized and spaced. Again, to reduce the number of unique combinations, consequential analysis units are combined where this differentiation resulted in little area (usually less than 5 ha) within an analysis unit. For future stands, fertilization of a proportion of S1CH ecosites (cedar-salal or “SCHIRP” sites) will be modelled (see section 8.6.2 for details) and it is assumed no juvenile spacing will take place.

7.2.5 Regeneration Method

Natural regeneration of western hemlock and balsam can be relied upon to meet regeneration requirements on some sites within TFL 6: typically the S1HA, S2, M1, M2 and M4 ecosites. Natural regeneration success has been more prevalent in the former Jeune Landing operation than elsewhere. For future stands, natural regeneration will be modelled on a proportion of the appropriate ecosites with the rest being modelled assuming regeneration by planting (see section 8.6.1 for details).

7.2.6 Analysis unit codes

A 4-digit code identifies the ecosite group, age class, leading species (for existing stands) or regeneration method (for future stands) and silviculture treatment for each analysis unit (Table 21 and Table 22).

Table 21 – Existing Stand Analysis Units Legend

First Digit Ecosite	Second Digit Age Class	Third Digit Leading Species	Fourth Digit Treatment
0 S3	1 1 – 10 years old	1 Ba	0 None
1 S1HA	2 11 – 50 years old	2 Cw	1 Fertilized
2 S4, S5	3 51 – 140 years old	3 Cy	2 Spaced
3 S12, S13	4 141+ years old	4 Fd	
4 S12F		5 Hw	
5 M1, M3, M5		6 Ss	
6 S1CH		7 Other / grouped	
7 S2, S6, S10			
8 S2F			
9 M2, M4			

Table 22 – Future Stand Analysis Units Legend

First Digit <i>Ecosite</i>	Second Digit <i>Age Class</i>	Third Digit <i>Regen Method</i>	Fourth Digit <i>Treatment</i>
0 S3	5 Future	0 Planted	0 None
1 S1HA		1 Natural	1 Fertilization
2 S4, S5			
3 S12, S13			
4 S12F			
5 M1, M3, M5			
6 S1CH			
7 S2, S6, S10			
8 S2F			
9 M2, M4			

For example, the code 1450 identifies the S1HA/141+ years old/Hw-leading/no silviculture treatment analysis unit.

8 GROWTH AND YIELD

This section describes the approach used to develop yield tables for managed and natural stands. The general approach is to develop yield tables for existing and future stands, thus specific yield tables are developed for:

- 1) Existing natural mature stands.
- 2) Existing natural immature stands.
- 3) Existing managed stands.
- 4) Future managed stands.

8.1 Site Index

Site index (the total height of the dominant trees in a stand at age 50 years) is the most common measure of forest site productivity and forest growth used in British Columbia. It provides a numeric description of site productivity that enables forest managers to predict forest stand growth and the yield of timber at harvest. Site index allows the comparison of productive potential between sites across a broad range of existing stand conditions.

Site indices for the main commercial tree species are the same as used in MP #9 and are based on a study completed by J.S. Thrower and Associates (now part of Teco Resource Management Group) in 1997 (see Appendix A). This study determined statistically-based average site indices by tree species and productivity groups created by grouping Lewis ecological units having similar site potential.

Table 23 shows the site index by species and productivity group and the THLB area-weighted average by species.

Table 23 - Site Index Values by Species and Productivity Group

Productivity Group	Ecosystems	Productive Forest (ha)	THLB (ha)	Site Index by species (m)				
				Ba	Cw / Cy	Fd	Hw	Ss
1	S3	6,955	3,384	30.1	27.0	37.5	33.7	35.2
2/3	S1HA, S4, S5, S12, S12F, S13, M1, M3, M5	106,957	81,043	26.2	23.8	31.4	28.0	31.0
4	S1CH, S2, S2F, S6, S10, M2, M4	33,147	23,383	20.7	19.2	22.6	20.1	25.0
Total / THLB area-weighted average		147,059	107,811	25.1	22.9	29.7	26.5	29.8

8.2 Utilization Levels

The utilization level is 12.5 cm DBH for all existing managed stands (stands less than 51 years old) and for future stands. Stump height for these stands is 30 cm and top diameter inside bark (DIB) is 10 cm. Utilization level for natural stands is 17.5 cm DBH, with stump height of 30 cm and top DIB of 15 cm

(Table 24). Operationally, natural immature stands (age 51 -140 years) are utilized to a 12.5 cm DBH; however, volumes for these stands are being estimated using VDYP and VDYP does not accept the smaller DBH standards for TFL 6.

Table 24 - Utilization levels

Age Class	Utilization			Firmwood Standard
	Minimum DBH (cm)	Stump Height (cm)	Top DIB (cm)	
Unmanaged (>50 years old)	17.5	30.0	15.0	50%
Managed (<51 years old)	12.5	30.0	10.0	50%

8.3 Operational Adjustment Factors

Managed stand yield tables are created using TIPSy. Unadjusted TIPSy output reflects the growth relationships observed in research plots that were generally located in fully-stocked, even-aged stands of uniform site and in forests of little or no pest activity. To reflect operational environments, two operational adjustment factors (OAFs) are applied to TIPSy outputs to reduce the potential yields. OAF 1 is constant across all ages and is intended to account for small, unmapped non-productive areas in a stand and competition from non-commercial tree species and brush. OAF 2 increases with age and is intended to reflect the impact of insects, disease and decay. Typically OAF 1 is 15%; however, results from random samples placed in second growth stands within the TFL indicate a low proportion of unmapped non-productive area such that an OAF 1 of 10% (as was used in MP #9) better reflects the stocking levels achieved on TFL 6.

Adjustments to managed stand volumes are the same as those used in MP #9:

1. OAF 1: 10%
2. OAF 2: 5%

8.4 Volume Reductions

8.4.1 Unmanaged Stands Volume

Unmanaged stands volumes are estimated using VDYP. VDYP normally computes the volume of standing wood less the amount that has decayed. Default VDYP waste and breakage values for TFL 6 will be applied to account for wood lost due to falling and yarding (breakage) and wood lost due to short log segments and other reasons (waste) plus the sound wood portion of a tree or log which has more than 50% decay (waste₂).

8.4.2 Managed Stands Volume

With the implementation of the Western Forest Strategy for conserving biological diversity (see section 6.14.2 for details) future stands volumes will be reduced to account for the growth impact of stand-level retention from the previous harvest. A volume reduction of 2% will be applied to all future stands in Enhanced Forestry Zones where stand-level retention is assumed to average at least 10%; a 3% volume

reduction will be applied to all future stands in General Management Zones where average stand-level retention will be at least 15%; and a 5% volume reduction will be applied to all future stands in Special Management Zones to account for stand-level retention of at least 20%. These reductions are based on similar assumptions used in previous analyses for TFLs 39 and 44. The general notion that growth impacts on forest regeneration increase with greater retention has been observed in early results from research being conducted by Dr. Nick Smith within TFLs 39 and 44.

The yield reduction will occur when individual stands are harvested during modelling. Yield curves are left unaltered.

8.5 Yield Tables for Unmanaged Stands

Unmanaged stands are those stands that are projected to be greater than 50 years old in the year 2011. The projected ages of these stands are based on the adjusted ages in the VRI database for 2001, which was the reference year for the statistical adjustment (see Appendix B). Adjusted site indices were used as heights in 2011 are not known. Finally the volume adjustment factors were applied. Crown closure values from the inventory were used except for those stands that were less than 50 years old in 1995 (the year of the photos used in Phase I of the VRI process and therefore the year of the crown closure estimate). In VDYP, stands younger than 50 years old use default crown closure values by species.

VDYP outputs were generated for every applicable resultant polygon and then area-weighted averages were determined for each analysis unit. Separate tables were generated for the THLB and the non-contributing landbase – only the THLB yields are included in this document.

8.5.1 Existing Mature Stand Volumes

The timber volume in existing mature stands (those > 140 years old in 2011) was determined for each analysis unit by calculating the area-weighted average inventory volumes and species composition (Table 25). The yield curves for each mature analysis unit are listed and shown in Appendix C.

Table 25 - Existing mature volume

Analysis Unit	THLB Area (ha) ¹	Weighted Avg Volume/ha (m ³ /ha) ²	Inventory Volume (m ³)	Ba %	Cw %	Cy %	Fd %	Hw %	Ss %	Other %
0410	43	1,055	45,221	56.2	1.9	3.8	0.3	35.1	2.5	0.2
0420	100	720	71,650	1.8	70.5	1.2	-	21.8	2.7	2.0
0450	247	938	231,610	16.3	6.7	0.1	-	68.7	6.8	1.4
0460	53	1,136	60,513	3.0	3.9	-	0.4	28.3	64.4	-
1410	210	1,069	224,656	57.8	4.7	0.4	-	37.0	0.1	-
1420	383	829	317,578	1.5	68.1	1.0	0.3	28.6	0.4	0.1
1450	7,891	924	7,295,531	16.5	10.6	0.3	0.2	70.7	1.7	-
1460	51	1,163	58,976	1.6	3.7	-	-	36.0	58.7	-
1470	55	689	38,068	1.1	10.7	40.5	17.5	29.4	0.4	0.4
2470	78	920	71,385	10.9	32.4	0.5	-	44.4	11.4	0.4

¹ THLB area is rounded to nearest hectare

² Average volume per hectare is rounded to zero decimal places

Analysis Unit	THLB Area (ha) ¹	Weighted Avg Volume/ha (m ³ /ha) ²	Inventory Volume (m ³)	Ba %	Cw %	Cy %	Fd %	Hw %	Ss %	Other %
3410	44	1,035	45,959	55.2	6.3	0.7	-	37.7	-	0.1
3420	500	839	419,012	3.7	63.4	3.3	-	29.1	0.3	0.2
3450	2,880	938	2,700,095	18.7	9.3	0.3	0.2	70.1	1.4	-
3470	45	989	44,340	0.1	4.0	5.7	25.2	28.0	35.3	1.7
4440	41	739	30,317	0.1	5.2	-	66.4	28.3	-	-
4450	127	852	108,059	6.8	3.5	-	12.6	74.8	2.2	0.1
5410	70	982	68,797	56.3	3.7	2.0	0.9	36.4	-	0.7
5420	497	752	373,235	3.9	62.0	7.6	0.2	24.8	0.2	1.3
5430	114	502	57,313	4.8	11.2	58.3	-	15.8	-	9.9
5450	3,302	805	2,657,625	20.1	9.1	3.6	-	65.8	-	1.4
6420	4,126	748	3,087,373	2.0	69.9	4.1	-	23.5	0.3	0.2
6430	147	622	91,383	1.2	28.6	49.8	-	20.2	-	0.2
6450	381	859	327,366	11.3	20.1	1.5	-	65.2	1.6	0.3
7420	1,379	615	847,624	1.3	77.9	1.0	-	17.7	0.6	1.5
7450	333	789	262,600	11.1	16.6	0.3	-	69.2	2.2	0.6
7470	36	762	27,427	2.7	20.6	24.5	-	12.0	38.1	2.1
8420	102	607	61,661	1.6	71.7	-	0.5	26.2	-	-
8470	39	762	29,510	7.8	13.7	1.3	14.5	57.9	4.8	-
9420	239	521	124,633	1.6	65.2	10.5	-	19.6	-	3.1
9430	16	480	7,448	7.4	16.7	47.5	-	24.0	-	4.4
9450	217	741	160,823	14.7	13.7	4.3	-	66.5	0.1	0.7
Total	23,744	840	19,947,787	13.8	25.6	2.2	0.5	55.7	1.7	0.4

8.5.2 Unmanaged Immature Stand Volumes

These are stands that are between 51 years old and 140 years old (in 2011). The assumption is that these stands are the results of natural regeneration following harvesting or natural disturbances. Volume in these stands is estimated using VDYP.

The yield curves for each analysis unit are listed and shown in Appendix D.

Table 26 - Unmanaged immature stand volumes

Analysis Unit	THLB Area (ha) ¹	Weighted Avg Volume/ha (m ³ /ha) ²	Inventory Volume (m ³)	Ba %	Cw %	Cy %	Fd %	Hw %	Ss %	Other %
0350	687	608	417,420	5.3	1.1	-	0.4	78.7	7.8	6.7
0352	231	560	129,356	10.5	2.1	-	-	79.0	3.6	4.8
0370	25	513	12,767	9.1	41.5	-	1.0	20.8	26.6	1.0
1310	109	805	87,730	60.3	0.4	0.1	-	39.2	-	-
1320	32	268	8,713	0.2	69.4	-	0.5	28.5	0.5	0.9
1340	43	630	27,404	3.0	1.9	-	52.9	37.3	2.3	2.6
1350	14,570	627	9,142,065	8.4	2.9	-	0.8	81.4	3.7	2.8
1351	48	321	15,482	14.6	11.0	-	5.8	66.4	1.3	0.9
1352	2,763	541	1,495,555	9.8	1.8	-	0.2	83.6	1.5	3.1
1360	32	757	24,496	0.2	1.2	-	-	24.1	73.1	1.4

¹ THLB area is rounded to nearest hectare

² Average volume per hectare is rounded to zero decimal places

Analysis Unit	THLB Area (ha) ¹	Weighted Avg Volume/ha (m ³ /ha) ²	Inventory Volume (m ³)	Ba %	Cw %	Cy %	Fd %	Hw %	Ss %	Other %
2370	46	608	28,014	7.5	4.9	-	0.2	74.5	9.8	3.1
3350	1,439	552	794,817	14.2	2.3	-	1.2	74.9	5.1	2.3
3351	50	486	24,413	1.3	0.4	-	23.2	73.1	0.4	1.6
3352	122	488	59,406	10.8	1.7	-	-	84.3	0.8	2.4
3370	26	395	10,383	8.8	22.4	-	28.4	34.7	4.7	1.0
4340	152	494	75,118	6.5	7.6	-	50.1	34.5	0.4	0.9
4350	1,349	606	818,243	5.9	3.4	-	5.2	78.4	6.2	0.9
4352	191	561	106,886	11.0	1.0	-	0.3	86.8	-	0.9
4370	30	688	20,346	8.3	8.1	-	2.7	24.1	56.8	-
5350	378	460	173,918	18.1	2.7	0.1	1.1	76.3	0.3	1.4
5370	58	503	29,363	45.1	12.3	-	3.0	38.6	-	1.0
6320	382	171	35,247	0.3	72.4	-	-	22.6	1.3	3.4
6321	37	166	6,161	0.1	54.0	-	-	33.2	5.2	7.5
6350	566	375	212,341	5.0	13.5	-	0.2	75.9	1.6	3.8
6351	157	360	56,321	5.1	25.3	-	-	66.5	0.1	3.0
6370	36	364	12,953	9.9	8.9	-	15.5	28.1	35.2	2.4
7320	210	150	31,486	0.2	71.7	-	-	23.6	2.2	2.3
7350	713	476	339,701	6.2	9.0	-	0.3	79.9	1.8	2.8
7360	10	475	4,759	6.0	10.4	-	7.6	30.5	41.9	3.6
8350	262	474	124,305	3.1	10.5	-	10.9	70.6	2.8	2.1
9370	46	445	20,417	13.3	19.9	-	1.5	64.1	0.6	0.6
Total	24,801	580	14,375,197	9.0	3.7	-	1.5	79.3	3.8	2.8

8.6 Yield Tables for Managed Stands

8.6.1 Stocking density

A significant planting program has existed in TFL 6 since the early 1960's. For the last 15 to 20 years most of the harvested area has been planted, typically at planting levels of around 1,200 sph and many areas also include substantial natural in-growth. TIPSy does not directly model planted stands with natural in-growth. Planted stands currently aged 1 to 10 years and future stands are modelled at initial densities such that the effect of the OAFs reduces stand densities to approximately the target densities at free-growing ages (generally 15 years). Initial densities for naturally regenerated stands reflect a generalization of densities found at regeneration delay and free-growing dates.

Stands currently aged 11 to 50 years are modelled with lower initial densities (relative to younger current and future stands) due to a higher reliance on natural regeneration.

8.6.2 Fertilization

Since 1986, nitrogen fertilization (post-establishment) has occurred on approximately 15,000 ha in TFL 6, with approximately 4,500 ha having received two treatments. The fertilization treatments have mostly occurred within the S1CH and S1HA ecosites. The treatments have typically been mid-late rotation of Douglas fir with urea (N200) or urea and phosphorous (N200/P75) applied at free-growing plus urea only

about 10 years later on the cedar sites (S1CH). The fertilization program has been contingent on government funding programs and is expected to continue in the next few years.

Fertilization is incorporated into the yield tables for current and future stands. The impact of fertilization is modelled for cedar, hemlock and spruce by using the site index from the next higher productivity group. The default TIPSy increases are used for Douglas fir with the assumption being that a single treatment is done at age 30 years. No volume gains are modelled for any other species. The gains for cedar, hemlock and spruce are supported by the most recent (fall of 2009) measurements (not yet published Master degree thesis) at the 'SCHIRP' (Salal Cedar Hemlock Integrated Research Program) site located within TFL 6 near Port McNeill.

8.6.3 Volumes for Existing Managed Stands Aged 11 - 50 Years

Silviculture assumptions for existing managed stands aged 11 – 50 years included a plantation regeneration method for all stands, species composition from the inventory database, establishment density based on inventory and free-growing stand data and expected relative stocking success. These silviculture assumptions and site index estimates by species were used as input into Batch TIPSy 4.1 with outputs generated for every applicable resultant polygon and then area-weighted averages determined for each analysis unit. See Table 27 for the average of the inputs by analysis unit. No genetic gain was applied to stands in this age range.

Table 27- TIPSy inputs for existing managed stands aged 11 – 50 years

AU	Initial SPH	Spaced SPH								Cw &					THLB Area (ha)
			Ba %	Cw %	Cy %	Fd %	Hw %	Ss %	Other %	Ba SI	Cy SI	Fd SI	Hw SI	Ss SI	
0210	900		54	-	-	-	25	21	-	30.1	27.0	37.5	33.7	35.0	178
0220	900		-	84	-	-	16	-	-	30.1	27.0	37.5	33.7	35.0	184
0250	900		7	6	-	-	77	10	-	30.1	27.0	37.5	33.7	35.0	772
0252	1,600	900	-	-	-	-	91	9	-	30.1	27.0	37.5	33.7	35.0	66
0260	900		-	8	-	-	25	67	-	30.1	27.0	37.5	33.7	35.0	481
0262	1,600	900	-	-	-	-	25	75	-	30.1	27.0	37.5	33.7	35.0	50
1210	900		64	-	-	-	36	-	-	26.2	23.8	31.4	28.0	31.0	567
1211	900		63	10	-	-	27	-	-	26.2	27.0	31.4	33.7	35.0	45
1220	900		-	75	-	-	25	-	-	26.2	23.8	31.4	28.0	31.0	475
1221	900		-	79	-	-	21	-	-	26.2	27.0	31.4	33.7	35.0	430
1230	900		-	9	60	-	31	-	-	26.2	23.8	31.4	28.0	31.0	29
1240	900		-	-	-	69	31	-	-	26.2	23.8	31.4	28.0	31.0	256
1241	900		-	-	-	68	32	-	-	26.2	27.0	31.4	33.7	35.0	109
1242	1,600	900	-	-	-	73	27	-	-	26.2	23.8	31.4	28.0	31.0	143
1250	900		9	8	-	-	83	-	-	26.2	23.8	31.4	28.0	31.0	17,840
1251	900		-	19	-	-	81	-	-	26.2	27.0	31.4	33.7	35.0	1,925
1252	1,600	900	8	-	-	-	92	-	-	26.2	23.8	31.4	28.0	31.0	3,660
1260	900		-	-	-	-	33	67	-	26.2	23.8	31.4	28.0	31.0	472
1262	1,600	900	-	-	-	-	27	73	-	26.2	23.8	31.4	28.0	31.0	58
2270	900		-	35	-	-	35	30	-	26.2	23.8	31.4	28.0	31.0	128
3210	900		68	-	-	-	32	-	-	26.2	23.8	31.4	28.0	31.0	103
3220	900		-	75	-	-	25	-	-	26.2	23.8	31.4	28.0	31.0	150
3221	900		-	75	-	-	25	-	-	26.2	27.0	31.4	33.7	35.0	167
3250	900		10	9	-	-	81	-	-	26.2	23.8	31.4	28.0	31.0	2,382
3251	900		1	13	-	-	86	-	-	26.2	27.0	31.4	33.7	35.0	190
3252	1,600	900	5	8	-	-	81	6	-	26.2	23.8	31.4	28.0	31.0	229
3270	900		-	6	22	21	21	30	-	26.2	23.8	31.4	28.0	31.0	71
4240	900		7	3	-	61	29	-	-	26.2	23.8	31.4	28.0	31.0	120

AU	Initial SPH	Spaced SPH								Cw &					THLB Area (ha)
			Ba %	Cw %	Cy %	Fd %	Hw %	Ss %	Other %	Ba SI	Cy SI	Fd SI	Hw SI	Ss SI	
4241	900		-	-	-	55	40	-	5	26.2	27.0	31.4	33.7	35.0	59
4250	900		9	8	-	14	64	-	5	26.2	23.8	31.4	28.0	31.0	104
5210	900		62	-	4	-	34	-	-	26.2	23.8	31.4	28.0	31.0	626
5220	900		-	75	-	-	25	-	-	26.2	23.8	31.4	28.0	31.0	204
5221	900		-	80	-	-	20	-	-	26.2	27.0	31.4	33.7	35.0	42
5230	900		16	8	52	-	24	-	-	26.2	23.8	31.4	28.0	31.0	168
5250	900		17	6	-	-	77	-	-	26.2	23.8	31.4	28.0	31.0	2,758
5251	900		12	8	-	-	80	-	-	26.2	27.0	31.4	33.7	35.0	69
5252	1,600	900	23	-	-	-	77	-	-	26.2	23.8	31.4	28.0	31.0	263
6210	1,200		80	7	-	-	13	-	-	20.7	19.2	22.6	20.1	25.0	47
6220	1,200		-	76	-	-	24	-	-	20.7	19.2	22.6	20.1	25.0	2,637
6221	1,200		-	84	-	-	16	-	-	20.7	23.8	22.6	28.0	31.0	3,398
6230	1,200		-	16	71	-	13	-	-	20.7	19.2	22.6	20.1	25.0	68
6240	1,200		-	22	-	70	8	-	-	20.7	19.2	22.6	20.1	25.0	43
6250	1,200		-	20	-	-	80	-	-	20.7	19.2	22.6	20.1	25.0	1,270
6251	1,200		-	26	-	-	69	5	-	20.7	23.8	22.6	28.0	31.0	977
6261	1,200		-	26	-	-	24	50	-	20.7	23.8	22.6	28.0	31.0	22
6270	1,200		-	15	-	-	27	46	12	20.7	19.2	22.6	20.1	25.0	54
7220	800		-	78	-	-	22	-	-	20.7	19.2	22.6	20.1	25.0	730
7221	800		-	88	-	-	12	-	-	20.7	23.8	22.6	28.0	31.0	715
7250	800		6	14	-	-	74	6	-	20.7	19.2	22.6	20.1	25.0	999
7251	800		-	24	-	-	68	8	-	20.7	23.8	22.6	28.0	31.0	266
7252	1,600	900	7	6	-	5	76	6	-	20.7	19.2	22.6	20.1	25.0	64
7260	800		-	13	-	-	26	61	-	20.7	19.2	22.6	20.1	25.0	72
7261	800		-	30	-	-	16	54	-	20.7	23.8	22.6	28.0	31.0	36
7270	800		-	10	39	11	16	-	24	20.7	19.2	22.6	20.1	25.0	95
8250	800		-	13	-	12	75	-	-	20.7	19.2	22.6	20.1	25.0	129
8270	800		-	53	-	26	21	-	-	20.7	19.2	22.6	20.1	25.0	48
9220	800		-	83	-	-	17	-	-	20.7	19.2	22.6	20.1	25.0	108
9221	800		-	88	-	-	12	-	-	20.7	23.8	22.6	28.0	31.0	35
9250	800		17	9	-	-	74	-	-	20.7	19.2	22.6	20.1	25.0	256
9270	800		38	5	28	-	29	-	-	20.7	19.2	22.6	20.1	25.0	16
Total / THLB area-weighted average			8	22	-	1	65	2	-	24.9	23.5	29.4	27.5	30.3	47,655

The yield curves for each existing managed age 11 – 45 years analysis unit are listed and shown in Appendix E.

8.6.4 Volumes for Existing Managed Stands Aged 1 - 10 Years

Silviculture assumptions for existing managed stands aged 1 – 10 years included a plantation regeneration method for all stands, species composition from the inventory database (simplified to eliminate minor species present) or future stand species assumptions where no current species information was available, and establishment density that reflects stocking success. These silviculture assumptions and site index estimates by species were used as input into Batch TIPSYS 4.1 with outputs generated for every applicable resultant polygon and then area-weighted averages determined for each analysis unit. Genetic gain for Cw, Cy, Fd and Hw was applied to stands in this age range based on average values for the most common seedlots planted in TFL 6 since 2000. Expected GW values for Hw are reduced to reflect a component of natural regeneration expected in the harvested stands.

Average TIPSYS inputs for existing managed stands aged 1 – 10 years are given in Table 28.

Table 28 - TIPSy inputs for existing managed stands aged 1 – 10 years

AU	SPH	Ba %	Cw %	Cy %	Fd %	Hw %	Ss %	Ba SI	Cw/ Cy SI	Fd SI	Hw SI	Ss SI	Genetic Worth %				THLB Area (ha)
													C w	C y	Fd	H w ¹	
0110	1,100	40	-	-	-	20	40	30.1	27.0	37.5	33.7	35.2	-	-	-	10	194
0150	1,100	11	16	-	-	73	-	30.1	27.0	37.5	33.7	35.2	-	-	-	10	44
1110	1,100	61	-	9	-	30	-	26.2	23.8	31.4	28.0	31.0	8	-	-	10	24
1120	1,100	-	71	-	-	29	-	26.2	23.8	31.4	28.0	31.0	8	-	-	10	154
1150	1,100	19	18	-	-	63	-	26.2	23.8	31.4	28.0	31.0	8	-	-	10	5,815
2120	1,100	1	57	-	-	2	40	26.2	23.8	31.4	28.0	31.0	8	-	-	-	15
3150	1,100	29	19	-	-	52	-	26.2	23.8	31.4	28.0	31.0	8	-	-	10	876
4140	1,100	-	24	-	63	13	-	26.2	23.8	31.4	28.0	31.0	8	-	10	10	384
5110	1,100	41	-	24	-	35	-	26.2	23.8	31.4	28.0	31.0	-	7	-	6	755
5150	1,100	23	10	5	-	62	-	26.2	23.8	31.4	28.0	31.0	-	7	-	6	153
6120	1,500	-	98	-	-	2	-	20.7	19.2	22.6	20.1	25.0	8	-	-	-	1,078
6150	1,500	13	10	-	-	77	-	20.7	19.2	22.6	20.1	25.0	8	-	-	-	56
7120	1,000	-	89	-	-	11	-	20.7	19.2	22.6	20.1	25.0	8	-	-	10	304
7150	1,000	9	17	-	-	74	-	20.7	19.2	22.6	20.1	25.0	8	-	-	10	24
8170	1,000	-	21	-	37	42	-	20.7	19.2	22.6	20.1	25.0	8	-	10	10	21
9150	1,000	2	1	37	-	60	-	20.7	19.2	22.6	20.1	25.0	-	7	-	6	48
Total / THLB area-weighted average		18	28	2	3	48	1	25.4	23.2	30.2	26.9	30.2	7	1	-	9	9,947

The yield curves for each existing managed age 1 – 10 years analysis unit are listed and shown in Appendix F.

8.6.5 Regeneration Delay

The regeneration delay refers to the average time between harvesting and establishment of the next rotation. The average regeneration delay as reported in TFL 6 Annual Sustainable Forest Management Plan Indicator Reports through until 2008 has decreased until it is now approximately 0.8 years.

The majority of harvested area is planted, with approximately 20% of harvested area being regenerated naturally. Planted seedlings are typically one year old. Hence the regeneration delay from harvest until germination of the next crop of planted trees is generally less than 1 year. Also, early seedling growth is assisted on some sites (e.g. cedar-salal sites) in TFL 6 by the practice of fertilization at time of planting.

A one year regeneration delay for planted future managed stands is incorporated into the yield tables that will be used in the analyses. For naturally regenerated stands a 3-year regeneration delay is incorporated into the corresponding yield tables.

¹ GW for Hw reduced from 14% in low elevation ecosites and from 10% in high elevation ecosites to reflect expected natural regeneration component in future harvested stands.

8.6.6 Future Stand Volumes

Ecologically-based silviculture strategies for future stands were developed by Western Forest Products staff based on current practice (Table 29). Initial densities were chosen such that the effect of the OAFs reduces stand densities to approximately the target densities at free-growing ages (generally 15 years) and to reflect both the continued practice to plant almost all harvested areas and the natural in-growth experienced on many sites. Species and stocking levels are portrayed at this broad average level to simplify modelling. It is recognized that this includes a range of specific prescriptions that for example might include establishment of alder on a small % of the landbase (for further discussion on this see *Hardwood Management in the Coast Forest Region* (MoFR, 2009)).

Projections of average Genetic Worth (GW) were developed from WFP's Saanich Forestry Centre seed inventory. Expected GW values for Hw are reduced to reflect a component of natural regeneration expected in the harvested stands. Average values for GW by species and analysis unit listed in Table 29 will be applied to future managed stands.

See Appendix G for the yield tables. In the timber supply model, future stands yields will be reduced to account for the impact on growth of the next crop by trees retained to meet stand-level retention requirements (see Section 8.4.2 for details).

Table 29 - TIPSy inputs for future managed stands

Future AU	SPH	Ba %	Cw %	Cy %	Fd %	Hw %	Ss %	Cw /Cy SI					Genetic Worth %				THLB Area (ha)
								Ba SI	Cw SI	Fd SI	Hw SI	Ss SI	Cw	Cy	Fd	Hw ¹	
0500	1,100	40	-	-	-	20	40	30.1	27.0	37.5	33.7	35.2	-	-	-	10	3,384
1500 ²	1,100	20	20	-	-	60	-	26.2	23.8	31.4	28.0	31.0	8	-	-	10	41,399
1510	3,000	20	-	-	-	80	-	26.2	23.8	31.4	28.0	31.0	-	-	-	-	17,743
2500	1,100	-	60	-	-	-	40	26.2	23.8	31.4	28.0	31.0	8	-	-	-	276
3500	1,100	30	20	-	-	-	50	26.2	23.8	31.4	28.0	31.0	8	-	-	10	9,375
4500	1,100	-	25	-	65	10	-	26.2	23.8	31.4	28.0	31.0	8	-	10	10	2,619
5500 ²	1,100	40	-	25	-	35	-	26.2	23.8	31.4	28.0	31.0	-	7	-	6	6,742
5510	2,000	30	-	10	-	60	-	26.2	23.8	31.4	28.0	31.0	-	-	-	-	2,889
6500 ³	1,500	-	100	-	-	-	-	20.7	19.2	22.6	20.1	25.0	8	-	-	-	7,810
6501	1,500	-	100	-	-	-	-	20.7	23.8	22.6	28.0	31.0	8	-	-	-	7,810
7500	1,000	-	90	-	-	10	-	20.7	19.2	22.6	20.1	25.0	8	-	-	10	6,109
8500	1,000	-	25	-	65	10	-	20.7	19.2	22.6	20.1	25.0	8	-	10	10	615
9500 ²	1,000	-	-	40	-	60	-	20.7	19.2	22.6	20.1	25.0	-	7	-	6	727
9510	2,000	20	-	10	-	70	-	20.7	19.2	22.6	20.1	25.0	-	-	-	-	312
Total / THLB area-weighted average		18	30	2	2	42	6	25.1	23.2	29.7	27.0	30.3	6	1	-	2	107,811

¹ GW for Hw reduced from 14% in low elevation ecosites and from 10% in high elevation ecosites to reflect expected natural regeneration component in future harvested stands.

² Assume 30% of total potential THLB area in these analysis units is managed for natural regeneration resulting in approximately 20% of the total THLB in TFL 6 being managed for natural regeneration.

³ Assume 50% of total potential THLB area in this analysis unit is fertilized. See Section 8.6.2 for details.

8.6.7 Not Satisfactorily Restocked (NSR) Areas

Due to the data set being updated for depletion to the end of 2008 but aged to 2011, it was assumed that all harvested areas have been regenerated using the applicable future stand assumptions (ignoring the natural regeneration and fertilization options). The assumption was that these stands were regenerated two years following harvest.

9 NON-RECOVERABLE LOSSES

Windthrow, insects, disease and fire can cause catastrophic losses of whole stands of trees. Over the long-term the probability of losses to such natural causes can be predicted. Where losses occur in merchantable stands some of the dead or dying timber may be salvageable. When modelling timber supply, unsalvaged losses are added to the desired harvest forecast and then subtracted from the forecast upon completion of the modelling exercise.

The analysis will model 7,000 m³/year of non-recoverable losses.

9.1 Windthrow

Historically, windthrow has occurred mainly in relatively small patches. Loss of single trees or small groups are mostly accounted for in inventory sampling for existing timber yield estimates and OAFs applied for young stands. A great deal of research has been done over the past 10 years to try to determine the variables that affect the amount of expected windthrow along cutblock edges following harvest and the effectiveness of various edge treatment techniques (e.g. pruning, topping, feathering) to reduce the amount of windthrow experienced. The results of this research aid in cutblock design and treatment prescriptions such that the amount of windthrow experienced due to endemic winds is greatly reduced.

Operational staff estimates for unsalvaged losses due to windthrow are in the range of 5 – 10 hectares per year. If the upper end of this range is assumed and an average volume of 700 m³/ha is used, the resulting losses are approximately 7,000 m³/year. In MP #9 non-recoverable losses from windthrow were estimated at 7,000 m³/year. The landbase for MP #10 is smaller due to removals from the TFL (see section 6.1) and the AAC is expected to decrease; therefore the amount of non-recoverable losses would logically decrease. However, counteracting those factors is increased stand-level retention due to Western's Forest Strategy (see section 6.14.2) that will create more patches and therefore edges of timber that will be subjected to winds. Therefore, an estimate of 7,000 m³/year for non-recoverable losses will be used again in this analysis.

9.2 Insects and Disease

The forests of TFL 6 have been relatively free of major insect or disease infestations and therefore no losses are associated. There have been no major catastrophic outbreaks causing significant unsalvaged mortality or volume losses. The main active agent has been Spruce weevil (*Pissodes strobe*) which has infested Sitka spruce plantations to the point where the species is a minor component in reforestation programs. WFP has a weevil resistant orchard producing seedlings with an average 86% resistance (meaning that on average 7% of the progeny from this orchard are projected to display attack in a given year). These highly resistant seedlings are used wherever spruce is a suitable species and site conditions indicate a high hazard of weevil infestation. Between 2003 and 2008 approximately 160,000 spruce seedlings were planted in high hazard cutblocks – all weevil resistant seedlings. The last notable outbreak by other insect species was in the late 1980's when an outbreak of black-headed budworm (*Acleris*

gloverana) was detected on approximately 7,000 ha, mainly in the vicinity of the head of Holberg Inlet. Spray trials of *Bacillus thuringiensis* (B.T.) were conducted in 1989 and 1990 with good success. By 1991 the population had collapsed and in most cases the damage done appeared to be temporary as new foliage was evident.

Hemlock dwarf mistletoe is widespread throughout merchantable sized stands. Sanitation treatments of advanced regeneration are sometimes required to prevent the spread in newly regenerated western hemlock stands. Usually regenerated stands are not impacted significantly by hemlock dwarf mistletoe.

Root diseases sometimes result in small pockets of mortality. These losses are assumed accounted for by the operational adjustment factors (OAFs) applied to yield curves. Impacts of root disease (*Armillaria ostoyae*) and laminated root rot (*Phellinus weirii*) are less than observed in other areas. Additional OAF allowances are not applied.

9.3 Fire

The risk of loss of timber due to fire is relatively low within the TFL. The bulk of the TFL has a wet climate characterized by cool, wet summers and fire suppression has been efficient; hence the likelihood of losses to forest fire is small.

10 INTEGRATED RESOURCE MANAGEMENT

The intent of this section is to give an overview of the resource inventories available and being used for the timber supply review. The section also describes other resource management information that is being utilized for planning within TFL 6.

10.1 Forest Resource Inventories

Table 30 summarizes the forest resource inventories currently being maintained for the TFL.

Table 30 - Forest resource inventory status

Item	Status
Forest Inventory	Vegetation Resource Inventory (VRI) completed between 2000 and 2001 (photo interpretation and field sampling) with final phase (statistical adjustment) completed in 2009.
Ecosystems and terrain	Completed by T. Lewis, Ph.D., P.Ag., P.Geo. between 1982 and 1985 with an update in 2007.
Terrain Stability	Completed by T. Lewis in 1992 (Block 2) and 1995 (Block 1). Block 2 inventory was updated to MoFR standards in 1998.
Recreation Inventory	Updated in January 2004 by RRL Recreation Resources Ltd. to 1998 MoFR standards.
Visual Landscape Inventory	VLI updated between 2003 and 2005 to 1997 MoFR standards by RRL Recreation Resources Ltd. Accepted by NICCFD in June 2010 and is being used as basis for GAR Order to establish Visual Quality Objectives for TFL 6 and Block 7 of the Pacific TSA.
Ungulate Winter Ranges (UWRs)	Revised UWRs for Columbian black tailed deer and Roosevelt elk (U-1-010) effective as of November 2004.
Wildlife Habitat Areas (WHAs)	Seven WHAs established – six for marbled murrelets, 1 for northern goshawk.
Old Growth Management Areas (OGMAs)	OGMAs have been established in the San Josef and Marble landscape units. Refinement of draft OGMAs is proceeding for the Holberg, Keogh, Mahatta, and Neroutsos LUs.
Stream Classification	Operational stream inventories.
Archaeological	An Archaeological Overview Assessment (AOA) for the former Port McNeill Forest District was completed in 1995 by I.R. Wilson Consultants.
Operability	Physical and economic operability inventory completed in 1999.
Caves and Karst	Overview inventory completed in 1979 by P. Griffith. Limestone terrain identified in Lewis ecosystem inventory.

10.2 Forest Cover Requirements

10.2.1 Visual Quality

The Visual Landscape Inventory (VLI) for TFL 6 was updated between 2003 and 2005. This updated inventory was accepted by the District Manager (DM) of the NICCFD in June 2010 and was used as the basis for undertaking the required process for establishing legal visual quality objectives (VQOs) within scenic areas under GAR. The proposed VQOs were incorporated into the data set.

On September 24, 2010, the DM signed the *Order to Establish Visual Quality Objectives for Tree Farm Licence 6 and Block 7 of the Pacific TSA within the North Island Central Coast Forest District*. The established VQO classes and VQO polygons differ slightly from the proposed VQOs due to comments received during the review process. The established VQO classes will be used in the analysis; however the polygon revisions will not be incorporated (as this requires the entire data set to be recreated). The minor polygon differences result in 97 ha of THLB that fall within an established VQO polygon not being modelled.

VQOs within scenic areas to be modelled in the timber supply analysis are Retention (R), Partial Retention (PR) and Modification (M). The amount of area that can be disturbed (i.e. has not achieved visually effective green-up) is 5%, 15% and 25% for each VQO respectively. These levels are set at the upper end of the % disturbance range for use in timber supply analyses as visual landscape design during cutblock layout has become common practice in sensitive viewscapes. Cutblock designs that follow the lines and forms of the viewscapes allow more timber to be removed and still meet the VQO when compared to unnatural cutblock shapes. Additionally, the use of the retention silviculture system can result in more timber removal in visually sensitive areas by strategically placing retention patches to act as visual screens. A sensitivity analysis in which these percentages are reduced to the mid-point of the range for each VQO (2.5%, 10% and 20% respectively) will indicate the sensitivity of timber supply to management of visual quality objectives and the design of cutblocks within visually sensitive areas.

A 5 m visually effective green-up (VEG) height is proposed for TFL 6. As Woodstock uses volume over age curves for yield tables, an age surrogate of 15 years old (3 periods) will be established to represent VEG height for each analysis unit.

Table 31 outlines the management assumptions for dealing with visual quality within the TFL.

Table 31 - Visual Quality Management Assumptions

Visual Quality Objective	Productive Forest (ha)	THLB Area (ha)	Disturbance %
M	8,080	6,382	25%
PR	11,354	7,674	15%
R	808	398	5%

10.2.2 Adjacent Cutblock Green-up

A 3 metre green-up height in General and Special Management Zones will be used for areas without established visual quality objectives. A 10 year (2 periods) age surrogate will be used within the model to represent the 3 m height. This is a reasonable approximation (on average) with prompt planting, planting of improved seedlings and fertilization at time of planting in some areas. As the Woodstock model applies cover class constraints by period and the analysis will use five-year periods it is believed ten years is more appropriate than fifteen years for the 3m requirement. A 1.3 metre green-up height in Enhanced Forestry Zones will be used for areas without established visual quality objectives. A 5 year (1 period) age surrogate will be used within the model to represent the 1.3 m height. Since Woodstock does not have the capability to spatially model adjacency requirements, a proxy will be used: a maximum of 25% of the THLB within a zone but outside of VQO polygons will be permitted to be less than the green-up age (5 or 10 years depending on the zone type).

10.2.3 Landscape Level Biodiversity

Landscape Units and Biodiversity Emphasis Options (BEOs) were designated through the *Order Establishing Provincial Non-Spatial Old Growth Objectives* effective June 30, 2004 (NSOG order). This order is in effect until Old Growth Management Areas (OGMAs) are spatially determined through Landscape Unit planning. For TFL 6 OGMAs have been established within the San Josef (Intermediate BEO) landscape unit (January 2005) and the Marble (Intermediate BEO) landscape unit (July 2010). Draft OGMAs in the Holberg (Low BEO), Keogh (Low BEO), and Neroutsos (Low BEO) landscape units have been identified to meet the NSOG order (refer to Table 11).

For the Mahattta (Low BEO) landscape unit draft OGMAs have been identified for the WFP tenure (TFL 6 and unregulated Timber Licences) portion of the landscape unit only. These draft OGMAs meet the old seral target within TFL 6 drawn down to 1/3 for the first rotation

These draft OGMAs will be used in the timber supply analysis but must complete a public and First Nations' review process before coming legal and may be revised prior to that review process.

For the forest types within TFL 6, old forest is defined as stands >250 years old. The old seral target is based on the combination of BEO, BEC variant, and the natural disturbance type (NDT) of the variant. The draft OGMAs for landscape units with a Low BEO identify enough area to meet the old seral target drawn down to 1/3 for the first rotation (80 years). Where the draft OGMAs do not meet the full target an old seral stage cover constraint will be applied. The constraint for the end of the second rotation (160 years) will be 2/3 of the full target, with the full old seral target being achieved by the end of the third rotation (240 years). Table 32 indicates the landscape biodiversity constraints that will be applied for old seral forest. For a breakdown of the current forest age by landscape unit and variant see Table 19.

Table 32 - Old Seral Constraints

Landscape Unit	BEO	BEC	NDT	Area (ha)		Old Seral constraint (% of productive)		
				Productive	THLB	1 st rotation	After 2 nd rotation	After 3 rd rotation
Holberg	Low	CWHvh1	1	4,928	3,763	OGMAs	8.7%	13%
		CWHvm1	1	19,370	15,274	OGMAs	8.7%	13%
		CWHvm2	1	1,535	1,214	OGMAs	8.7%	13%
Keogh	Low	CWHvm1	1	12,522	10,153	OGMAs	8.7%	13%
		CWHvm2	1	1,508	1,224	OGMAs	8.7%	13%
Mahatta	Low	CWHvm1	1	18,620	13,583	OGMAs	OGMAs	13%
		CWHvm2	1	3,591	2,340	OGMAs	OGMAs	13%
		MHm1	1	3	0	N/A	N/A	N/A
Neroutsos	Low	CWHvm1	1	16,929	12,289	OGMAs	OGMAs	13%
		CWHvm2	1	5,407	3,458	OGMAs	OGMAs	OGMAs

10.2.4 Community Watersheds

There is one designated community watershed (CWS) within TFL 6 - Calbick Creek (930.003) is a 64 ha CWS (with 55 ha of THLB) located between Quatse Lake and Coal Harbour. Even though this CWS is no longer being used as a source for drinking water by the Quatsino First Nation (the water licence holder), a forest cover constraint will be applied so no more than 10% of the productive area within the watershed will be covered with stands less than 10 years old.

10.2.5 Fisheries Sensitive Watersheds

There are no established Fisheries Sensitive Watersheds (FSWs) within TFL 6.

10.2.6 Rate-of-harvest Restrictions

The FSP for TFL 6 contains a strategy for addressing Objective 8 of the VILUP HLP order (mitigate adverse hydrological impacts of practices permitted in EFZ's). The FSP strategy includes preparing a watershed assessment and to the extent practicable, implementing the recommendations in such an assessment. An assessment was completed in 2007 from which a set of watershed management strategies were developed (October 2007).

The intent of the watershed management strategies is to allow for continued recovery in watersheds impacted by past practices, particularly by reducing the incidence of landslides. From a timber supply analysis perspective, the strategies include limiting the rate-of-harvest on "steep terrain" in four watersheds: Colonial, Goodspeed, Hathaway and Koprino. "Steep terrain" is defined as areas classified in a terrain stability inventory as Class IV (potentially unstable) or Class V (unstable) plus areas where the slope gradient is greater than 60%.

No specific harvest rate limits were recommended in the original management strategies document. Later follow-up with the professional who conducted the watershed assessment and developed the watershed management strategies resulted in a recommendation of limiting harvest on steep terrain in the Colonial watershed to approximately 135 ha over a 10 year period. The 135 ha equates to roughly 1/6 of the "steep" THLB estimated to be within the Colonial watershed for this analysis.

As the frequency of past landslides in the other three watersheds is similar to that in the Colonial watershed, the same ratio (1/6) will be used for the rate-of-harvest limits for steep terrain in the Goodspeed, Hathaway and Koprino watersheds (Table 33).

Table 33 - Forest Cover Constraints for “Steep Terrain”

Applicable Landbase	Total Productive Area (ha)	“Steep” Productive Area (ha)	“Steep” THLB Area (ha)	Maximum area of “steep terrain” harvested over a 10 year period (ha)
Colonial Watershed	3,463	1,202	822	135
Goodspeed Watershed	7,993	1,554	1,177	196
Hathaway Watershed	3,840	770	506	84
Koprino Watershed	5,480	1,027	441	73

Higher Level Plans

The order establishing Resource Management Zones and Resource Management Zone objectives within the area covered by the Vancouver Island Land Use Plan came into effect as of December 1, 2000. Each Special Management Zone (SMZ) established by the order has an objective of maintaining mature seral forest over one quarter to one third of the forested area of the SMZ.

There are portions of two SMZ’s within TFL 6 – SMZ 2 (West Coast Nahwitti Lowlands) and SMZ 4 (Koprino). For this analysis, a constraint will be incorporated that maintains 25% of the productive forest land base in the mature and/or old seral stage within these SMZ’s.

10.3 Timber Harvesting

10.3.1 Minimum Harvestable Age

Minimum harvestable ages are simply minimum criteria for use in the timber supply model. While actual harvesting may occur in stands at or below the minimum requirements, in order to meet forest-level objectives (i.e. maintaining overall timber flows), other stands will not be harvested until well past the minimum ages because consideration of other resource values may take precedence or timber may be in ample supply.

In the previous two analyses minimum harvestable ages were selected as the age closest to the time when average stand diameter-at-breast-height (DBH) reached a threshold diameter that varied by site productivity class (see Table 34). This criteria was selected on the basis of values and cost structures associated with current harvesting and manufacturing systems.

Table 34 – Minimum harvest criteria used in past two analyses

Site Productivity Class	Minimum Average DBH used in MP #8	Minimum Average DBH used in MP #9
Good (PG 1)	45 cm	42 cm
Medium (PG 2/3)	40 cm	37 cm
Poor (PG 4)	35 cm	30 cm

This analysis will again use minimum harvest ages based on average stand diameters; however, this time the average diameters will vary by harvesting system rather than site productivity - the notion being that larger diameters in general reflect higher values. To have an economically sustainable harvesting program, average stand values must be greater than average harvesting costs. Average harvesting costs are lowest for ground-based systems (e.g. skidder, “hoe-chucking”) and are highest for helicopter, with cable systems (e.g. grapple yarding) costs falling between those.

The data set prepared for the analysis includes logging system (ground, cable or heli) based on the combination of operability class (see section 6.6) and slope class. Conventionally operable areas with a slope between 0% and 30% are assumed harvestable by ground-based systems and all conventionally operable areas on steeper slopes are assumed harvestable by cable systems. Helicopter operable areas are found across all slope classes. Table 35 indicates the minimum stand-average DBH that will be used in the analysis and the resulting range of ages. The younger ages are on higher productivity sites while the older ages are on lower productivity sites.

Table 35 - Minimum Harvest Criteria

Harvest System	Minimum Average DBH	Age range (years)	Area-weighted Average Age (years)
Ground	30 cm	40 – 115	70.4
Cable	37 cm	60 – 205	98.8
Heli	42 cm	70 - 290	123.8

10.3.2 Initial Harvest Rate

The current AAC for the analysis area includes 1,243,958 m³ for WFP and 11,578 m³ for First Nations; a total of 1,255,536 m³. The TFL 6 MP #9 analysis projected a 5.8% reduction in harvest between 2001 and 2011 and 9.5% by the 2016-2020 period – an average reduction of 7.6% for the 10-year period 2011-2020. When a 7.6% reduction is applied to 1,255,536 m³ the result is 1,160,000 m³ (rounded to nearest thousand). The latter value will be used as the initial harvest rate.

This “initial harvest rate” provides a starting point for the analysis. However, the timber harvesting landbase and other assumptions have changed since the MP #9 analysis. Therefore, the initial harvest level may subsequently be adjusted to achieve harvest flow objectives over the entire planning horizon.

10.3.3 Harvest Rules

The analysis will be undertaken with the model Woodstock, using optimization to project harvest schedules. With optimization, the model determines harvest order to achieve the defined objective. This is different from a simulation approach where rules are specified for harvest priority. Harvest constraints will, however, be applied to model the transition from old-growth harvest to second-growth harvest and to reflect performance within the non-conventional portion of the THLB.

The analysis will be done using 5-year periods to accommodate the assumptions for cutblock adjacency (green-up) and visually effective green-up; however since AAC's are now determined for up to 10 years, the model will be constructed to achieve equal harvest levels over 10-year intervals (i.e. Period 1 and Period 2 harvest levels will be equal; Period 3 and Period 4 harvest levels will be equal; etc.)

Recent harvest numbers and short-term plans indicate continued harvesting of second-growth in TFL 6. Second-growth harvest in the base case option will commence at approximately 20% of the total harvest and will gradually increase over time until the transition to second-growth harvest is largely complete (small volumes of old-growth harvest may continue because of the scheduling impacts of cover class constraints).

Recent harvest (1997 – 2008) within the non-conventional (helicopter) portion of the THLB has been approximately 1.0% of the total harvest area whereas it represents approximately 2.9% of the THLB area and 5.6% of the THLB volume. The level of performance in the non-conventional THLB is not anticipated to increase significantly in the near future; therefore, a 12,000 m³/year constraint will be applied in the timber supply model. This value represents approximately 1.0% of the initial harvest level. Sensitivity analyses will explore the impact to timber supply of varying the amount of non-conventional timber available for harvesting (see Section 3.2). These analyses may result in WFP recommending acknowledgement of the economic uncertainty of non-conventional harvesting in a manner different than the constraint to be used in the base case.

10.3.4 Silviculture Systems

Nearly all of the harvest within TFL 6 has been done with clearcut or clearcut-with-reserves silviculture systems. With the recent (2006) amalgamation of companies (WFP, Canfor (TFL 37), and Cascadia Forest Products (formerly Weyerhaeuser and prior to that MacMillan Bloedel)) with different forest management strategies, WFP has reviewed its forest strategy. Included in this is a program for conserving biodiversity on company tenures. The approach is to vary the use of retention systems and the amount of stand level retention by Resource Management Zones of the Vancouver Island Land Use Plan and by ecosection and is being phased in over the next few years. For this analysis it will be assumed the WFP strategy is fully implemented from the beginning of the analysis period. Refer to section 6.14.2 for details on the retention targets and THLB area impacts and section 8.4.2 for growth and yield impact assumptions.

10.3.5 Harvest Flow Objectives

Harvest level projections will maximize volume harvested over the entire analysis period (250 years) subject to the following constraints:

- Gradually adjust harvest levels towards the current estimate of the long-term stable harvest level; and,
- Achieve a stable long-term growing stock.

10.4 Reporting on Cedar

Traditional and cultural uses of cedar are important to First Nations. Opportunities for accessing and managing cedar have been increased through the allocation of AAC to First Nations (refer to Section 6.1).

The descriptions of analysis units in Section 8 show that cedar is a significant component of both old and recently regenerated forest.

Reporting on the base case will include projections of cedar volumes in the forest inventory.

11 Glossary

Allowable Annual Cut (AAC)	The rate of timber harvest permitted each year from a specified area of land, usually expressed as cubic metres per year.
Analysis Unit (AU)	A grouping of forest types – for example, by biogeoclimatic zone, site productivity, leading tree species, and age - done to simplify analysis and the generation of timber yield tables.
Base case harvest forecast (Current Management Option)	The timber supply forecast which illustrates the effect of current forest management practices on the timber supply using the best available information, and which forms the reference point for sensitivity analysis.
Biodiversity (biological diversity)	The diversity of plants, animal and other living organisms in all their forms and levels of organization, including the diversity of genes, species and ecosystems, as well as the evolutionary and functional processes that link them.
Biogeoclimatic zones and variants (BEC)	A large geographic area with broadly homogeneous climate and similar dominant tree species.
Cutblock	A specific area, with defined boundaries, authorized for harvest.
Cutblock adjacency	The desired spatial relationship among cutblocks. Most adjacency restrictions require that recently harvested cutblocks must achieve a desired condition (green-up) before nearby or adjacent areas can be harvested.
Forest inventory	An assessment of timber resources. It includes computerized maps, a database describing the location and nature of forest cover, including size, age, timber volume, and species composition, and a description of other forest values such as recreation and wildlife habitat.
Forest and Range Practices Act	Legislation that governs forest and range practices and planning, with a focus on ensuring management of all forest values.
Forest type	The classification or label given to a forest stand, usually based on tree species composition.
Free-growing	An established seedling of an acceptable species that is free from growth-inhibiting brush, weeds and excessive tree competition.

11 Glossary

Geographic Information System (GIS)	A geographic information system, also known as a geographical information system or geospatial information system, is a system for capturing, storing, analyzing and managing data and associated attributes which are spatially referenced to the Earth.
Green-up	The time needed after harvesting for a stand of trees to reach a desired condition (usually expressed as a specific height) - to ensure maintenance of water quality, wildlife habitat, soil stability, or aesthetics – before harvesting is permitted in adjacent areas.
Growing stock	The volume estimate for all standing timber at a particular time.
Harvest forecast	The potential flow of timber harvest over time. A harvest forecast is usually a measure of the maximum timber supply that can be realized over time for a specified land base and a set of management practices. It is a result of forest planning models and is affected by the size and productivity of the land base, the current growing stock, and management objectives, constraints and assumptions.
Inoperable areas	Areas defined as unavailable for timber harvest for terrain-related or economic reasons. Operability can change over time as a function of changing harvesting technology and economics.
Integrated resource management (IRM)	The identification and consideration of all resource values, including social, economic and environmental needs in resource planning and decision-making.
Karst features	Karst is a distinctive topography that develops as a result of the dissolving action of water on carbonate bedrock (usually limestone, dolomite or marble). Karst features include fluted rock surfaces, vertical shafts, sinkholes, sinking streams, springs, complex sub-surface drainage systems and caves.
Landscape-level biodiversity	The <i>Landscape Unit Planning Guide</i> and the <i>Order Establishing Provincial Non-Spatial Old Growth Objectives</i> provide objectives for maintaining biodiversity at the landscape level and stand level. At the landscape level, objectives are provided for the maintenance of old growth.

11 Glossary

Landscape unit	A planning area based on topographic or geographic features, that is appropriately sized (up to 100,000ha), and designed for application of landscape-level biodiversity objectives.
Long-term harvest level	A harvest level that can be maintained indefinitely given a particular forest management regime (which defines the timber harvesting land base, and objectives and guidelines for non-timber values) and estimates of timber growth and yield.
Management assumptions	Approximations of management objectives, priorities, constraints and other conditions needed to represent forest management actions in a forest planning model. These include, for example, the criteria for determining the timber harvesting land base, the specifications for minimum harvestable ages, utilization levels, and integrated resource management and silviculture and pest management programs.
Model	An abstraction and simplification of reality constructed to help understand an actual system. Forest managers and planners have made extensive use of models, such as maps, classification systems and yield projections, to help management activities.
Natural disturbance type (NDT)	An area that is characterized by a natural disturbance regime, such as wildfires and wind, which affects the natural distribution of seral stages. For example areas subject to less frequent stand-initiating disturbances usually have more old forests.
Non-recoverable losses	The volume of timber killed or damaged annually by natural causes (e.g. fire, wind, insects and disease) that is not harvested.
Operability	Classification of an area considered available for timber harvesting. Operability is determined using the terrain characteristics of the area as well as the quality and quantity of timber on the area.
Riparian area	Areas of land adjacent to wetlands or bodies of water such as swamps, streams, rivers or lakes.
Riparian habitat	The stream bank and flood plain area adjacent to streams or water bodies.

11 Glossary

Sensitivity analysis	A process used to examine how uncertainties about data and management practices could affect timber supply. Inputs to an analysis are changed and the results are compared to a baseline or the base case.
Site index	A measure of site productivity. The indices are reported as the average height, in metres, that the tallest trees in a stand are expected to achieve at 50 years (age is measured at 1.3 metres above the ground).
Site Index by Biogeoclimatic Ecosystem Classification site series (SIBEC)	Site index estimates for tree species according to site units of the Biogeoclimatic Ecosystem Classification system of British Columbia.
Stocking	The proportion of an area occupied by trees, measured by the degree to which the crowns of adjacent trees touch, and the number of trees per hectare.
TIPSY (Table Interpolation Program for Stand Yields)	A BC Forest Service computer program used to generate yield projections for managed stands based on interpolating from yield tables of a model (TASS) that simulates the growth of individual trees based on internal growth processes, crown competition, environmental factors and silvicultural practices.
Timber harvesting land base (THLB)	Forest land within the TFL where timber harvesting is considered both acceptable and economically feasible, given objectives for all relevant forest values, existing timber quality, market values and harvesting technology.
Timber supply	The amount of timber that is forecast to be available for harvesting over a specified time period, under a particular management regime.
Tree farm licence (TFL)	Provides rights to harvest timber, and outlines responsibilities for forest management, in a particular area.
Ungulate	A hoofed herbivore, such as a deer.
VDYP (Variable Density Yield Projection)	A BC Forest Service computer program used to generate yield projections for natural stands based on temporary inventory sample and permanent growth sample data.
Volume estimates (yield projections)	Estimates of yields from forest stands over time. Yield projections can be developed for stand volume, stand diameter or specific products.

11 Glossary

Waste ₂	Timber volume lost to short logs and other reasons <u>plus</u> the soundwood portion of a tree or log which has more than 50% decay.
Watershed	An area drained by a stream or river. A large watershed may contain several smaller watersheds (basins).
Wildlife tree	A standing live or dead tree with special characteristics that provide valuable habitat for wildlife.

APPENDICES

Appendix A: Second Growth Site Index

Appendix B: TFL 6 VRI Statistical Adjustment

Appendix C: Mature Yield Tables

Appendix D: Unmanaged Immature Yield Tables

Appendix E: Existing Managed Aged 11 – 50 Years Yield Tables

Appendix F: Existing Managed Aged 1 – 10 Years Yield Tables

Appendix G: Future Managed Yield Tables

Appendix A - Second Growth Site Index

Second Growth Site Index of TFL 6

Prepared by
J.S. Thrower & Associates Ltd.

Project: WPV-091-002 (96-08-WPa)

March 19, 1997

Executive Summary

The primary objective of this study was to develop statistically based estimates of average site index for the major commercial tree species and ecosystems on Western Forest Products Limited Quatsino Tree Farm (TFL 6). This was achieved by sampling four Productivity Groups created by grouping Lewis ecological classification units having similar site potential. The average site index for hemlock was estimated from 61 randomly located plots within these four groups. Field sampling was restricted to hemlock-leading polygons aged 21-140 years to ensure good estimates of site index. Estimates for the average site index for balsam, cedar, and Sitka spruce were derived from equations relating the average site index of hemlock to the averages of these other species. The equations were derived from samples taken on the random plots and thus provide a representative sample of the TFL. An equation developed by the Ministry of Forests was used to estimate the average site index for Douglas-fir.

The average site indices computed in this study differ only slightly from the estimates derived previously by Western Forest Products when compared by Productivity Groups. The overall average site index for the TFL is virtually identical to the previous estimate. The results present a matrix of average site indices recommended for use in timber supply analyses for TFL 6.

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1.0 Introduction

1.1 Background

Western Forest Products (WFP) was one of the first companies in BC to use ecosystem classification for estimating site index in timber supply analyses. This was done in the last Management Plans for TFL 6 on northern Vancouver Island, and TFL 24 on the Queen Charlotte Islands. The average site indices in these projects were computed from data collected primarily from subjectively located permanent sample plots. This project aims to fulfil WFP's commitment to the Ministry of Forests (MOF) to update and refine the ecosystem-based estimates of site index using statistically-based sampling for the TFL 6 Management Plan 9 due in December 1999.

1.2 Objectives

The primary objective of this project was to:

Develop statistically-based estimates of average site index for the major commercial tree species and ecosystems on TFL 6 for use in timber supply analyses.

A secondary objective was to collect basal area information from a cluster of point-sample plots around randomly located site index plots to supplement forest inventory information.

This report was prepared for Dr. John Barker, *R.P.F.* at WFP to describe and document the methods and procedures used to develop the site index estimates.

2.0 TFL 6 Landbase

2.1 Geographic Location and TFL Area

The WFP Quatsino Tree Farm (TFL 6) is located on northern Vancouver Island. The TFL encompasses the area from Port Hardy on the east coast, to south of Cape Scott Park on the west coast, and south to Quatsino Sound, Neroutsos Inlet, and Victoria Lake. The total area of the TFL is 170,194 ha; over 93% of which is productive forest land (Table 36).

Table 36. Landbase summary for TFL 6.

Land Type	Area (ha)	
Non-Productive Land	11,483	6.7 %
Productive Land	158,711	93.3 %
NSR Polygons	4,448	2.6 %
Roads	2,164	1.3 %
Forested Polygons	152,099	89.4 %
Total TFL 6 Area	170,194	100 %

2.2 Forest Cover

About two-thirds of the 152,099 ha of forested area in the TFL is leading western hemlock (Hw) (102,664 ha) (Figure 3, Appendix A.1). Red cedar is the second largest component (18.8%) followed by yellow cedar (5.4%), Sitka spruce (2.8%), and balsam (2.3%). Nearly one-half of the forested area in the TFL is age class 8 (141 yrs+) (Figure 4, Appendix A.1) (WFP Age Class 8 includes MOF Classes 8 and 9). Other major components are age class 1 (1-20 yrs - 19%) and age class 2 (21-40 yrs - 15%).

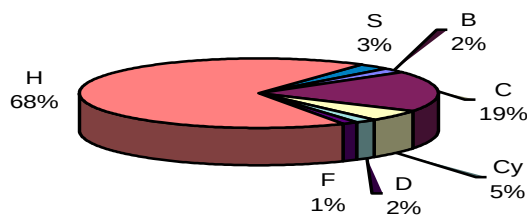


Figure 3. Species distribution of TFL 6.

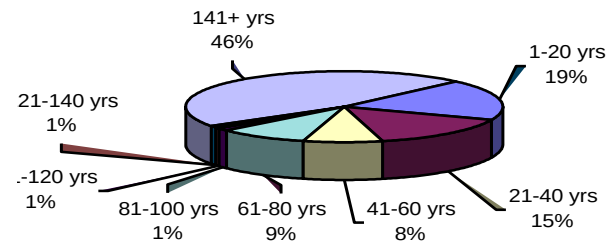


Figure 4. Age distribution of forested land in TFL 6.

2.3 Ecological Classification

2.3.1 Lewis Ecosystem Units

An ecological inventory and terrain classification was completed on the TFL in 1985 by Dr. Terry Lewis.¹ These eco-units are not named the same as BEC units; however, there is a very close relation between the two systems and for most units, boundaries are identical.² The Lewis system was designed for TFL 6 and thus recognizes local ecosystems not included in BEC.

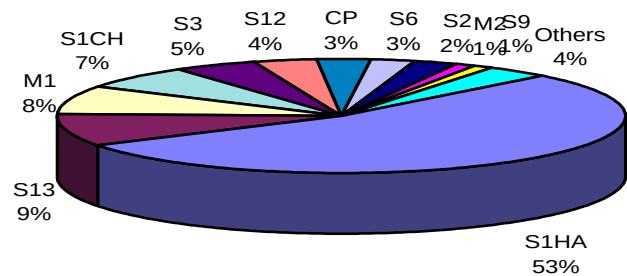


Figure 5. Distribution of Lewis ecosystem units on TFL 6

Approximately one-half of the forested area in the TFL is the S1HA ecosystem (Figure 5, Appendix A.2). This equates to the zonal CWHvm1/01 (Appendix A.3). Other major ecosystems are the S13 (composite of CWHvm1/05 and 07), the M1 (CWHvm2/01), and the S1CH (salal phase of CWHvm1/01).

¹ Lewis, T. 1985. Ecosystems of Quatsino Tree-Farm Licence (TFL 6). Contract Rep. to WFP. 77 pp.

² Personal communication with Bob Green, R.P.F., March 1996.

2.3.2 BEC Conversion

The Lewis eco-units were converted to BEC units by Bob Green, *R.P.F.* of B.A. Blackwell and Associates Ltd. (Figure 6, Appendix A.3). This was not necessary to complete this project; however, it provided a common link to other site studies allowing us to refine our Productivity Groups. This conversion was based on Mr. Green's extensive knowledge of the area, familiarity with Dr. Lewis' classification system, and documentation of the classification on the TFL including detailed descriptions of the ecosystems.

The Lewis classification system identifies some sites unique to TFL 6. These special sites are mostly related to non-forested and wetland ecosystems and did not affect the sampling plan for this project. The S13 eco-unit is a composite of the CWHvm1/05 and 07; these two BEC site series are very close in productivity and would be grouped into the same sampling unit for any forest-level sampling program.

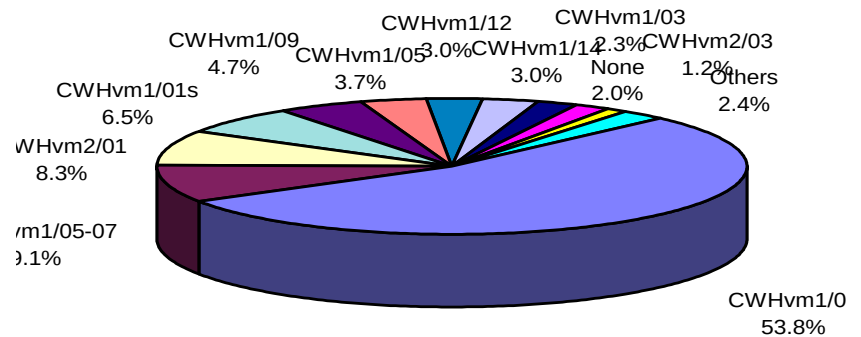


Figure 6. Distribution of BEC site series on TFL 6 converted from Lewis ecosystem map-units.

3.0 Sampling Plan

3.1 Sampling Approach

The primary objective of this project was to develop estimates of average site index for the major commercially important species on the TFL, including Hw, Ba, Cw, Ss, and Fdc. However, lack of sufficient area of species other than Hw across the various ecosystems on the TFL precluded observing site index for these species in the random plots. Therefore, the objective of developing estimates of average site index for all species on all major ecosystems on the TFL was achieved in a two-phase process:

- Random Sampling for Hemlock Site Index – where statistically-based estimates of average site index were developed for Hw from plots randomly located across the TFL in stands aged 21-140 years.
- Sampling for Site Index Conversion Equations – where pairs of site index estimates were measured from the random plots and nearby areas to develop equations to predict the average site index of Ba, Cw, and Ss from the average site index of Hw from the random samples.

3.2 Productivity Groups

3.2.1 Overview

The field sampling and analyses focused on four Productivity Groups (PGs). These groups provided an ecologically-based stratification of the TFL aggregating areas of similar productivity. This provided the stratification necessary to reduce within-group variation in site index to increase sampling efficiency. This stratification also provides a link to current management and planning systems on the TFL based on the Lewis ecosystem classification. These PGs formed the basis for sampling and will form the basis for applying the results.

3.2.2 Original Sampling Plan

The previous estimates of average site index for the TFL were developed by Dr. John Barker using Lewis ecosystem units to create seven PGs (Table 37). These Barker PGs were used in the previous Management Plan 8 for yield prediction. The original sampling plan¹ for this project combined six of the Barker PGs into three PGs for our project. This original plan aimed to reduce sampling costs by reducing the number of sampling strata. However, the original plan was changed to add another PG to recognize the higher productivity of the S3 eco-unit. This was achieved by allocating plots proportionally to the two PGs and confirming the difference in interim analysis.

Table 37. The four Productivity Groups (PGs) related to Barker/Lewis groups and BEC ecological units.

	Barker PG	Lewis Unit	BEC Site Series	ha in TFL	% of TFL	% of PG
PG 1	EE	S3	CWHvm1/09	7,407	5.0	100
<i>Subtotal</i>				7,407	5.0	
PG 2	EG+	S4	CWHvm1/10	113	0.1	0.8
		S5	CWHvm1/14	693	0.5	4.7
		S13	CWHvm1/05+07	13,862	9.3	94.5
<i>Subtotal</i>				14,668	9.9	100
PG 3	EG	S1-ha	CWHvm1/01	81,730	54.8	80.8
		S12	CWHvm1/05	5,651	3.8	5.6
	EG-	M1	CWHvm2/01	12,561	8.4	12.4
		M3	CWHvm2/05	106	0.1	0.1
		M5	CWHvm2/08	1,127	0.8	1.1
<i>Subtotal</i>				130,511	67.9	100
PG 4	EM	S1-ch	CWHvm1/01s	9,838	6.6	48.4
	EL	S2	CWHvm1/03	4,364	2.9	21.5
		S6	CWHvm1/14	3,913	2.6	19.3
		M2	CWHvm2/03	1,878	1.2	9.2
		M4	CWHvm2/11	332	0.3	1.6
<i>Subtotal</i>				281,347	13.6	100
Total				143,485	94.3	
PG 5	EVL	S7	CWHvm1/13	61	0.0	1.1
		CP	CWHvm1/12	4,622	3.1	83.6
		MH	MH/01	845	0.6	15.3
<i>Subtotal (not sampled)</i>				711,707	3.7	100

3.2.3 Final Productivity Groups

The four PGs used in this project are aggregations of the existing Barker groups (Table 2). Three of the former seven Barker PGs were retained in the original form. The lowest productivity group (PG 5) represented only 3.7% of the landbase and thus was not sampled. The other two PGs in

¹ J.S. Thrower and Associates Ltd. 1996. Sampling Plan: Second Growth Site Index Inventory in Western Forest Products TFL 6. Confidential Contract Report to Dr. John Barker, R.P.F. and Western Forest Products Ltd. May 21, 1996. 26 pp.

this project were combinations of two of the original Barker groups. About 3,000 ha (2.0%) of the forested polygons (avalanche, wetlands, bog, and park areas) were not assigned to a Barker PG and were not included in our PGs before creating the additional group.

3.3 Sample Size

The target number of plots detailed in the sampling plan was achieved in the field sampling (Table 38). The small sample size in PG 1 (8 plots) was the result of splitting the original highest productivity PG in the preliminary sample plan into two separate groups as noted above. The ecosystems in this PG were very homogeneous and thus a small standard error was achieved even with the small sample size.

Table 38. Preliminary target and actual sample size for the randomly located site index plots.

Prod. Group	Lewis Eco-Unit	Target Initial Sample Size	Actual Sample Size
1	S3	7	8
2	S4, S5, S13	13	13
3	S1HA, S12, M1, M3, M5	25	25
4	S1CH, S2, S6, M2, M4	15	15
5	S7, CP, MH	0	0
Total		60	61

3.4 Random Sampling for Hemlock Site Index

3.4.1 Target and Sample Populations

The target population for application of the results of this project is the total land area within each PG. The sample population is the land area in each PG suitable for sampling. The sample population is always smaller than the target population in forest-level site productivity sampling. This is because stands of all species suitable for estimating site index do not occur everywhere across the landscape. The sample population was defined as polygons with leading Hw forest cover aged 21-140 years (age class 2-7) (Table 39).

Table 39. Area of target population, sample population, and sampling intensity.

Prod. Group	Plots	Target Population			Sample Population		
		Area (ha)	Sample Intensity (ha/plot)		Area (ha)	Sample Intensity (ha/plot)	
1	8	7,407	5.0 %	926	2,155	29.1 %	269
2	13	14,668	9.9 %	1,128	2,052	14.0 %	158
3	25	101,175	67.9 %	4,047	37,280	36.8 %	1,491
4	15	20,325	13.6 %	1,355	3,873	19.1 %	258
5	0	5,528	3.7 %	NA	44	0.8 %	NA
Total	61	149,103	100 %		45,404	100 %	

3.4.2 Statistical Inference

The sampling plan was designed to select random points on the landbase to measure Hw site index within each of the four PGs. The only restriction to the randomization and application of statistical theory in this design is that Hw stands suitable for estimating site index do not occur on all areas in the TFL. Theoretically, this imposes a bias in the sampling design. However, most of the potential

for bias from this restriction is removed through the ecological classification system. The potential effect of this randomization restriction is addressed through the assumption that the productive potential of the land within a given PG is the same for areas that could be sampled as for those that could not be sampled. This is a reasonable assumption and should not be of major concern in the application of these estimates.

3.4.3 Polygon and Plot Selection

Random sample points within each PG were selected in a two-phase process. First, polygons were selected randomly with probability proportional to area (with replacement) from a list of all polygons in each PG. Points were then randomly located on maps within the selected polygons and transferred to air photos for field sampling.

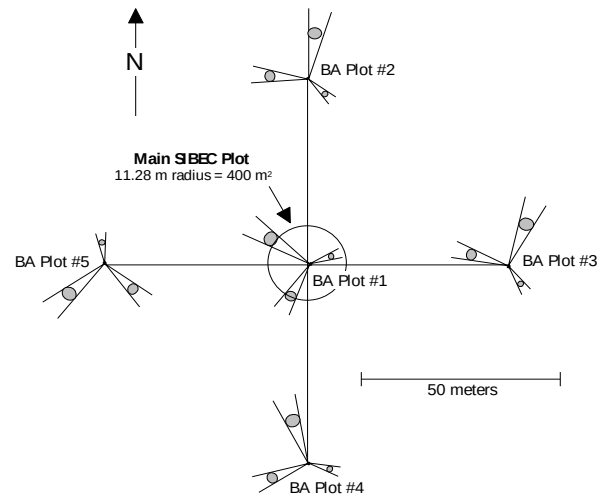


Figure 7. Sample plot cluster showing main SIBEC plot and the five basal area plots.

3.5 Sampling for Site Index Conversion Equations

Equations to predict the site index of Ba, Cw, and Ss from the site index of Hw were developed from field measurements. The intent was to develop these Hw-Ba, Hw-Cw, and Hw-Ss relationships from trees that adequately represent the range of conditions on the TFL. To achieve this representative sample of the land base, site index measurements were taken from three sources:

1. Trees measured in the randomly located Hw sample plots.
2. Trees measured in the basal area plots systematically located around each random plot.
3. Trees measured from areas adjacent to the random plots.

3.6 Basal Area Sampling

A cluster of five point-sample plots was sampled in conjunction with each randomly located site index plot (Figure 7). The cluster contained a central plot located at the random site index plot center, and four satellite plots located at 50 m in the cardinal directions. The objective was to obtain additional information about the basal area and volume of these randomly sampled second-growth stands to supplement WFP's existing forest cover inventory and to aid in the development of growth and yield models.

4.0 Field Methods

4.1 Plot Establishment

4.1.1 Plot Location and Suitability Assessment

The random site index plots were established as closely as possible to the point indicated on the photos and maps by compassing and chaining from the nearest photo tie point. The first step upon reaching a plot was to ensure the plot was in the target forest-cover and ecosystem polygon. The second step was to determine if the stand condition was suitable for estimating site index. When stand conditions were not suitable for estimating site index, plots were systematically offset in a pre-determined 25 m grid pattern. Suitable plots contained at least three healthy, dominant or co-dominant, unsuppressed Hw site trees. Approximately one-third of the plots (21 of 61) were offset because the original location did not contain suitable site trees.

4.1.2 Tie Points and Monuments

Tie points to sample plots were located using metal tags indicating the plot number, date, crew initials, bearing, and distance to the plot. A double band of blue and yellow ribbon was located above and below each metal tag. Compass lines to the plots were flagged using blue ribbon. A bearing tree was located close to each plot center identified in the same manner as the tie points, including bearing and distance to plot center written on a metal tag. Plot centers were marked with a 24" wooden survey stake. Site trees on the plot were marked with orange flagging tape. Trees measured for site index pairs were marked with yellow flagging tape. Original maps and air photos showing plot locations were submitted to Dr. John Barker at WFP head office.

4.2 Random Plots For Hemlock Site Index

Three types of information were taken on each of the randomly located sample plots: site tree, mensurational, and ecological.

4.2.1 Site Tree Measurements

The Hw site trees were selected as the largest diameter, dominant or co-dominant, healthy, and unsuppressed trees within the 400 m² sample plots. Four Hw site trees were measured where possible, however, three trees were considered an adequate sample if one tree was not suitable for estimating site index. Total heights were measured and ages at breast height determined from increment cores (all samples included the 1996 growing season). Site index for the Hw site trees were computed in the field and later confirmed with computer analyses using Wiley's site index equation.¹ The bearing and distance from plot center to each site tree was recorded to facilitate possible future relocation of the samples.

Site index was estimated using an age correction to account for measurements taken during the growing season. The growing season for height was assumed to occur between May 15 and July 15 (62 days). The age of a tree for estimating site index was thus the ring count minus the portion of the growing season for height missing when the tree was measured.

¹ Wiley, K.N. 1978. Site index tables for western hemlock in the Pacific Northwest. Weyerhaeuser Company, For. Res. Center For. Paper. 28 pp.

4.2.2 Stand Mensurational Data

Measurements were taken from all trees in the sample plots to describe the general stand conditions from which site indices were estimated. These measurements included quadrant (starting north and progressing clockwise), diameter, species, and live/dead code for all trees. In dense stands where it was estimated that plots contained more than 50 trees and the stand was relatively uniform, randomly selected half plots were measured.

4.2.3 Ecological Data

Each 400 m² sample plot was classified to the BEC site series level. Classification was based on detailed descriptions of the vegetation, soil, and environmental conditions.

4.3 Site Index Conversion Measurements

Site index pairs of Hw with the other three target species were taken using standard measurement and site tree selection procedures. To reduce site variation between pairs, the sample trees were located within 10 m of each other, and on the same BEC site series. Site indices for all trees were computed in the field and later confirmed with computer analyses. Site indices were estimated using the curves and equations for coastal species presented by Thrower and Nussbaum.¹

4.4 Basal Area Measurements

The basal area factor (BAF) for each plot cluster was selected at the main plot and used in all plots in the cluster (Figure 7, Figure 6). Randomly selected half sweeps were used when an initial sweep indicated the plot tally would exceed 12 trees. Measurement of the selected trees included diameter, species, and the live/dead code. Three additional Hw trees were measured from across the range of diameters in each cluster to provide data to construct height-diameter equations for volume estimation.

5.0 Results and Discussion

5.1 Hemlock Site Index

5.1.1 Average Site Index and Ecological Trends

The results showed the expected trend where the average Hw site indices increased from PG 4 to PG 1 (Table 40, Figure 8, Appendix A.4). Productivity Group 1 showed the highest site indices and the lowest standard error. This was expected as the alluvial ecosystems in this PG are relatively homogeneous and are among the highest in productivity on northern Vancouver Island. Productivity Group 2 also showed a relatively narrow range of observed site index compared to PGs 3 and 4 (Figure 8). The wide range of site indices in these two broad groups reflects the wide ecological amplitude of Hw and also the variation inherent in these PGs. Further work could be done to narrow the range of productivity in these groups by more closely examining ecological conditions and trends for use in future studies.

Table 40. Site index statistics (using corrected ages) by Productivity Group.

Prod. Group	No. Plots	Avg. SI (m)	Min SI (m)	Max SI (m)	SD SI (m)	SE SI (m)
1	8	33.7	30.9	38.5	2.73	0.96
2	13	29.3	22.0	34.5	4.15	1.15
3	25	27.9	16.3	37.2	7.12	1.42
4	15	20.1	6.0	30.6	8.22	2.12

¹ Thrower, J.S. and Nussbaum, A.F. 1991. Site index curves and tables for British Columbia. BC Min. For. Res. Br. Lnd. Mgmt. Hdbk. Field Guide Insert 3.

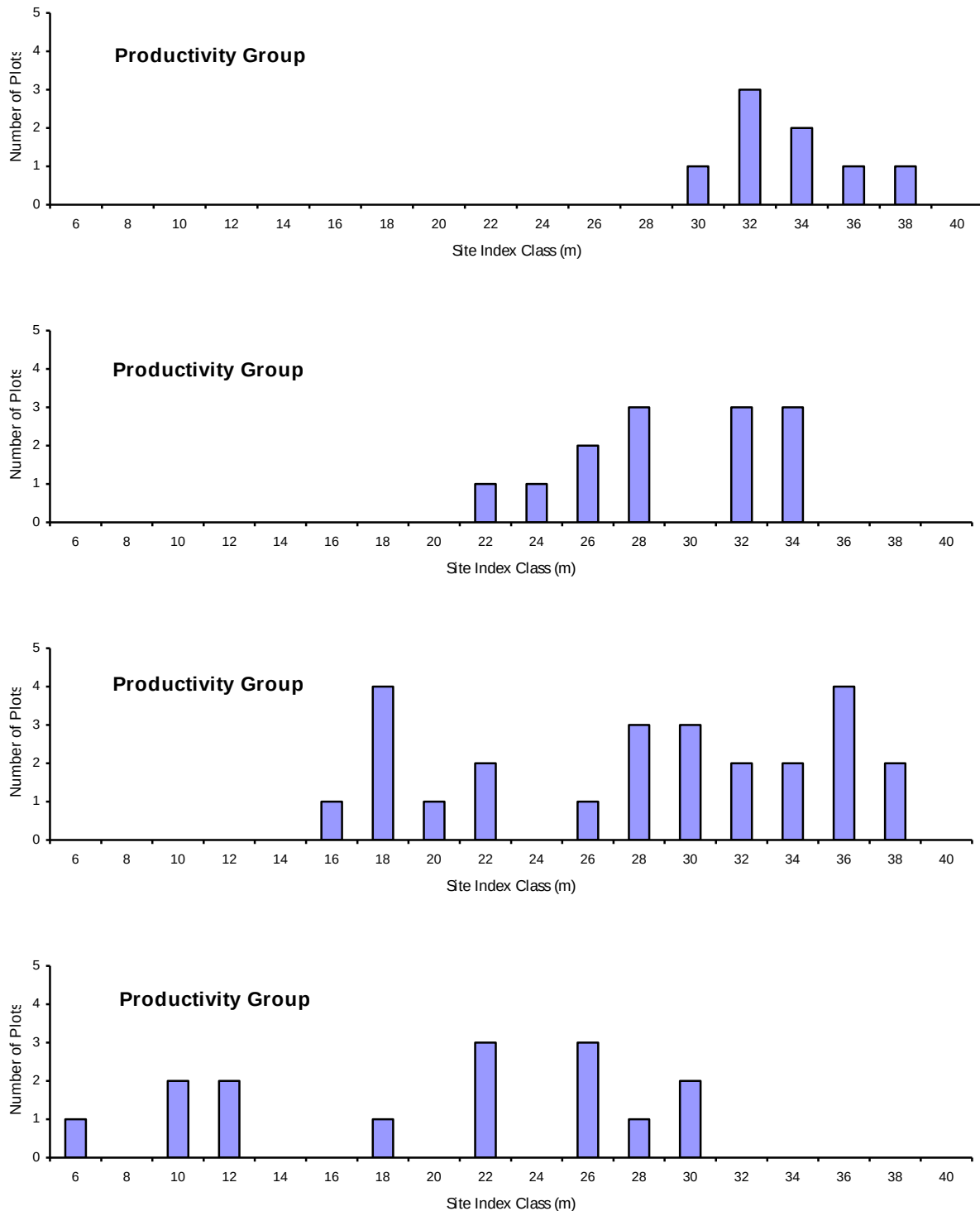


Figure 8. Distribution of sample plots by site index and Productivity Group.

5.1.2 Combining Productivity Groups 2 and 3

Considerable variation in site index was observed in the PGs, and increased from PG 1 through PG 4. This wide range of site index within relatively homogeneous ecological units is a characteristic of Hw not shared by other coastal species such as Douglas-fir. This suggests that some of the PGs could be combined. For example, PG 2 and PG 3 differ in site index by only 1.4 m. This difference is probably not significant, and combining the two PGs would simplify timber supply analysis procedures. The combined PG represents about 78% of the TFL. Maintaining these two groups as separate units probably will not provide significant advantages given the small difference in average site index. We thus recommend combining these two groups.

5.1.3 Western Hemlock Age Correction

The site index of each plot was calculated with and without the age correction accounting for the measurement of trees during the period of active height growth. Overestimating age by half a growing season on average resulted in site index underestimation of 0.4 m on average. As the final results are rounded to the nearest meter of site index, this additional adjustment resulted in the rounding up of site index for only Hw in PG 1.

5.2 Site Index Conversion Equations

A linear equation was used to express the relationship between the site index of Hw and the other target species:

$$SI = B_0 + B_1 SI_{Hw}$$

The intercept and slope were estimated using the least-squares technique.

The relationship between the site index of the Hw and neighboring Ba and Cw site trees was relatively strong (Table 41, Figure 9, Figure 10), and the relationship with Ss was somewhat weaker (Table 41, Figure 11).

Table 41. Regression statistics for the site index conversion equations.

Equation	No. Site Index Pairs	Equation Intercept (m)	Equation Slope	R ²	Standard Error (m)
Hw-Ba	49	6.8	0.691	77.2 %	2.58
Hw-Cw	29	7.7	0.572	70.4 %	3.48
Hw-Ss	28	9.8	0.756	44.8 %	3.47

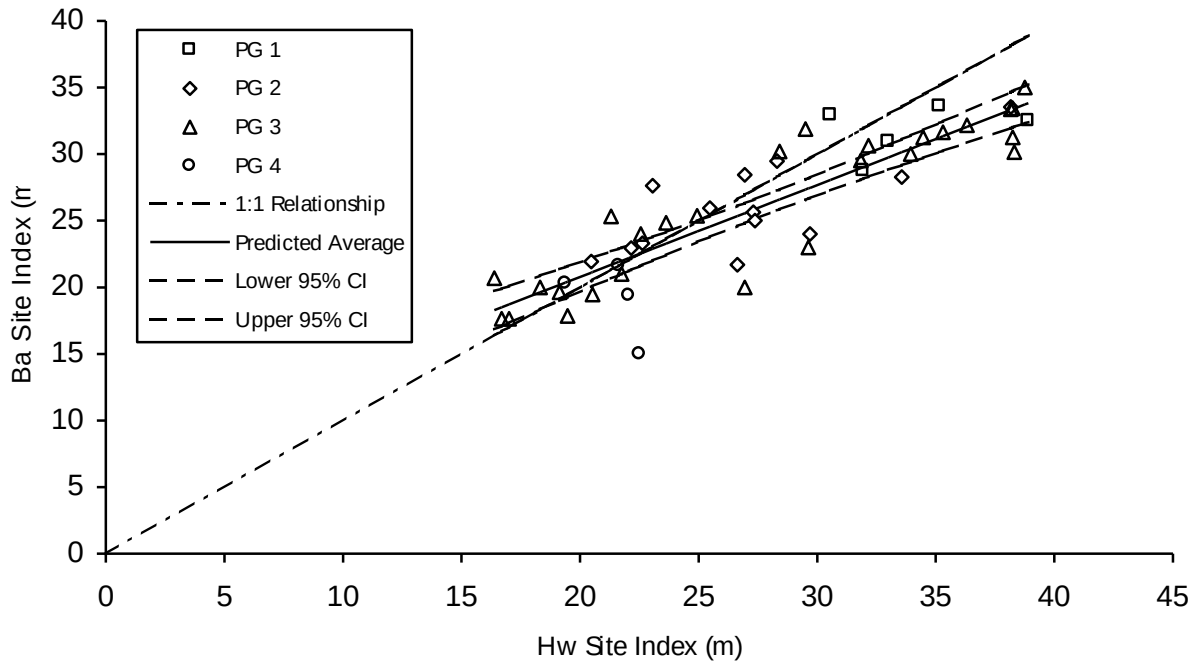


Figure 9. Relationship between Hw and Ba site index.

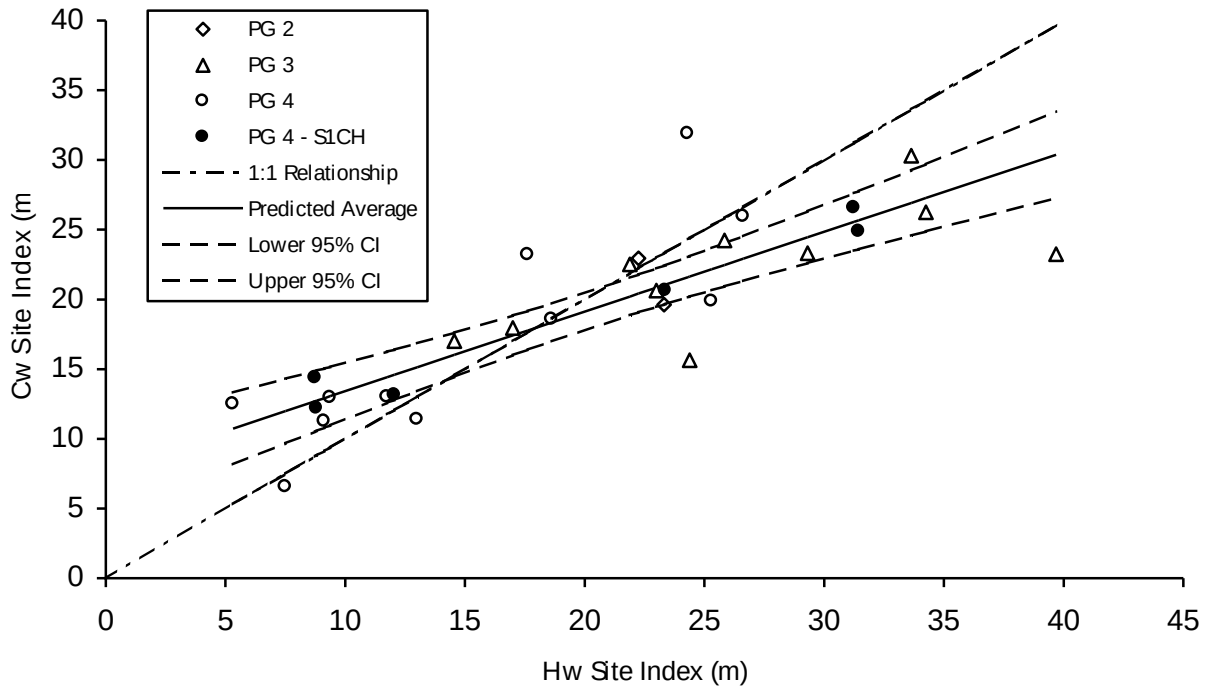


Figure 10. Relationship between Hw and Cw site index.

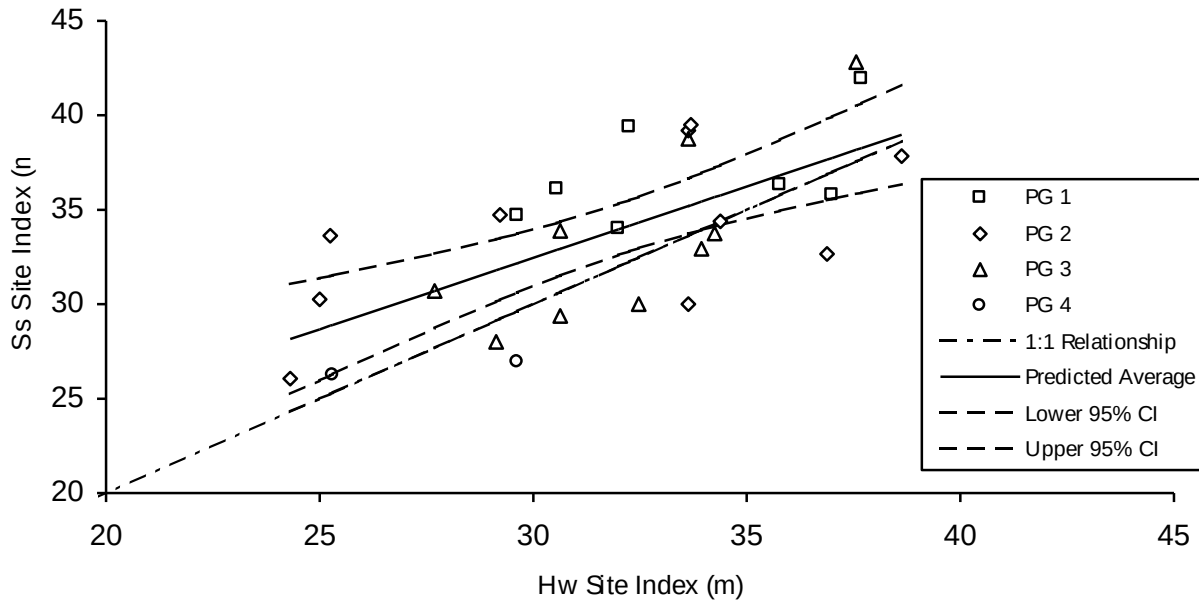


Figure 11. Relationship between Hw and Ss site index.

5.3 Site Index Estimates for All Target Species

The average site indices for Ba, Cw, and Ss were estimated with the conversion equations developed in this project (Table 41) and for Fdc using the equations developed by the MOF.¹ Together with the estimated averages from the random Hw plots, this provides estimates for the four major commercial species and ecosystems on the TFL (Table 42). These estimates apply to over 96% of the forest landbase.

Table 42. Average site index (m) for the five target species by Productivity Group.

Prod. Group	Hw Random Sampling Estimate	Ba TFL 6 Conversion Equation	Cw TFL 6 Conversion Equation	Ss TFL 6 Conversion Equation	Fdc Nigh's Conversion Equation
1	33.7	30.1	27.0	35.2	37.5
2	29.3	27.1	24.5	31.9	32.7
3	27.9	26.1	23.7	30.9	31.2
2+3	28.0	26.2	23.8	31.0	31.4
4	20.1	20.7	19.2	25.0	22.6

¹ Site Index (Fdc) = 0.48 + 1.11 Site Index (Hw). Nigh, G.D. 1995. Site index conversion equations for mixed species stands. BC Min. For. Res. Rep. No. 1. 20 pp.

5.4 Comparison with Previous Estimates

The estimated average Hw site indices in this study are very similar to the previous estimates derived by Dr. Barker (Table 43). The largest proportion of the landbase is in PG3 (68%) where the new estimated average Hw site index is higher than the previous estimate by only 0.4 m. The largest difference in Hw site index is in PG2 (10% of the landbase) where the new estimate is lower by 1.7 m. The average site index weighted by the proportion of area in each PG is only 0.3 m higher than the previous estimates (Table 44).

Table 43. Comparison of Hw site indices.

Prod. Group	Barker Group	Area in TFL (ha)	Prop. of TFL (%)	Barker Site Index (m)	New Site Index (m)	Diff. from Barker (m)
1	EE	7,405	5.0 %	33.0	33.7	+ 0.7
2	EG+	14,668	9.9 %	31.0	29.3	- 1.7
3	EG	87,381		28.0		
	EG-	13,795		24.0		
	<i>Total</i>	<i>101,175</i>	<i>67.9 %</i>	<i>27.5</i>	<i>27.9</i>	<i>+ 0.4</i>
4	EM	9,838		20.0		
	EL	10,485		18.0		
	<i>Total</i>	<i>20,323</i>	<i>13.6 %</i>	<i>19.0</i>	<i>20.1</i>	<i>+ 1.1</i>

The average site index for Ba, Cw, Ss, and Fdc predicted by the conversion equations differed from the previous estimates more than Hw. Overall, the new estimates are about 1.3 m lower for Ba and about 0.4 m lower for Cw. The new estimates for Fdc and Ss were higher overall. However, the averages shown in Table 44 are weighted by the proportion of area in each PG. Thus, the overall averages are not meaningful for Fdc and Ss, which do not occur on all the PGs.

Table 44. Comparison of new estimates of average site index (m) with estimates from Management Plan 8.

Prod. Group	Hw Random Sampling			Ba TFL 6 Conversion Equations			Cw TFL 6 Conversion Equations			Ss TFL 6 Conversion Equations			Fdc Nigh Conversion Equations		
	New SI	MP 8 SI	Change	New SI	MP 8 SI	Change	New SI	MP 8 SI	Change	New SI	MP 8 SI	Change	New SI	MP 8 SI	Change
1	33.7	33.0	+0.7	30.1	33.0	-2.9	27.0	30.0	-0.2	35.2	36.0	-0.8	37.5	36.0	+1.5
2	29.3	31.0	-1.7	27.1	31.0	-3.9	24.5	27.0	-0.2	31.9	33.0	-1.1	32.7	33.0	-0.3
3	27.9	27.5	+0.4	26.1	27.5	-1.4	23.7	24.5	-0.5	30.9	29.3	+1.6	31.2	29.3	+1.9
2+3	28.0	27.9	+0.1	26.2			23.8			31.0			31.4		
4	20.1	19.0	+1.1	20.7	19.0	1.7	19.2	15.0	+0.6	25.0	18.5	+6.5	22.6	18.5	+4.1
Wtd Avg	26.3	26.0	+0.3	24.7	26.0	-1.3	22.5	22.8	-0.4	29.3	27.5	+1.8	29.4	27.5	+1.9

5.5 Basal Area

Average basal area in PG 2 was significantly higher than the other three PGs (Table 45). However, because stand density is generally lower on higher sites, the quadratic mean diameter was larger in PG 1. These results show excellent stocking in all PGs, which was also noted during fieldwork. Statistics for the 61 plots are given in Appendix A.5 and Appendix A.6.

The height-diameter relationships for Hw and Ba were very strong (Table 46, Appendix A.7). Individual tree volume was predicted using the MOF logarithmic volume equations¹ using diameter at breast height and the predicted heights. For minor species, the Hw equation was used to estimate volume. The volume/basal area ratio (VBAR) was about 8 m³/m² for PGs 1, 2, and 3, and 5.5 m³/m² for PG 4. This reflects the generally shorter trees in the lower productivity PG 4.

Table 45. Average inventory attributes from the random sample plots.

Prod. Group	Average Top Ht (m)	Average BH Age (yrs)	Average Site Index (m)	Average Density (no/ha)	Average Basal Area (m ² /ha)	Average DBHq (cm)	Total Gross Volume (m ³ /ha)	Average VBAR (m ³ /m ²)
1	21.6	27.6	33.7	2,431	46.6	15.6	374.3	8.0
2	22.5	37.0	29.3	4,363	58.2	13.0	490.5	8.4
3	21.0	35.8	27.9	4,383	47.4	11.7	373.9	7.9
4	12.3	28.0	20.1	3,498	27.1	9.9	150.8	5.6

Table 46. Height-diameter equations.

Spp	No. Trees	HT-DBH Equation	R ²	SEE (m)
Ba	60	$Ht = 1.3 + 83.347(1 - e^{-0.004516DBH^{1.19949}})$	90.9%	2.58
Cw	44	$Ht = 13 + 25.045(1 - e^{-0.036DBH^{0.915}})$	80.1%	1.86
Hw	371	$Ht = 1.3 + 95.045(1 - e^{-0.005548DBH^{1.121847}})$	85.5%	3.54
Ss	38	$Ht = 1.3 + 0.6543DBH - 0.00179DBH^2$	68.0%	4.42

¹ Ministry of Forests. 1976. Whole Stem Cubic Metre Volume Equations.

6.0 Recommendations

This study provides statistically-based estimates of average site index for the major commercial tree species and ecosystems on TFL 6. We suggest the following recommendations for application of these estimates:

- Productivity Groups 2 and 3 should be combined for estimating average site index for timber supply analyses (Table 47). The increased simplicity in analyses from the reduced number of PGs is more significant than the very small difference in the average site index between these two groups (Table 42).

Table 47. Site indices recommended for timber supply analyses for TFL 6.

	HW	Ba	Cw	Ss	Fdc
Prod. Group	Random Sampling (m)	TFL 6 Conversion Equation (m)	TFL 6 Conversion Equation (m)	TFL 6 Conversion Equation (m)	Nigh's Conversion Equation (m)
1	34	30	27	35	38
2+3	28	26	24	31	31
4	20	21	19	25	23

- These average site indices are applicable for growth and yield estimation in timber supply analyses. The estimates were developed independently for each PGs and should be applied to the landbase in the same manner.
- These PGs should be saved as a separate layer in the GIS. This will provide a framework for application of the estimates that is consistent with the sampling used in their derivation. This is needed in the event that future modifications or adjustments are made to map boundaries for the ecological classification.
- These estimates are not applicable for predicting the growth and yield of old-growth stands. The appropriate method of yield prediction should reflect actual growth in these stands, and be linked to the stand-level inventory attributes.
- These site indices should be applied to existing post-harvest regenerated stands where the current inventory site index was derived from a pre-harvest (old-growth) stand.

The site indices for timber supply for areas not sampled in this project should be assigned the values used in the previous analyses for Management Plan 8.

Appendix A.1 - TFL 6 Area by species and age class

Table 48. Area (ha) by species and WFP age class.

Species	Age Class								Total	% Total
	1	2	3	4	5	6	7	8		
A	1	0	0	0	0	0	0	0	1	0.0 %
Ac	4	0	0	0	0	0	0	0	4	0.0 %
B	1,201	147	105	175	88	80	26	1,619	3,442	2.3 %
Bg	3	0	0	0	0	0	0	0	3	0.0 %
Bp	1	0	0	0	0	0	0	0	1	0.0 %
C	5,232	230	69	44	9	14	9	23,000	28,608	18.8 %
Cy	201	1	0	9	0	0	0	8,001	8,212	5.4 %
D	96	901	1,155	476	106	0	1	0	2,735	1.8 %
Dc	0	0	0	0	0	0	0	0	0	0.0 %
Ds	1	0	0	0	0	0	0	0	1	0.0 %
F	891	598	300	67	2	0	68	225	2,151	1.4 %
H	18,969	19,458	10,424	13,139	1,465	928	765	37,516	102,664	67.5 %
HS	2	0	0	0	0	0	0	0	2	0.0 %
Pl	9	0	2	0	0	0	0	0	12	0.0 %
S	1,702	1,610	25	10	10	7	5	890	4,261	2.8 %
Total	28,319	22,945	12,080	13,920	1,680	1,030	875	71,251	152,099	100.0 %
% Total	18.6 %	15.1 %	7.9 %	9.2 %	1.1 %	0.7 %	0.6 %	46.8 %	100.0 %	

Appendix A.2 - TFL 6 Ecological Classification

Table 49. Area (ha) by species and Lewis Ecosystem unit.

Spp	A	Ac	B	Bg	Bp	C	Cy	D	Dc	Ds	F	H	HS	PI	S	Total	% Total
A						4	29	2				88				123	0.1 %
CP						2,951	1,581					90				4,622	3.0 %
M1			1,194		0	1,407	1,099	3			0	8,844			14	12,561	8.3 %
M2			37			471	712					657				1,878	1.2 %
M3			7			4	3					93				106	0.1 %
M4			3			74	144					111				332	0.2 %
M5			18			278	136	3				686			6	1,127	0.7 %
M6						1										1	0.0 %
MH			57			115	205					468				845	0.6 %
NV								1				8			7	17	0.0 %
P			18			125	501					510				1,154	0.8 %
PM							1									1	0.0 %
S11			5			31	3	10			3	49			13	114	0.1 %
S12			23			417	13	162			539	4,491			6	5,651	3.7 %
S13			318			1,790	217	212			100	10,982		0	243	13,862	9.1 %
S1CH			47			5,187	646	34			68	3,629		4	224	9,838	6.5 %
S1HA		1	1,574		1	10,804	897	1,134	0	1	1,301	64,626		4	1,386	81,730	53.7 %
S2			20			1,371	282	2			11	1,733			57	3,477	2.3 %
S2F			2			185	3	17			86	574			13	880	0.6 %
S2P						1					1	2		3		7	0.0 %
S3		2	102	3		564	11	1,035			18	3,702	2	1	1,776	7,216	4.7 %
S3B			2			6		10				58			113	189	0.1 %
S4			1			1	1	29				63			18	113	0.1 %
S5	1	1	7			193		74			1	181			235	693	0.5 %
S6			7			2,381	360	6			24	988			147	3,913	2.6 %
S7						33	26					2				61	0.0 %
S9						215	1,342					19				1,575	1.0 %
W						0		3				6			2	11	0.0 %
None			0			0	0					2			0	3	0.0 %
Total	1	4	3,442	3	1	28,608	8,212	2,735	0	1	2,151	102,664	2	12	4,261	152,099	100.0 %
% Total	0.0 %	0.0 %	2.3 %	0.0 %	0.0 %	18.8 %	5.4 %	1.8 %	0.0 %	0.0 %	1.4 %	67.5 %	0.0 %	0.0 %	2.8 %	100.0 %	

Appendix A.3 - BEC-Lewis Conversions

Table 50. BEC-Lewis conversions.

Lewis Unit	BEC Unit	Name	Comments
CWHvm1			
S1-ha	01	HwBa – Blueberry	
S1-ch	01s	HwBa – Blueberry: Salal phase	
S2	03	HwCw- Salal	
S2 ^F and S2 ^P	02	HwPI – Cladina	
S3	09	Ss – Salmonberry (high bench)	Inactive floodplains may be 07 BaCw –Salmonberry
S3 ^B	04	CwHw – Swordfern	May include 05 BaCw - Foamflower if SMR is fresh
S4	10	Act -Red osier dogwood	
S5	14	CwSs - Skunk cabbage	Rich alluvial Skunk cabbage
S6	14	CwSs- Skunk cabbage	Poor non-alluvial Skunk cabbage
S7	13	PI – Sphagnum	
S8	13	PI – Sphagnum	
S9	na	no site unit	Non-forested bog
S10	06s	HwBa - Deer fern: salal phase	Moist, poor CwHw salal site
S11	na	no site unit	Non-forested fen
S12	05	BaCw – foamflower	Limestone influence evident
S12 ^{CH}	01	HwBa –Blueberry	Limestone influence masked by thick for. floor
S12 ^F	05	BaCw –Foamflower	Limestone influence evident
S13	05 & 07	BaCw –Foamflower & BaCw – Salmonberry	
S15	06	HwBa - Deer Fern	Not described in TFL 6 report
CP	12	CwYc – Goldthread	
CWHvm2			
M1	01	HwBa – Blueberry	
M2	03	HwCw – Salal	
M3	05	BaCw – Foamflower	
M4	11	CwSs _ Skunk cabbage	
M5	08	BaSs -Devil's club	
MH & AT			
MH or MH1	01	HmBa – Blueberry	No information provided in reports
MH2	03	HmBa – Mountain heather	No information provided in reports
MH4	05	HmYc - Deer cabbage	No information provided in reports
P	na	Parkland	
A	na	Snow avalanche	
W	na	Wetlands	
AT	na	Alpine Tundra	

Appendix A.4 - Site index statistics by Barker Productivity Groups, Lewis Ecosystem Units, and BEC Site Series

Table 51. Site Index statistics by Barker Productivity Groups.

Prod. Group	Random Sample Plots	Average Site Index (m)	Minimum Site Index (m)	Maximum Site Index (m)	Standard Deviation (m)	Standard Error (m)
EE	8	33.7	30.9	38.5	2.73	0.96
EG+	13	29.3	22.0	34.5	4.15	1.15
EG	17	31.5	17.1	37.2	5.29	1.28
EG-	8	20.3	16.3	27.3	3.56	1.26
EL	8	19.6	6.0	30.6	9.31	3.29
EM	7	20.7	9.8	28.4	7.46	2.82

Table 52. Site index statistics by Lewis Ecosystem Units.

Lewis Eco. Unit	Random Sample Plots	Average Site Index (m)	Minimum Site Index (m)	Maximum Site Index (m)	Standard Deviation (m)	Standard Error (m)
S1CH	7	20.7	9.8	28.4	7.46	2.82
S1HA	15	32.1	25.4	37.2	3.87	1.00
S2	1	17.3	17.3	17.3		
S3	8	33.7	30.9	38.5	2.73	0.96
S5	1	33.5	33.5	33.5		
S6	6	19.5	6.0	30.6	10.90	4.45
S12	2	26.8	17.1	36.5	13.74	9.71
S13	12	29.0	22.0	34.5	4.13	1.19
M1	8	20.3	16.3	27.3	3.56	1.26
M2	1	22.3	22.3	22.3		

Table 53. Site index statistics by Lewis units converted to BEC site series.

Converted BEC Site Series	Random Sample Plots	Average Site Index (m)	Minimum Site Index (m)	Maximum Site Index (m)	Standard Deviation (m)	Standard Error (m)
CWHvm1/01	15	32.1	25.4	37.2	3.87	1.00
CWHvm1/01s	7	20.7	9.8	28.4	7.46	2.82
CWHvm1/03	1	17.3	17.3	17.3		
CWHvm1/05	2	26.8	17.1	36.5	13.74	9.71
CWHvm1/05-07	12	29.0	22.0	34.5	4.13	1.19
CWHvm1/09	8	33.7	30.9	38.5	2.73	0.96
CWHvm1/14	7	21.5	6.0	33.5	11.27	4.26
CWHvm2/01	8	20.3	16.3	27.3	3.56	1.26
CWHvm2/03	1	22.3	22.3	22.3		

Table 54. Site index statistics by field classified BEC site series.

Field Class. BEC Site Series	Random Sample Plots	Average Site Index (m)	Minimum Site Index (m)	Maximum Site Index (m)	Standard Deviation (m)	Standard Error (m)
CWHvm1/01	21	29.5	17.3	37.2	5.22	1.14
CWHvm1/01s	2	18.2	11.0	25.4	10.15	7.17
CWHvm1/05	9	33.2	26.6	38.5	3.68	1.23
CWHvm1/06	5	25.7	11.0	32.1	8.60	3.85
CWHvm1/06s	2	15.5	9.8	21.3	8.11	5.73
CWHvm1/07	9	32.0	27.4	37.2	3.79	1.26
CWHvm1/13	2	9.2	6.0	12.4	4.49	3.17
CWHvm1/14	1	26.9	26.9	26.9		
CWHvm2/01	6	20.4	17.1	22.6	2.32	0.95
CWHvm2/05	1	18.6	18.6	18.6		
CWHvm2/06	1	17.4	17.4	17.4		
CWHvm2/07	1	27.3	27.3	27.3		
CWHvm2/11	1	16.3	16.3	16.3		

Appendix A.5 - Plot Statistics

Table 55. Inventory statistics for the 61 sample plots.

PG	Plot No.	Lewis Unit	Map BEC Site Series	Barker PG	No. BA Plots	Ht (m)	BH Age (yrs)	Site Index (m)	Density All Stems (no/ha)	BA (m ² /ha)	Total Gross Vol. (m ³ /ha)	DBHq (cm)
1	1-01	S3	CWHvm1/09	EE	3	23.0	29.0	34.4	245	32.0	388.2	40.8
1	1-02	S3	CWHvm1/09	EE	4	28.3	34.3	36.8	632	54.0	608.6	33.0
1	1-04	S3	CWHvm1/09	EE	2	19.1	24.5	33.1	7,310	85.0	444.9	12.2
1	1-05	S3	CWHvm1/09	EE	4	25.1	35.0	32.5	2,002	56.0	484.7	18.9
1	1-06	S3	CWHvm1/09	EE	5	16.6	21.7	32.3	2,214	48.0	308.6	16.6
1	1-07	S3	CWHvm1/09	EE	4	14.7	19.9	31.1	3,351	40.5	220.0	12.4
1	1-08	S3	CWHvm1/09	EE	5	27.2	30.6	38.5	747	41.6	388.6	26.6
1	1-09	S3	CWHvm1/09	EE	5	18.4	25.7	30.9	4,740	34.8	236.8	9.7
2	2-01	S13	CWHvm1/05-07	EG+	5	41.3	67.3	34.5	2,108	92.8	933.3	23.7
2	2-03	S13	CWHvm1/05-07	EG+	5	43.9	83.2	32.5	604	80.0	1,189.6	41.1
2	2-04	S13	CWHvm1/05-07	EG+	4	19.2	32.9	26.6	2,172	45.0	290.4	16.2
2	2-05	S13	CWHvm1/05-07	EG+	2	29.3	44.1	31.9	2,000	60.0	541.2	19.5
2	2-07	S13	CWHvm1/05-07	EG+	3	14.0	18.0	32.5	6,441	58.7	270.6	10.8
2	2-08	S13	CWHvm1/05-07	EG+	2	12.6	19.7	27.4	1,828	8.0	37.3	7.5
2	2-10	S13	CWHvm1/05-07	EG+	3	16.5	26.5	27.4	1,041	34.7	295.9	20.6
2	2-11	S13	CWHvm1/05-07	EG+	5	35.7	53.6	34.2	835	68.0	711.7	32.2
2	2-12	S13	CWHvm1/05-07	EG+	5	16.2	31.3	23.5	13,931	80.0	491.9	8.6
2	2-13	S13	CWHvm1/05-07	EG+	4	13.6	20.4	28.6	4,342	22.0	93.9	8.0
2	2-14	S5	CWHvm1/14	EG+	4	19.1	24.1	33.5	615	44.0	379.3	30.2
2	2-15	S13	CWHvm1/05-07	EG+	5	17.7	38.0	22.0	5,532	70.8	379.9	12.8
2	2-16	S13	CWHvm1/05-07	EG+	4	13.6	22.4	26.4	12,210	38.0	143.1	6.3
3	3-01	S1HA	CWHvm1/01	EG	4	29.1	49.8	29.4	881	33.0	300.7	21.8
3	3-02	S1HA	CWHvm1/01	EG	5	15.0	24.1	27.2	6,220	45.6	195.9	9.7
3	3-03	S12	CWHvm1/05	EG	5	27.8	33.9	36.5	740	52.8	595.7	30.1
3	3-04	S1HA	CWHvm1/01	EG	3	29.8	37.6	36.2	698	56.0	661.4	32.0
3	3-06	S1HA	CWHvm1/01	EG	4	14.2	18.5	32.1	14,207	54.0	267.4	7.0
3	3-07	S1HA	CWHvm1/01	EG	4	26.4	37.8	32.3	751	38.0	365.3	25.4
3	3-09	S1HA	CWHvm1/01	EG	4	20.7	34.7	27.4	899	48.0	449.2	26.1
3	3-10	S1HA	CWHvm1/01	EG	5	18.0	19.7	37.2	2,365	52.4	349.7	16.8
3	3-11	M1	CWHvm2/01	EG-	4	16.5	34.1	22.3	1,632	58.0	465.6	21.3
3	3-12	S1HA	CWHvm1/01	EG	5	26.5	34.1	34.8	2,566	27.6	235.7	11.7
3	3-16	S12	CWHvm1/05	EG	5	10.9	29.2	17.1	4,098	23.2	89.9	8.5
3	3-19	S1HA	CWHvm1/01	EG	5	10.9	15.8	29.5	1,677	8.8	31.7	8.2
3	3-20	S1HA	CWHvm1/01	EG	5	17.9	32.2	25.4	5,546	41.6	198.4	9.8
3	3-26	S1HA	CWHvm1/01	EG	5	22.6	25.5	37.2	2,635	51.6	359.9	15.8
3	3-27	S1HA	CWHvm1/01	EG	5	31.7	43.0	35.0	727	60.0	681.2	32.4
3	3-28	S1HA	CWHvm1/01	EG	5	39.1	59.9	35.0	464	48.0	580.0	36.3
3	3-30	S1HA	CWHvm1/01	EG	3	38.6	78.3	29.2	1,014	96.7	1,126.2	34.8
3	3-31	S1HA	CWHvm1/01	EG	5	30.8	43.5	33.8	1,688	70.4	630.5	23.0
3	3-53	M1	CWHvm2/01	EG-	4	19.2	31.9	27.3	14,899	49.0	278.0	6.5
3	3-64	M1	CWHvm2/01	EG-	3	8.3	16.2	22.6	6,949	21.3	80.4	6.3
3	3-71	M1	CWHvm2/01	EG-	2	8.0	18.9	18.6	5,609	33.0	113.9	8.7
3	3-72	M1	CWHvm2/01	EG-	4	12.6	31.6	18.5	19,258	77.5	331.2	7.2
3	3-75	M1	CWHvm2/01	EG-	4	27.7	83.9	19.6	2,222	67.5	642.4	19.7
3	3-76	M1	CWHvm2/01	EG-	4	11.5	30.6	17.4	11,315	41.0	222.5	6.8
3	3-79	M1	CWHvm2/01	EG-	4	11.2	31.3	16.3	4,881	40.0	192.4	10.2
4	4-02	S1CH	CWHvm1/01s	EM	5	5.4	24.0	9.8	523	6.4	34.4	12.5
4	4-03	S1CH	CWHvm1/01s	EM	2	20.2	44.0	22.3	1,986	39.0	233.9	15.8
4	4-04	S2	CWHvm1/03	EL	3	13.9	38.0	17.3	3,852	42.7	218.6	11.9
4	4-05	S6	CWHvm1/14	EL	5	14.0	20.2	30.1	1,708	26.4	151.5	14.0
4	4-08	S6	CWHvm1/14	EL	4	7.4	27.0	12.4	4,417	26.0	126.4	8.7



February 2011

PG	Plot No.	Lewis Unit	Map BEC Site Series	Barker PG	No. BA Plots	Ht (m)	BH Age (yrs)	Site Index (m)	Density All Stems (no/ha)	BA (m ² /ha)	Total Gross Vol. (m ³ /ha)	DBHq (cm)
4	4-09	M2	CWHvm2/03	EL	4	11.8	23.5	22.3	12,448	34.0	119.3	5.9
4	4-11	S6	CWHvm1/14	EL	4	22.9	40.5	26.9	3,888	58.0	445.2	13.8
4	4-12	S1CH	CWHvm1/01s	EM	3	16.9	26.1	28.4	3,832	18.0	90.9	7.7
4	4-15	S1CH	CWHvm1/01s	EM	4	6.6	26.5	11.0	1,942	12.0	49.7	8.9
4	4-18	S1CH	CWHvm1/01s	EM	1	9.8	20.3	21.3	430	8.0	46.0	15.4
4	4-28	S6	CWHvm1/14	EL	5	4.2	31.9	6.0	909	15.2	83.5	14.6
4	4-31	S1CH	CWHvm1/01s	EM	4	18.7	31.4	26.9	6,169	57.5	294.3	10.9
4	4-32	S1CH	CWHvm1/01s	EM	3	7.9	13.4	25.4	8,258	32.0	152.1	7.0
4	4-54	S6	CWHvm1/14	EL	5	6.3	26.0	11.0	2,636	16.8	69.5	9.0
4	4-55	S6	CWHvm1/14	EL	5	19.0	26.8	30.6	565	21.6	172.5	22.1

APPENDIX A.6 – Main Plot Attributes in Relation to Top Height

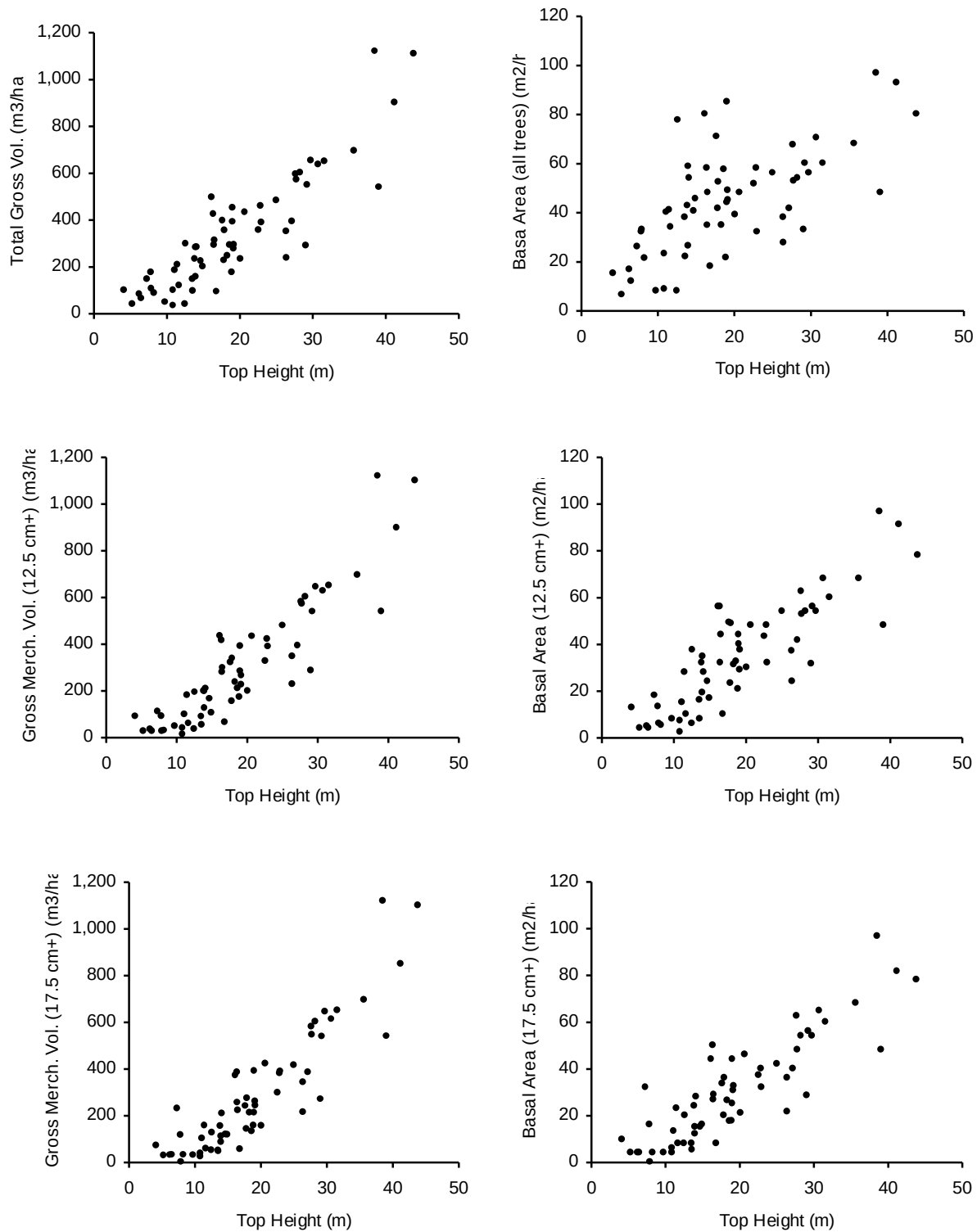


Figure 12. Plot volume and basal area in relation to top height (total, 12.5 cm+, and 17.5 cm+).

Appendix A.7 - Height-Diameter Curves

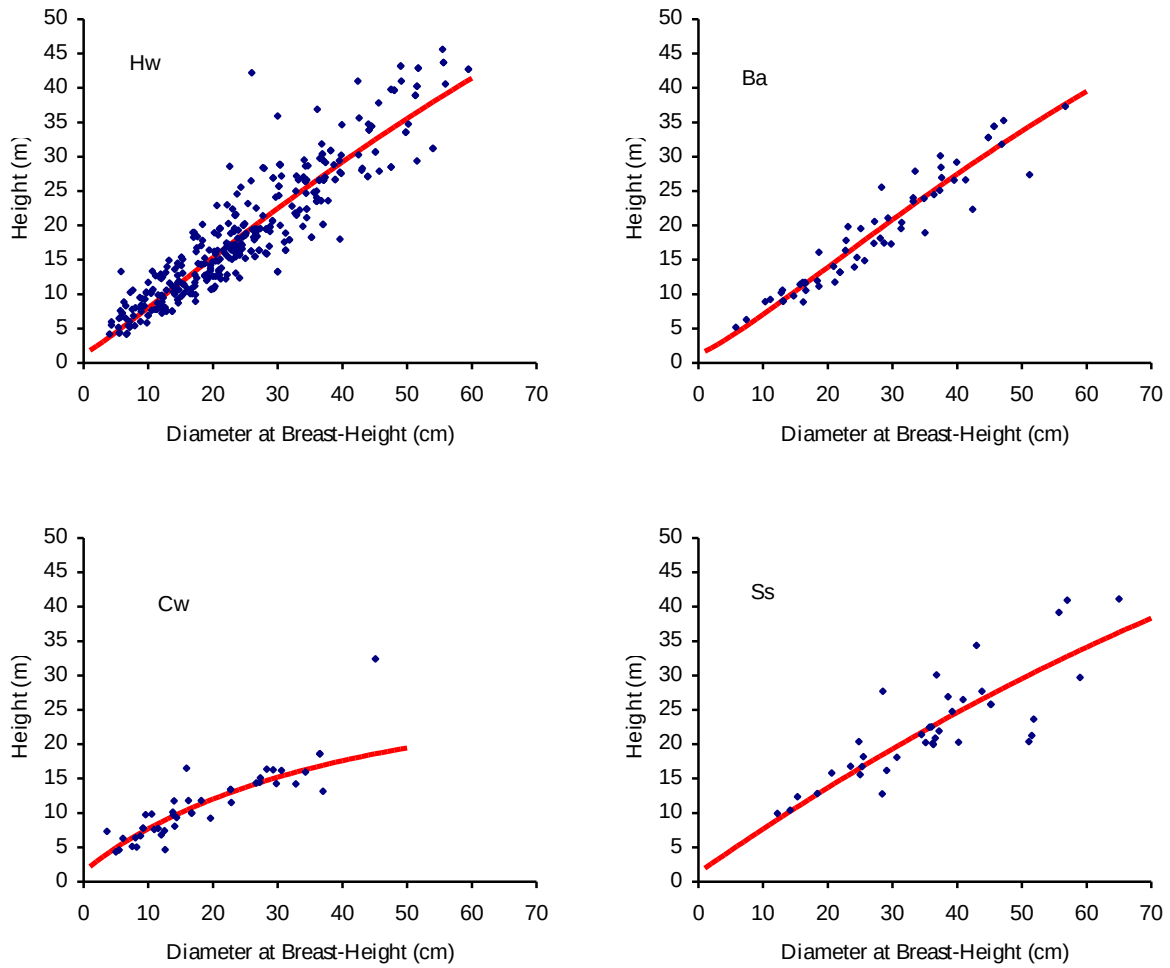


Figure 13. Height-diameter curves for Hw, Ba, Cw, and Ss.

Appendix B - TFL 6 VRI Statistical Adjustment

TFL 6
VEGETATION RESOURCES INVENTORY
STATISTICAL ADJUSTMENT

Prepared by:
Timberline Natural Resource Group Ltd.

Project Number: BC0108834

December 2009

EXECUTIVE SUMMARY

Western Forest Products Inc. (Western) initiated a Vegetation Resources Inventory (VRI) program in 2001 on Tree Farm License (TFL) 6 to Ministry of Forests and Range (MFR) inventory standards. The Phase II program was completed in the 2001 field season.

In May 2007, Timberline Natural Resource Group Ltd. was asked to complete the inventory adjustments in TFL 6 in preparation for Timber Supply Review (TSR). Height, age, and total live net merchantable volume (17.5+ cm) were adjusted following MFR inventory methods.

The target population, where the adjustment was applied, is the Vegetated Treed (VT) (BC Landcover Classification Scheme) portion of the TFL over 30 years of age (in 2001), excluding private lands, parks and other officially protected areas. The target population covers 137,688 ha.

Following adjustment, the TFL 6 inventory volume increased by approximately 14%. Height and age increased by 1% and 12%, respectively and site index decreased by 0.4%. The recommendations from this report are that Western apply the adjusted estimates of height, age, and volume into the upcoming TSR.

This version of the report incorporates the comments provided by MFR on July 27, 2009.

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1 Introduction

1.1 Background

1.1.1 Vegetation Resources Inventory Overview

The Vegetation Resources Inventory (VRI) is the Ministry of Forests and Range's (MFR) forest inventory standard on public lands in BC. Where possible, forest licensees must use the VRI standard in their Data Package submission for Timber Supply Review (TSR).

The VRI is a four-step process (Figure 14):

1. Phase I (unadjusted inventory data) – Estimates of polygon attributes are derived for the target population, usually from photo-interpretation.
2. Phase II (ground sample data) – Measurements are taken from randomly located ground samples in the target population.
3. Net Volume Adjustment Factor (NVAF) sampling – Random trees are selected for stem-analysis from the Phase II samples to develop adjustment ratios that correct taper and decay estimation bias.
4. Adjustment Phase – The Phase I estimates are adjusted using the NVAF-corrected Phase II ground samples to provide an adjusted unbiased estimate of forest inventory attributes. The final product is an adjusted VRI database (Section 3.4).

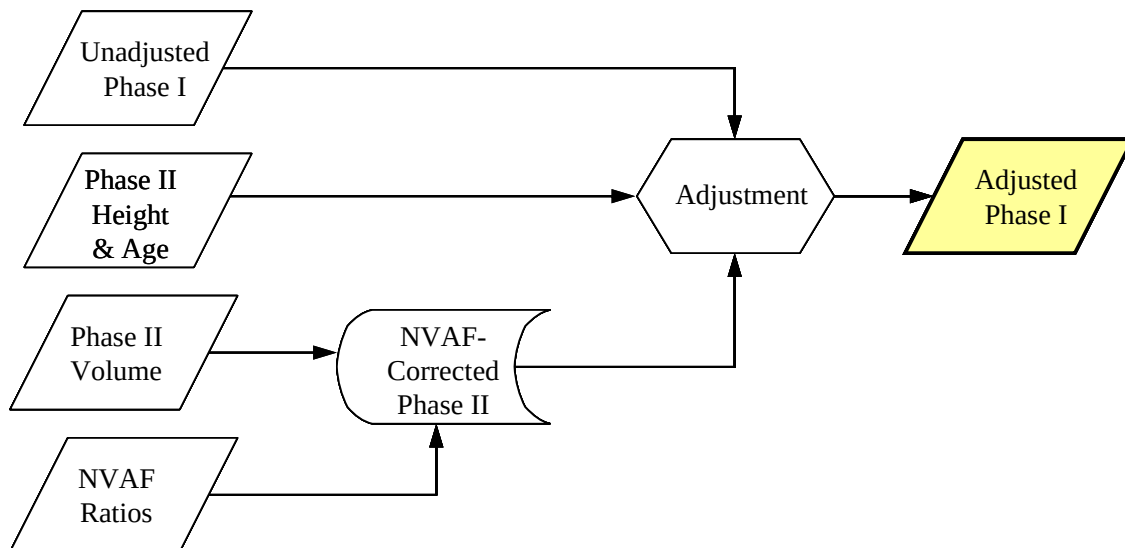


Figure 14: VRI program overview

1.1.2 VRI Program Background

Dave Byng, *RPF* led implementation of Western Forest Products Inc. (Western) Tree Farm Licence (TFL) 6 VRI Phase II program. In March 2006, Guillaume Thérien, *PhD* of Timberline Natural Resource Group Ltd. (Timberline) assisted Dave by completing statistical analysis of the data and developing preliminary adjustment factors for TFL 6.¹ Patrick Bryant, *RPF* of Western approached Timberline in March 2008 seeking to update the inventory with NVAF data, readjust the inventory according to the most current VRI statistical adjustment standards (i.e. only adjust polygons greater than 30 years), and to document the adjustment results.

1.2 Project Objectives

The objective of this project was to complete a statistical adjustment of the TFL 6 Phase I VRI to the most recent MFR standards using Phase II and NVAF data and report on the results.²

1.3 Terms of Reference

Timberline prepared this report for Patrick Bryant of Western. Stephanie Ewen, *RPF* was the lead analyst and prepared the report. Technical support was provided by Guillaume Thérien and the project manager was Hamish Robertson, *RPF*. This report will be provided to the MFR Forest Analysis and Inventory Branch (FAIB) for review and comment prior to its use in TSR.

¹ J.S. Thrower & Associates Ltd. 2005 Contract for Western Forest Products Ltd. (Project no. WPC-006).

² This analysis was completed in the spring of 2008 using VDYP (version 6.6d).

2 Data

2.1 Landbase

TFL 6 covers 205,839 ha in the northern portion of Vancouver Island (Figure 15). The TFL is administered by the North Island – Central Coast Forest District, which is part of the Coast Forest Region. TFL 6 is predominantly within the Coastal Western Hemlock biogeoclimatic zone. Western hemlock (Hw)-leading forests are most common in the TFL; also present are western redcedar (Cw), balsam (Ba), Sitka spruce (Ss), Douglas-fir (Fd) and red alder (Dr).

TFL 6 is bordered by the Kingcome Timber Supply Area (TSA), Strathcona TSA, TFL 39 and Cape Scott Provincial Park. A forest management license covering the TFL area was originally issued in 1950. In 1998, a portion of TFL 25 (Block 4 near Port McNeill) was added to TFL 6.

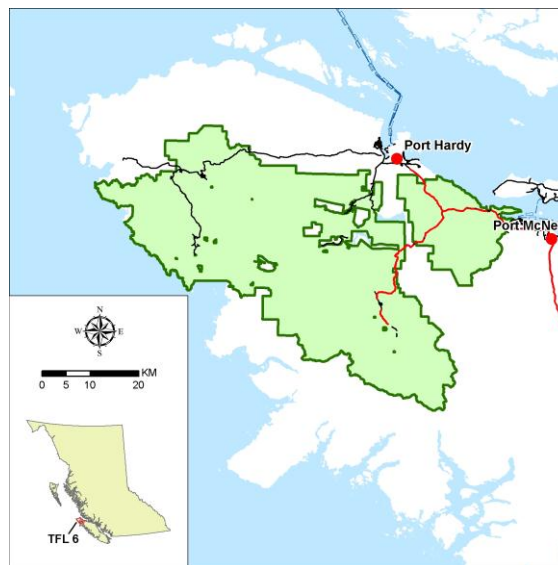


Figure 15: Location of TFL 6

2.2 Target Population

The TFL is 205,839 ha of which 137,688 ha are in the target population (Table 56).¹ The target population is the Vegetated Treed (VT) (BC Landcover Classification Scheme) portion of the TFL over 30 years of age (in 2001), excluding private lands, parks and other officially protected areas. The Phase I data provided the basis of units to be sampled. The main tree species in the target population are Hw (64%) and Cw (23%), reported by area as a leading species.

Table 56: TFL 6 netdown summary

Landclass	Area (ha)	%
Total Area	205,839	100
Leading Species Unknown	6,174	3
Productivity Group	5,477	
Unknown		2.7
Under 30 Years Old	56,500	27.4
Target Population	137,688	66.9

¹ The target population was identified from the 2006 Forest Cover database where records existed in the “treelayer” table, an age (at time of sampling) in the inventory ≥ 30 years, a populated leading species attribute (spp1), and a populated productivity group (prod_group) with a value ≤ 4 . The assumption is made that all private lands, parks and other officially protected areas were excluded from the total area of the TFL in the initial GIS data.

2.3 Stratification

2.3.1 Area

The Phase I population was stratified based on age class (Table 57). “Young” stands were established after 1861 (< 140 years in 2001), while “Old” stands were established in or before 1861 ($\geq$ 140 years in 2001). Each strata was sub-stratified into species groups based on Phase I leading species.¹

Table 57: TFL 6 stratification summary

Stratum	Sub-Stratum	Area (ha)	% Pop.	% Stratum
Young	Hw	52,969	38.5	87.6
Young	Cw	7,472	5.4	12.4
Young	Total	60,441	43.9	100
Old	Hw	42,054	30.5	54.4
Old	Cw	35,193	25.6	45.6
Old	Total	77,247	56.1	100
Total	Total	137,688	100	

Adjustment ratios were calculated at the strata level. Sub-strata were used to distribute samples.

2.3.2 Phase I (Photo-Interpretation) Inventory Statistics

Overall average net merchantable volume (17.5 cm utilization) in the unadjusted Phase I population was 507.7 m³/ha as projected to 2001 (Table 58). Average site index (SI) was approximately 26 m and 15 m in the “Young” and “Old” strata, respectively. Average age was approximately 59 years and 285 years for the “Young” and “Old” strata, respectively.

Table 58: Unadjusted inventory statistics for the TFL 6 target population

Stratum	Area (ha)	Mean Age (yrs)	Mean Height (m)	Mean SI (m)	Mean Vol. 17.5cm+ (m ³ /ha)
Young	60,441	59	25.4	25.5	382.3
Old	77,247	285	35.0	15.2	605.8
Total	137,688	186	30.8	19.7	507.7

Note: Phase I (photo-interpretation) volume is net merchantable volume as predicted from VDYP version 6.6d.

¹ The “Cw” species group includes stands that are Cw, Yc (yellow cedar), Dr (red alder), Pl (shore pine), Pw (white pine) or Ac (black cottonwood) leading. The “Hw” species group includes stands that are Hw, Ss, Ba, Hm (mountain hemlock), or Fd (Douglas-fir) leading.

2.4 Phase II (Ground Sampling)

2.4.1 Actual Sample Size

One hundred (100) plots were intended to be established in TFL 6.¹ Ninety-eight (98) plots were installed after 2 were dropped because they were unsafe.² Of the 98 plots, 4 were established outside the initial target population and 14 were located inside the target population, but were less than 30 years of age (Phase I). The total actual sample size was 80 plots (Table 59).

Table 59: Plot distribution by land class

Land Class	n	(%)
Harvested post-2001 Productivity Group	3	3.1
Unknown	1	1.0
Under 30 Years Old	14	14.3
Target Population	80	81.6

The plots covered the entire target population and their distribution is shown in Figure 16.

2.4.2 Sampling Weights

Sampling weights were calculated using the total actual number of plots sampled from within the target population. The sample plan³ notes that samples were selected at the sub-stratum level, and therefore weights were also calculated at the sub-stratum level (Table 60).

Table 60: Sampling weights for Phase II plots

Stratum	Sub-Stratum	Area (ha)	n	Area/n
Old	Cw	35,192.5	18	1,955.1
Old	Hw	42,054.4	23	1,828.5
Young	Cw	7,472.0	5	1,494.4
Young	Hw	52,969.4	34	1,557.9

² Plots 69 and 93 were dropped because they were unsafe.

³ Western Forest Products Limited. 2001. Tree Farm Licence 6 Quatsino Sound – North Vancouver Island Timber Emphasis VRI Ground Sampling Plan. Unpublished Report, February 2001. 16 pp.

2.4.3 Sample Statistics

The Phase II plot statistics showed that on average, the “Young” stands were 27 m tall, 62 years of age, had a site index of 28 m, and produced approximately 400 m³/ha of merchantable volume. Conversely, on average, the “Old” stands were 33 m tall, 326 years of age, had a site index of 15 m, and produced approximately 740 m³/ha (Table 61). In general, the average unadjusted Phase I heights appear similar to the average Phase II heights, and ages appear under-predicted (Table 62). In the “Young” stratum, the site index is under-predicted while volumes are slightly over-predicted. In the “Old” stratum, the site index is slightly over-predicted, while volumes are under-predicted.

Table 61: Phase II statistics for the TFL 6 samples

Stratum	Sub-Stratum	Height (m)	Height Sample Size (n)	Age (yrs)	Age Sample Size (n)	SI (m)	SI Sample Size (n)	Vol. 17.5cm+ (m ³ /ha)	Vol. Sample Size (n)
Young	Cw	29.8	4	67.3	4	28.0	4	242.5	5
Young	Hw	27	32	61.8	30	28.5	30	426.1	34
Young	Total	27.3	36	62.4	34	28.4	34	403.4	39
Old	Cw	23.6	14	370.2	16	10.9	14	589.6	18
Old	Hw	40.8	18	288.2	20	18.5	16	868.7	23
Old	Total	33	32	326	36	14.9	30	741.6	41
Total		30.5	68	210.3	70	20.9	64	593.1	80

Note: Phase II (ground sampling) volume was whole-stem volume less tops, stumps, NVAF-corrected cruiser-called decay, waste, and breakage.

Table 62: Phase I statistics for the TFL 6 samples

Stratum	Sub-Stratum	Height (m)	Age (yrs)	SI (m)	Vol. 17.5cm+ (m ³ /ha)
Young	Cw	28.8	58.8	27.4	345.9
Young	Hw	23.9	55.1	24.8	421.1
Young	Total	24.4	55.5	25.1	411.8
Old	Cw	26.8	310.8	12.8	416.5
Old	Hw	41	276	17.7	729.1
Old	Total	34.5	292.1	15.5	586.7
Total		30.1	188.2	19.7	509.9

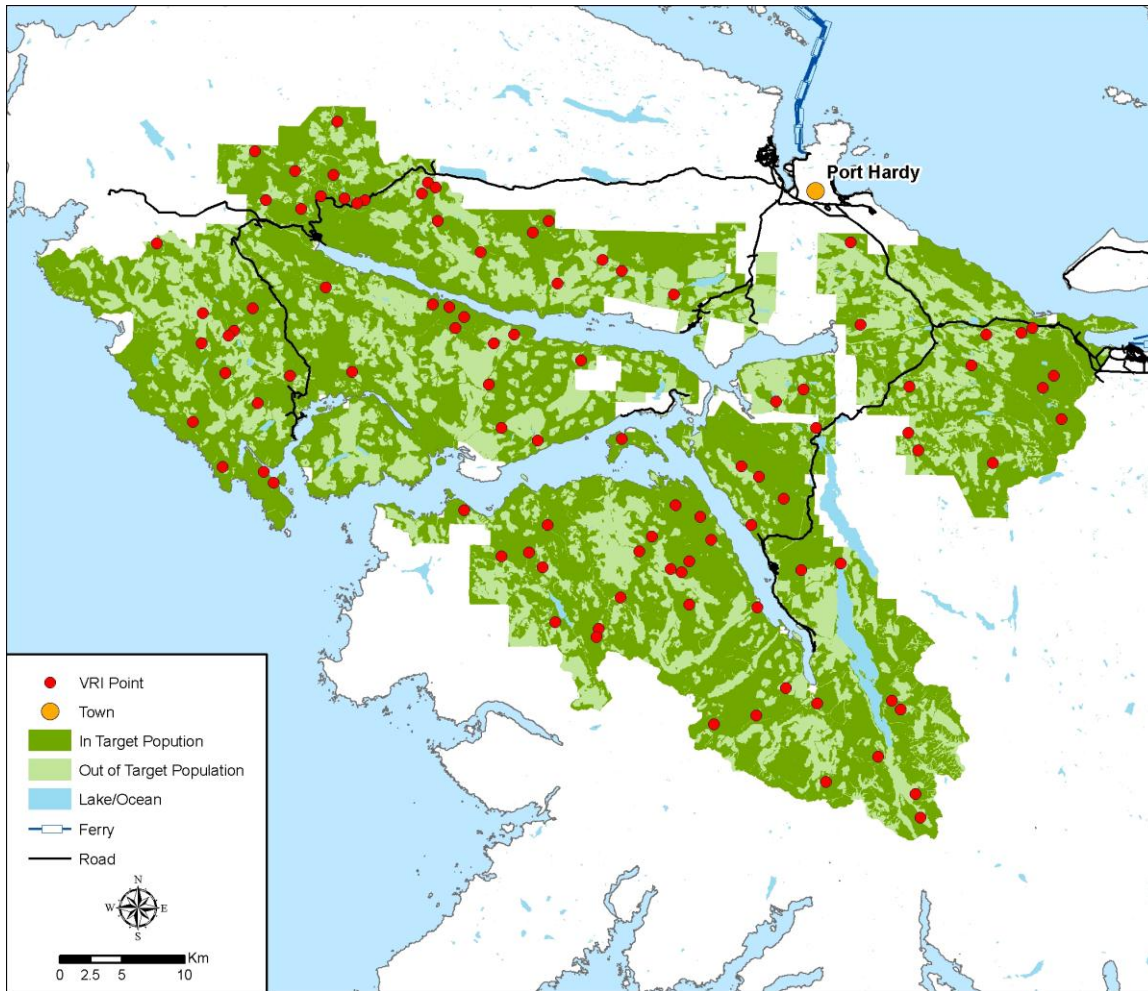


Figure 16: VRI plot locations in the target population for TFL 6

3 Methods

3.1 Unadjusted Phase I Population

The last full-scale inventory completed for TFL 6 was based on photos taken in 1967 and standardized in 1970. The inventory was partially updated for second growth stands in 1998. The Phase I data used in this analysis also includes denudation and regeneration updates up to and including 2004. The photo-interpreted age was projected to 2001¹ by adding or subtracting the required number of years. The photo-interpreted height, stocking class, and corresponding net merchantable volume were projected to 2001 using VDYP *version 6.6d*. All other VDYP inputs (species composition, crown closure, forest inventory zone, and public sustained yield unit) were not modified.

3.2 NVAF

NVAF ratios were generated by Will Smith, MFR and provided to Western for the adjustment analysis (Table 63).²

Table 63: NVAF ratios as supplied by Western

Live / Dead	Maturity	Species Group	NVAF Ratio
Live	Immature	All	0.94064
Live	Mature	Cw	1.29029
Live	Mature	Hw	0.94511
Dead	All	All	0.92527

3.3 Phase II Compilation and Data Screening

The Phase II data was compiled using the MFR SAS VRI Phase II compiler (June 27, 2002 version). Dead trees (standing and fallen) were recorded in all auxiliary plots. The received NVAF ratios were then applied to the compiled Phase II volumes. The SAS compiler has built-in error checking and validation routines to identify potential problems in the Phase II field data. No outstanding errors were encountered in the compilation.

3.4 Statistical Adjustment

The most recent MFR VRI statistical adjustment standards³ were used to adjust height, age, and live net merchantable volume. The MFR adjustment procedures assume that the unadjusted (Phase I) inventory volume is biased due to two sources of error:

1. An attribute bias associated with the photo-interpreted height and age; and
2. A model bias inherent to the growth and yield model used to estimate volume (*VDYP version 6.6d*).

Three attributes needed for volume prediction are not directly adjusted in this process. A new stocking class is derived by *VDYP* using adjusted age, while there are no acceptable standards for species composition and crown closure adjustment. Leaving these attributes unadjusted is assumed to create a negligible bias.

¹ 2001 was the year of sampling.

² Downloaded from Western's FTP site March 17, 2008.

³ VRI Procedures and Standards for Data Analysis Attribute Adjustment and Implementation of Adjustment in a Corporate Database, Version 2.0, March 2004. The statistical adjustment was completed in May 2008, prior to the release of *VDYP7* as a MFR standard.

The attribute adjustment procedure (Figure 14) is a two-step process called the Fraser method (Figure 17) and is described as follows:

Step 1: Phase I height and age bias are corrected using an adjustment ratio of means (ROM) calculated from the Phase I (height or age) and the Phase II plots. An attribute-adjusted volume is then estimated using VDYP with the adjusted height and age.

Step 2: An adjustment ROM estimated from the attribute-adjusted volume and the NVAF-corrected Phase II volume is calculated, and this ratio is used to correct the model bias in the attribute-adjusted volume.

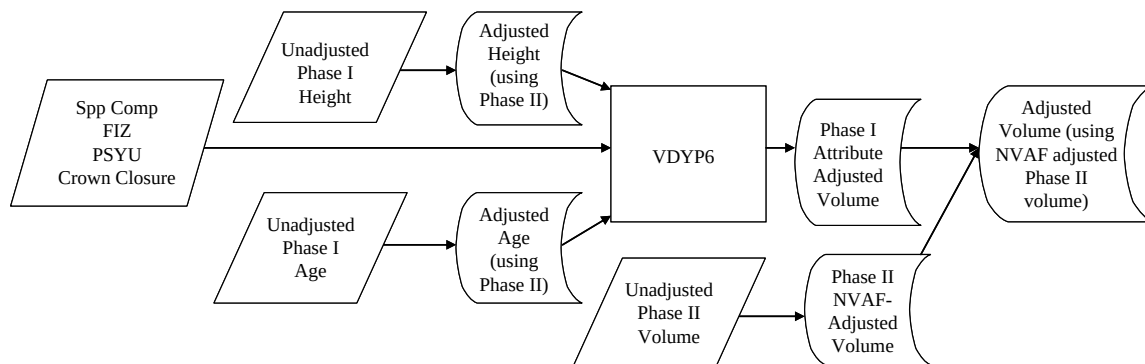


Figure 17: Fraser method

Although the ratios below are shown for each species sub-strata, the adjustment ratios were applied at the strata level that is maturity.

4 Results

4.1 Height

Twelve (12) plots were dropped from the analysis because their top height tree measurements did not match the sample leading species in the inventory, leaving 68 plots for analysis. Of these 68 plots, 36 and 32 plots were in the “Young” and “Old” strata, respectively. On average, inventory height was slightly biased (under-estimation of 1%, Table 64). Inventory height was under-estimated in the “Young” stratum (12%) and over-estimated in the “Old” stratum (5%). The 95% sampling error was 3.9% (Figure 18).

Table 64: Height adjustment statistics for the TFL 6 target population

Stratum	Sub-Stratum	Unadj. Pop.		n	Sample			Adj. Population		
		Area (ha)	Avg. (m)		Phase I (m)	Phase II (m)	ROM	Adj. Avg. (m)	95% E	
									(m)	(%)
Young	Cw	7,472	19.1	4	28.8	29.8	1.036	19.7		
Young	Hw	52,969	26.3	32	23.9	27.0	1.130	29.7		
Young	Total	60,441	25.4	36	24.4	27.3	1.118	28.4	1.8	6.2
Old	Cw	35,193	29.6	14	26.8	23.6	0.881	26.1		
Old	Hw	42,054	39.5	18	41.0	40.8	0.994	39.3		
Old	Total	77,247	35.0	32	34.5	33.0	0.955	33.4	1.7	5.2
Total	Total	137,688	30.8	68			1.014	31.2	1.2	3.9

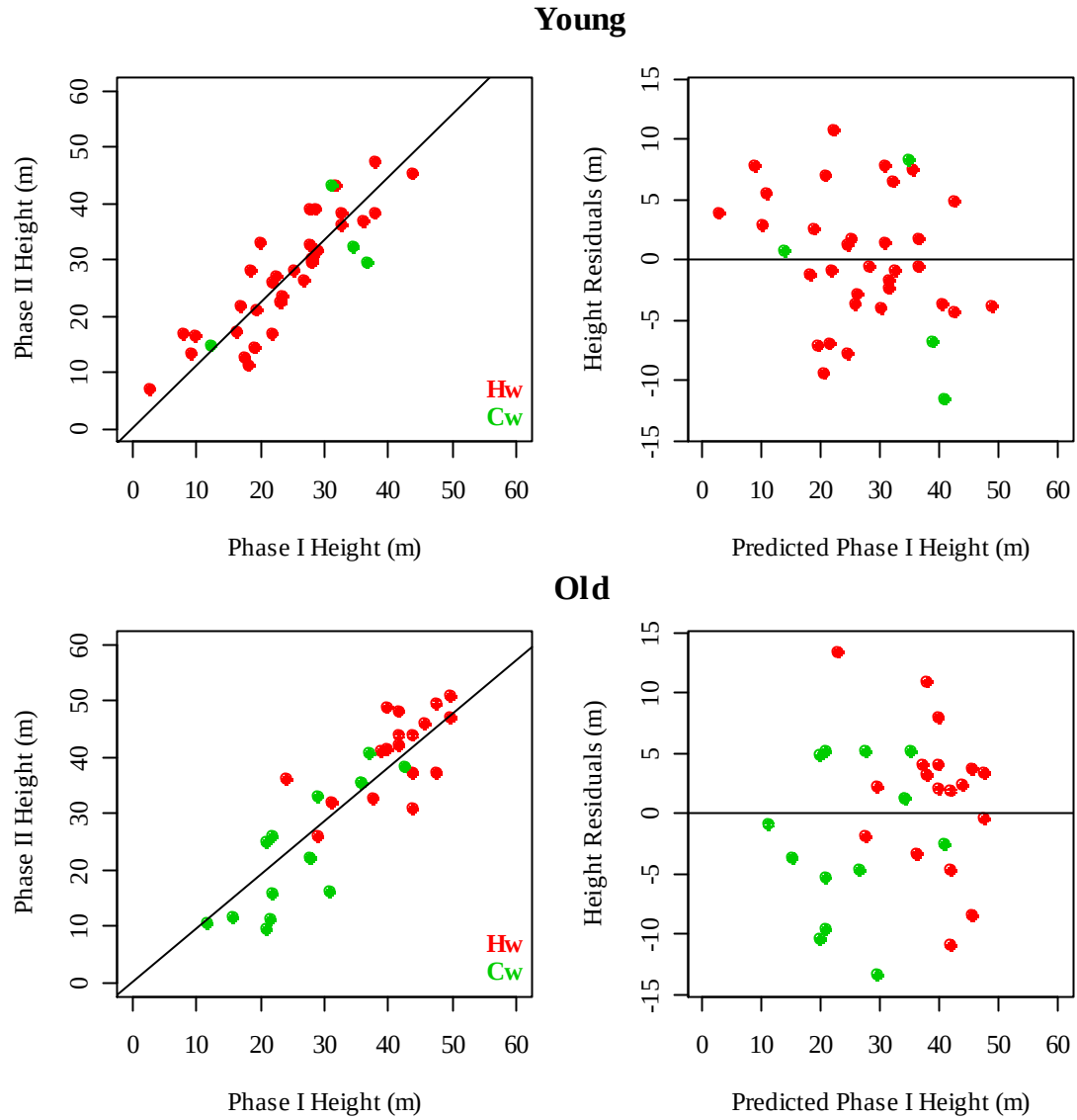


Figure 18: Height scatterplots

4.2 Age

Ten (10) plots were dropped from the analysis because their top height tree measurements did not match the sample leading species in the inventory, leaving 70 plots for analysis. Of these 70 plots, 34 and 36 plots were in the “Young” and “Old” strata, respectively. On average, inventory age was under-estimated by approximately 12% (Table 65). This under-estimation was common to both strata. There is one outlier¹ in the “Old” stratum; however, it only contributes to 1.1% of the sampling error. The 95% sampling error was higher than the sampling error for height (11.2% vs 3.9%). This may be associated with age class mid-pointing in the Phase I age estimation process. Figure 19 shows the same Phase I age estimate for a range of measured Phase II ages in the “Old” stratum.

Table 65: Age adjustment statistics for the TFL 6 target population

Stratum	Sub-Stratum	Unadj. Pop.		n	Sample		Adj. Population			
		Area (ha)	Avg. (yrs)		Phase I (yrs)	Phase II (yrs)	ROM	Adj. Avg. (yrs)	95% E	
Young	Cw	7,472	44.2	4	58.8	67.3	1.146	50.7		
Young	Hw	52,969	61.6	30	55.1	61.8	1.121	69.1		
Young	Total	60,441	59.5	34	55.5	62.4	1.124	66.8	5.0	7.5
Old	Cw	35,193	310.2	16	310.8	370.2	1.191	369.4		
Old	Hw	42,054	263.8	20	276.0	288.2	1.044	275.5		
Old	Total	77,247	284.9	36	292.1	326.0	1.116	318.1	41.9	13.2
Total	Total	137,688	186.0	70			1.117	207.8	23.2	11.2

¹ Inventory age for the leading species (Yc) was recorded as 161 years old, with a field-observed age of 560 years for the ground leading species (Yc).

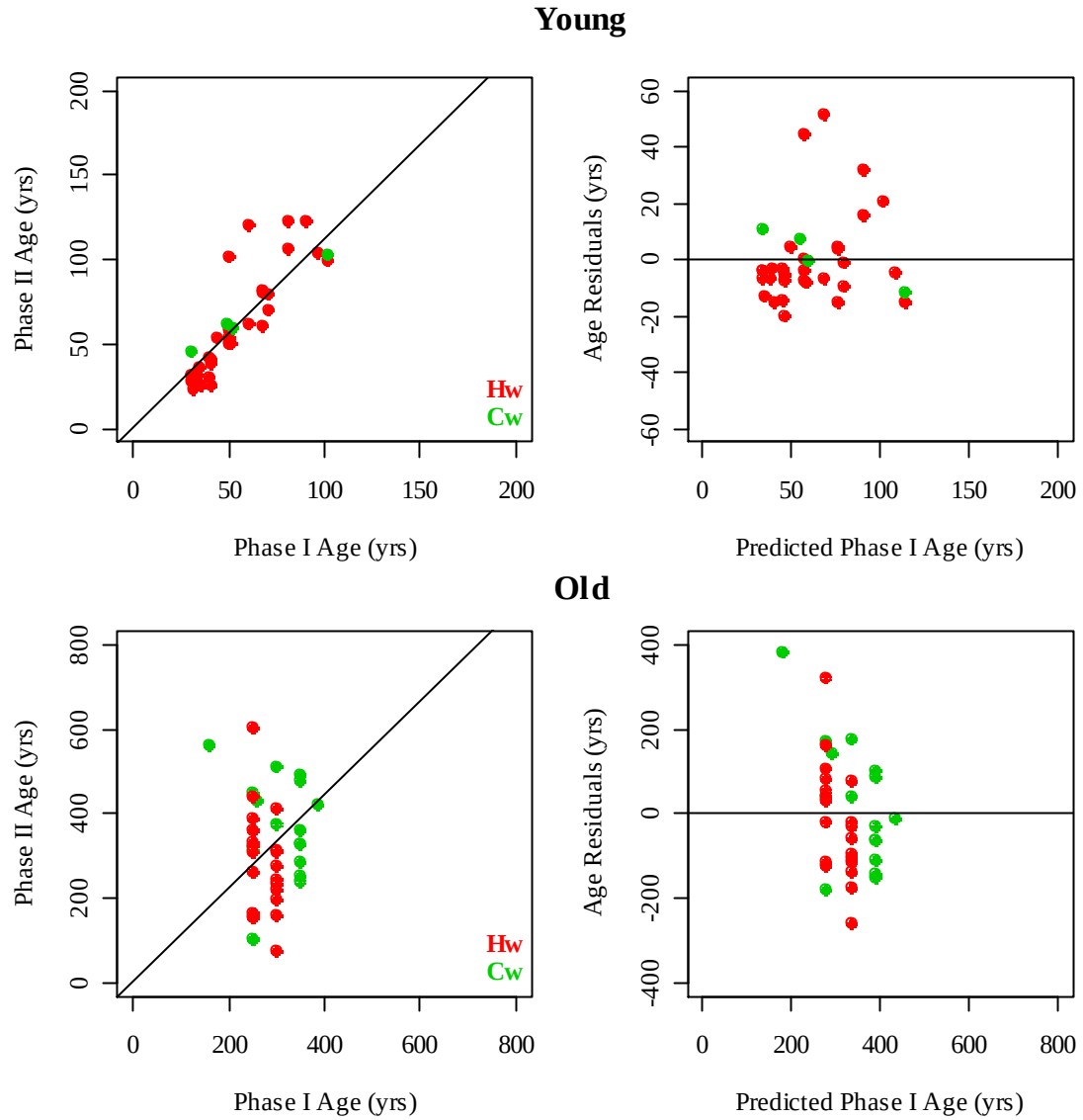


Figure 19: Age scatterplots

4.3 Attribute-Adjusted Volumes for the TFL 6 Target Population

VDYP volumes were re-estimated using the adjusted height and age inputs. Attribute-adjusted volumes increased by 15% and decreased by 9% in the “Young” and “Old” strata, respectively, when compared to the Phase I volumes (Table 66). Overall, volumes increased by 0.5% relative to the unadjusted volumes. The reduced volume in the “Old” stratum reflects a decrease in height averaging 1.6 m (the main driver of volume) and an average age increase of 33 years. The increased volume in the “Young” stratum reflects the increased heights and ages.

Table 66: Volume change in the TFL 6 target population due to attribute adjustments

Stratum	Area (ha)	Unadjusted Inventory	Attribute-Adjusted Inventory	Difference (m)	(%)
Young	60,441	382.3	451.8	69.5	15.4
Old	77,247	605.8	556.3	-49.5	-8.9
Total	137,688	507.7	510.4	2.7	0.5

4.4 Site Index

Site index is not directly adjusted in the VRI standard statistical adjustment. Instead, an adjusted site index is derived from the adjusted height and age. The average inventory site index decreased by approximately 0.4% after attribute-adjustment (Table 67). The minimal overall change in site index is due to the site index decreasing in the “Old” stratum, while increasing in the “Young” stratum. Site index increased in the “Young” stratum because of the proportionally higher increases in height than age. Similar to the volume changes described in Section 4.3, site index decreased in the “Old” stratum because of increased ages and decreased heights.

Table 67: Site index change due to input attribute adjustment

Stratum	Area (ha)	Site Index (m)	Adj. Site Index (m)	Difference (%)
Young	60,441	25.5	26.2	0.7
Old	77,247	15.2	13.9	-1.3
Total	137,688	19.7	19.3	-0.4

4.5 Live Net Merchantable Volume

All Phase II observations were used to compute the volume ratios. The live net merchantable volume increased by 15% after adjustment (Table 68). The target sampling error (10%) was met for the overall target population at the 17.5 cm utilization levels (Figure 20).

Table 68: Live merchantable volume (17.5+ cm) adjustment statistics for the TFL 6 target population

Stratum	Sub-Stratum	Attr.Adj. Vol.		n	Sample			Adj. Population		
		Area (ha)	Avg. (m ³ /ha)		Phase I (m ³ /ha)	Phase II (m ³ /ha)	ROM	Adj. Avg. (m ³ /ha)	95% E (m ³ /ha)	(%)
Young	Cw	7,472	191.4	5	345.9	242.5	0.701	134.2		
Young	Hw	52,969	488.5	34	421.1	426.1	1.012	494.3		
Young	Total	60,441	451.8	39	411.8	403.4	0.980	442.6	68.8	15.5
Old	Cw	35,193	434.5	18	416.5	589.6	1.416	615.1		
Old	Hw	42,054	658.3	23	729.1	868.7	1.191	784.3		
Old	Total	77,247	556.3	41	586.7	741.6	1.264	703.2	83.5	11.9
Total	Total	137,688	510.4	80			1.153	588.8	54.9	9.3

Note: Phase I volume is the attribute-adjusted net merchantable volume as predicted from VDYP version 6.6d using adjusted heights and ages. Phase II volumes have been adjusted with the appropriate NVAF ratios to remove bias from cruiser-called decay values.

4.6 Unadjusted vs. Adjusted Volume

After adjustment, the live inventory volume increased by approximately 14% when compared to the unadjusted inventory for the TFL 6 target population (Table 69).

Table 69: Volume change due to input attribute adjustment

Stratum	Area (ha)	Unadjusted Inventory (m ³ /ha)	Adjusted Inventory (m ³ /ha)	Difference	(%)
Young	60,441	382.3	442.6	60.4	13.6
Old	77,247	605.8	703.2	97.4	13.9
Total	137,688	507.7	588.8	81.2	13.8

Note: calculated at the 17.5 cm utilization level

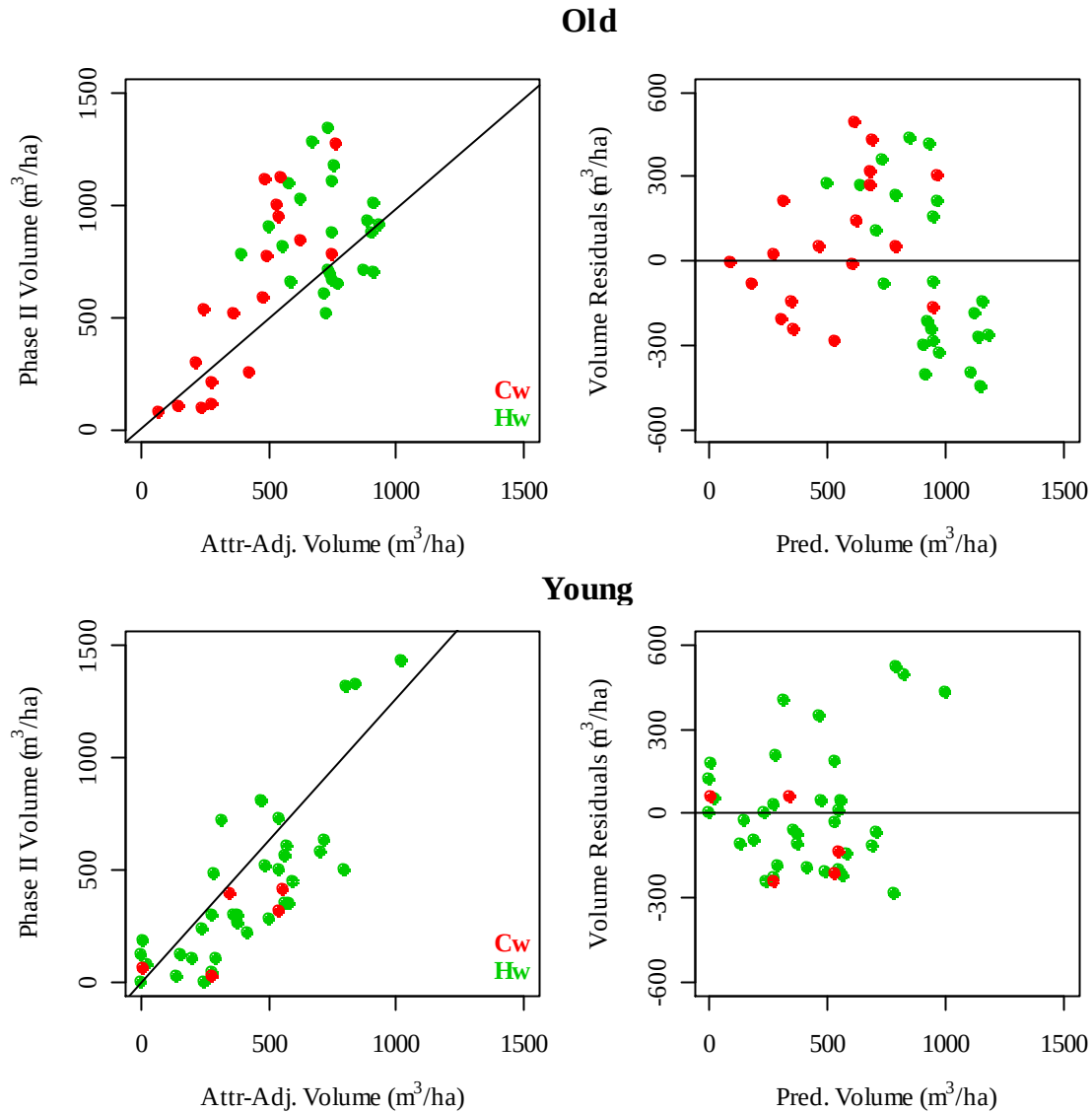


Figure 20: Volume scatterplots (Phase II vs. Attribute-Adjusted Phase I)

5 Discussion

5.1 Accuracy and Precision

The inventory adjustment provides unbiased estimates for the TFL 6 target population. This means unbiased estimates at the stratum level. There is always a possibility that local bias exists within a stratum. It would be inappropriate to try to estimate sub-stratum bias given the small sample size provided at a smaller scale.

The MFR-recommended precision for adjusted average volume at the management unit level is a sampling error of $\pm 10\%$ (95% probability). The overall sampling errors achieved in this project were smaller than this target (9.3%, Table 68). This means that the inventory adjustment provides the appropriate level of confidence for timber supply analysis.

5.2 Risks and Uncertainties

5.2.1 Age Trend

The statistical adjustment removes the bias in each stratum. In other inventory programs, age-related trends have existed within the VRI data that have led to concerns in the TSR process. To determine whether this is the case for TFL 6, residual errors for each adjusted attribute were plotted against stand age to identify any age-related trends. None of the attributes of interest showed an age-related trend in the residuals. Volume, the most important attribute, did not show any age-related trend in the residuals.

5.2.2 Age Adjustment

The age adjustment of the “Old” stratum was done using input data where Phase I ages had been mid-pointed (i.e., all stands within a given age class were assigned the appropriate mid-point age). Ages of the “Old” stratum increased by approximately 33 years and reflect the fact that the stands sampled had a higher average age than the mid-points used to represent the age classes. The adjustment process does not allow for ranges to be computed, only for existing ages to be updated. Therefore, the resulting adjusted database will still have a single adjusted age to represent each age class.

6 Conclusions & Recommendations

A statistical adjustment was completed for TFL 6 using standard MFR methodology. Unbiased estimates of height, age, and volume were obtained due to the design of the VRI statistical adjustment methods. These estimates represent the best estimates available at present. Therefore, we recommend that

Western apply the adjusted estimates of height, age, and volume in the upcoming TSR.

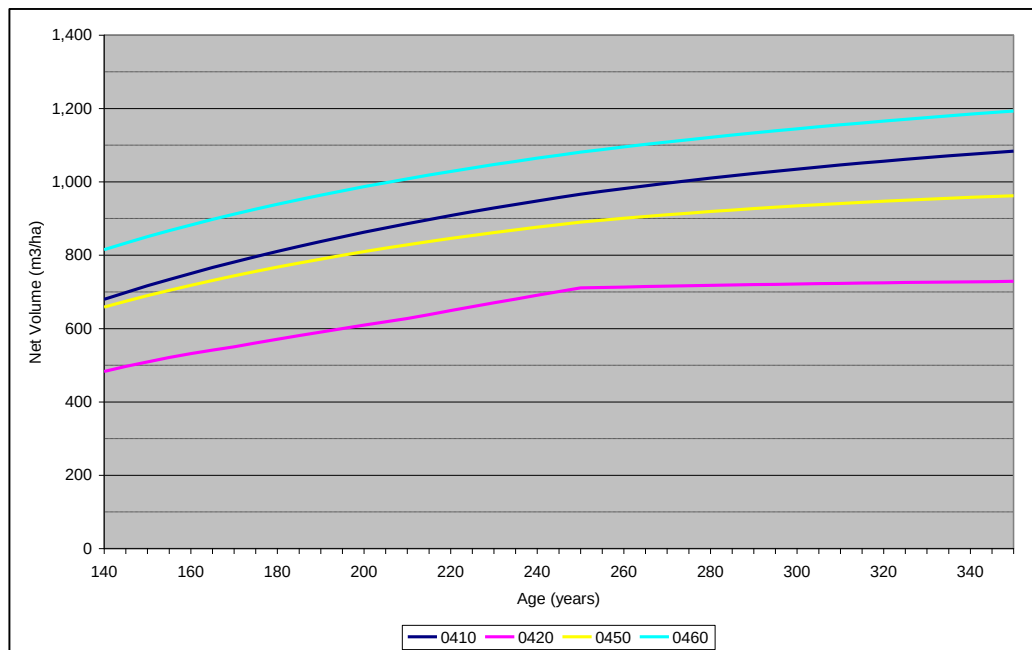
Appendix C - Mature Yield Tables

Net Merchantable Volume Yield Tables

Mature Stands

S3 Ecosite

Age	Analysis Units			
	0410	0420	0450	0460
140	680	483	658	815
145	698	496	674	833
150	716	509	690	850
155	734	521	704	867
160	750	531	718	882
165	766	541	731	897
170	781	550	744	912
175	796	560	756	925
180	810	571	767	939
185	824	581	779	951
190	837	590	789	964
195	850	600	800	975
200	862	609	810	987
205	874	618	819	998
210	886	627	828	1,008
215	897	638	837	1,019
220	908	649	846	1,028
225	918	659	854	1,038
230	929	670	861	1,047
235	938	680	869	1,056
240	948	691	876	1,064
245	957	701	883	1,073
250	966	711	890	1,081
255	974	712	895	1,088
260	982	713	900	1,095
265	989	714	905	1,102
270	996	716	910	1,109
275	1,003	717	915	1,115
280	1,010	718	919	1,121
285	1,017	719	923	1,127
290	1,023	720	927	1,133
295	1,029	721	931	1,139
300	1,035	721	934	1,145
305	1,040	722	938	1,150
310	1,046	723	941	1,155
315	1,051	724	944	1,161
320	1,056	725	947	1,166
325	1,061	725	950	1,171
330	1,066	726	953	1,175
335	1,071	727	955	1,180
340	1,075	727	958	1,185
345	1,080	728	960	1,189
350	1,084	729	962	1,193

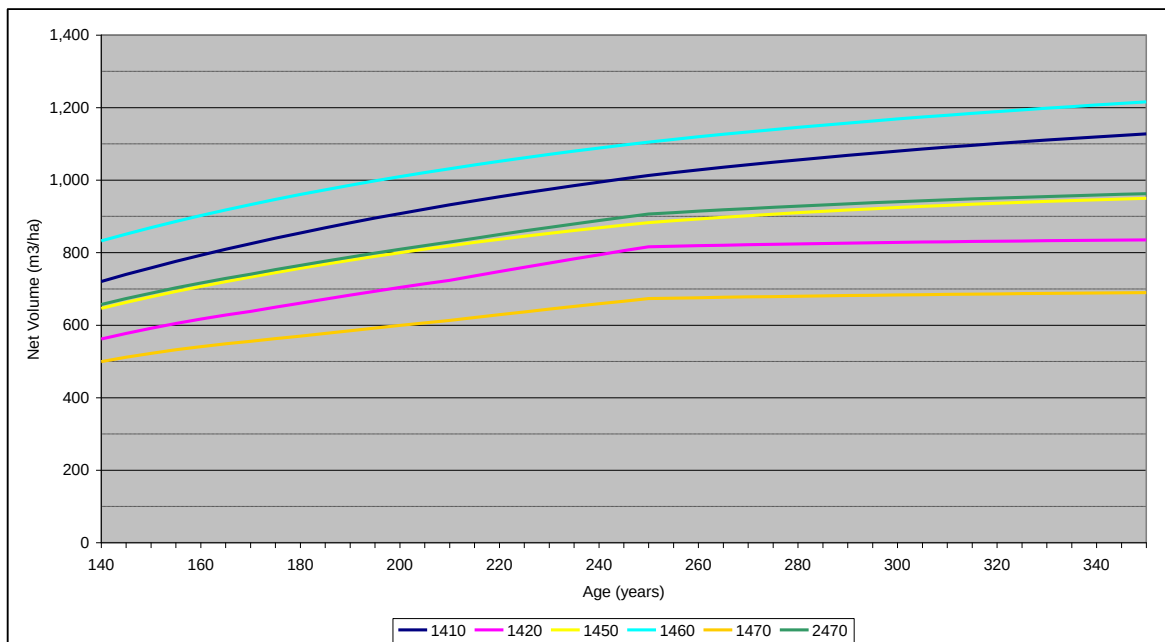


Net Merchantable Volume Yield Tables

Mature Stands

S1HA, S4, S6 Ecosites

Age	Analysis Units					
	1410	1420	1450	1460	1470	2470
140	720	562	647	832	499	657
145	740	577	663	851	511	673
150	758	591	678	869	522	688
155	776	605	693	886	532	703
160	793	617	707	902	541	716
165	809	628	720	918	549	729
170	825	638	732	932	555	741
175	840	649	745	947	563	753
180	854	661	757	960	570	765
185	869	672	768	973	577	776
190	882	683	779	986	585	788
195	895	693	790	998	592	799
200	908	704	800	1,010	599	809
205	920	714	810	1,021	606	819
210	932	724	819	1,032	613	829
215	943	736	828	1,042	621	840
220	954	748	837	1,052	629	850
225	965	760	845	1,062	636	860
230	975	771	853	1,071	644	870
235	985	783	861	1,080	652	879
240	995	794	869	1,089	659	888
245	1,004	805	876	1,097	666	897
250	1,013	816	883	1,105	673	906
255	1,021	818	888	1,112	674	910
260	1,028	819	893	1,120	676	914
265	1,035	820	897	1,126	677	918
270	1,043	822	902	1,133	678	922
275	1,049	823	906	1,139	679	925
280	1,056	824	910	1,146	680	928
285	1,062	825	914	1,152	681	932
290	1,068	826	918	1,157	682	935
295	1,074	827	921	1,163	683	937
300	1,080	828	924	1,169	683	940
305	1,086	829	928	1,174	684	943
310	1,091	830	931	1,179	685	946
315	1,096	831	933	1,184	686	948
320	1,101	831	936	1,189	686	950
325	1,106	832	939	1,194	687	953
330	1,110	833	941	1,198	687	955
335	1,115	833	944	1,203	688	957
340	1,119	834	946	1,207	689	959
345	1,123	835	948	1,212	689	961
350	1,127	835	950	1,216	690	963

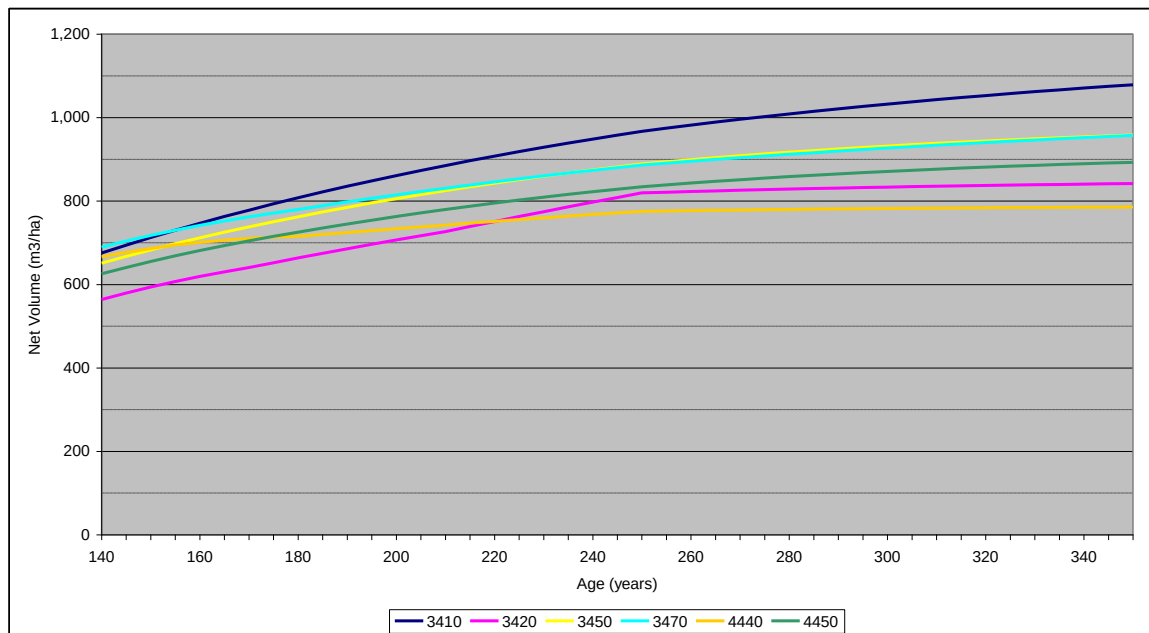


Net Merchantable Volume Yield Tables

Mature Stands

S12, S13, S12F Ecosites

Age	Analysis Units					
	3410	3420	3450	3470	4440	4450
140	675	564	651	690	666	626
145	694	579	667	704	677	641
150	712	594	683	718	686	655
155	730	607	698	730	694	669
160	747	619	712	742	701	681
165	763	630	725	752	706	693
170	778	640	738	762	710	705
175	793	652	750	771	713	715
180	808	664	762	780	715	725
185	822	675	774	789	720	735
190	835	686	785	798	724	745
195	848	696	795	806	729	754
200	861	707	806	815	734	763
205	873	717	816	823	738	772
210	885	727	825	831	742	780
215	897	739	834	838	747	788
220	908	751	843	846	751	795
225	918	763	851	853	756	802
230	929	775	859	860	760	809
235	939	786	867	867	764	816
240	948	797	875	874	768	822
245	958	809	882	880	772	828
250	967	819	889	886	775	834
255	975	821	894	891	776	839
260	982	823	899	895	777	843
265	989	824	904	900	778	847
270	996	826	908	904	779	851
275	1,003	827	913	908	779	855
280	1,009	829	917	912	780	858
285	1,015	830	921	916	780	862
290	1,021	831	925	920	781	865
295	1,027	832	928	923	782	868
300	1,032	833	932	927	782	871
305	1,038	834	935	930	783	874
310	1,043	835	938	934	783	876
315	1,048	836	941	937	783	879
320	1,053	837	944	940	784	881
325	1,057	838	947	943	784	883
330	1,062	839	949	946	784	886
335	1,066	840	952	949	785	888
340	1,071	841	954	952	785	889
345	1,075	841	956	955	785	891
350	1,079	842	958	958	786	893

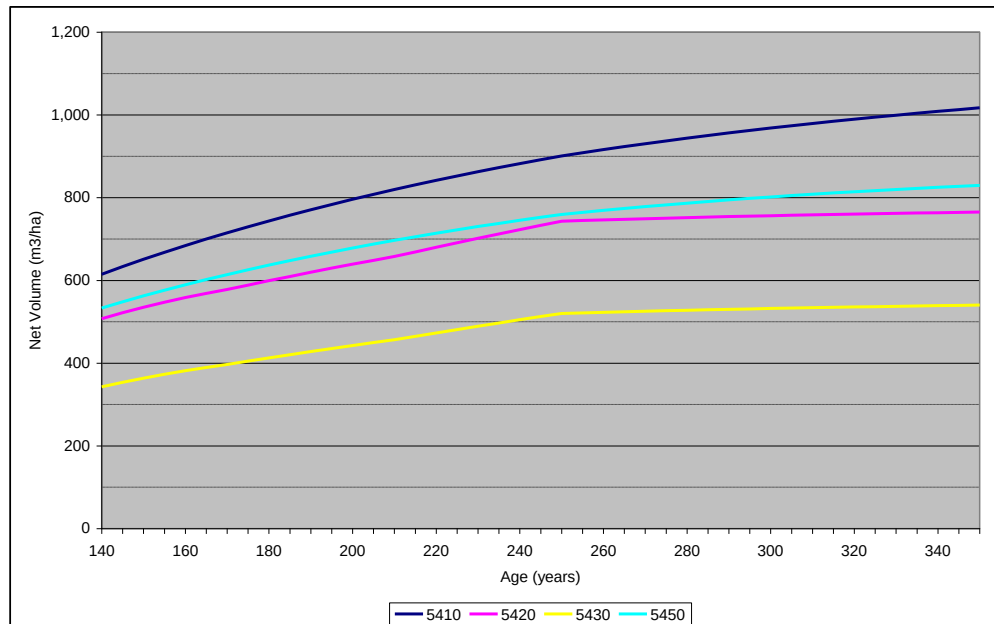


Net Merchantable Volume Yield Tables

Mature Stands

M1, M3, M5 Ecosites

Age	Analysis Units			
	5410	5420	5430	5450
140	615	507	343	533
145	633	522	354	548
150	651	535	364	563
155	668	547	373	576
160	684	558	382	589
165	700	569	389	602
170	715	578	397	614
175	730	589	405	625
180	744	599	413	637
185	757	610	420	648
190	771	620	428	658
195	784	629	435	668
200	796	639	443	678
205	808	648	450	688
210	820	658	457	697
215	831	669	465	705
220	842	680	473	714
225	852	691	481	722
230	863	702	489	730
235	873	712	497	738
240	882	723	505	745
245	892	733	513	752
250	901	743	520	759
255	909	745	521	764
260	916	746	523	769
265	923	748	524	774
270	930	749	525	778
275	937	750	527	783
280	944	752	528	787
285	950	753	529	791
290	956	754	530	795
295	962	755	531	798
300	968	756	532	802
305	974	757	533	805
310	979	758	534	808
315	985	759	535	811
320	990	760	536	814
325	995	761	537	817
330	999	762	537	820
335	1,004	763	538	822
340	1,009	764	539	825
345	1,013	764	540	827
350	1,017	765	540	830

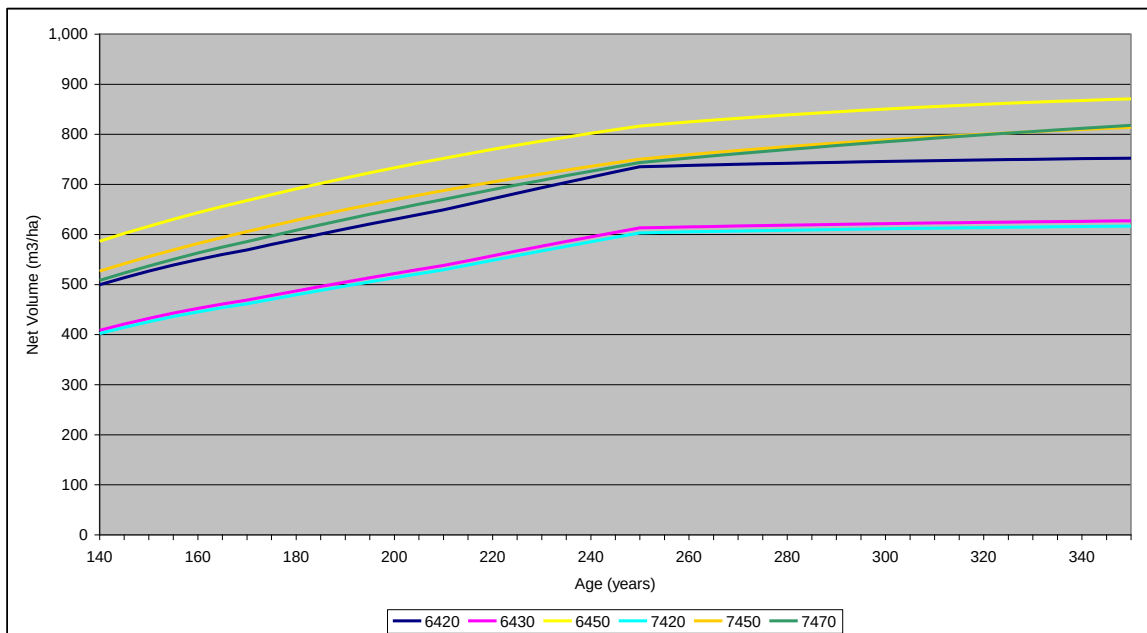


Net Merchantable Volume Yield Tables

Mature Stands

S1CH, S2, S6, S10 Ecosites

Age	Analysis Units					
	6420	6430	6450	7420	7450	7470
140	499	408	586	402	527	508
145	513	421	602	414	542	523
150	527	432	616	425	556	537
155	539	443	630	436	569	550
160	550	452	643	445	582	563
165	560	461	656	454	594	575
170	569	469	668	461	606	586
175	580	478	679	470	617	597
180	590	487	691	479	628	608
185	601	496	702	488	639	619
190	611	505	713	497	649	630
195	621	513	723	505	659	640
200	630	522	733	513	669	650
205	640	530	743	522	678	660
210	649	538	752	529	687	670
215	660	548	761	539	696	680
220	671	557	770	548	705	689
225	682	567	778	558	713	699
230	693	576	786	567	721	708
235	704	586	794	576	729	717
240	714	595	802	585	736	726
245	725	604	809	594	743	735
250	735	613	817	603	750	743
255	736	614	821	604	755	748
260	738	615	825	605	759	753
265	739	616	828	606	763	757
270	740	617	832	607	768	761
275	741	618	835	608	771	765
280	742	619	839	608	775	769
285	743	619	842	609	779	773
290	744	620	845	610	782	777
295	745	621	848	611	786	781
300	746	621	850	611	789	785
305	747	622	853	612	792	789
310	747	623	855	613	795	792
315	748	623	858	613	798	796
320	749	624	860	614	800	799
325	749	625	862	614	803	802
330	750	625	864	615	805	806
335	751	626	866	615	808	809
340	751	626	868	616	810	812
345	752	627	869	616	812	815
350	752	627	871	617	814	818

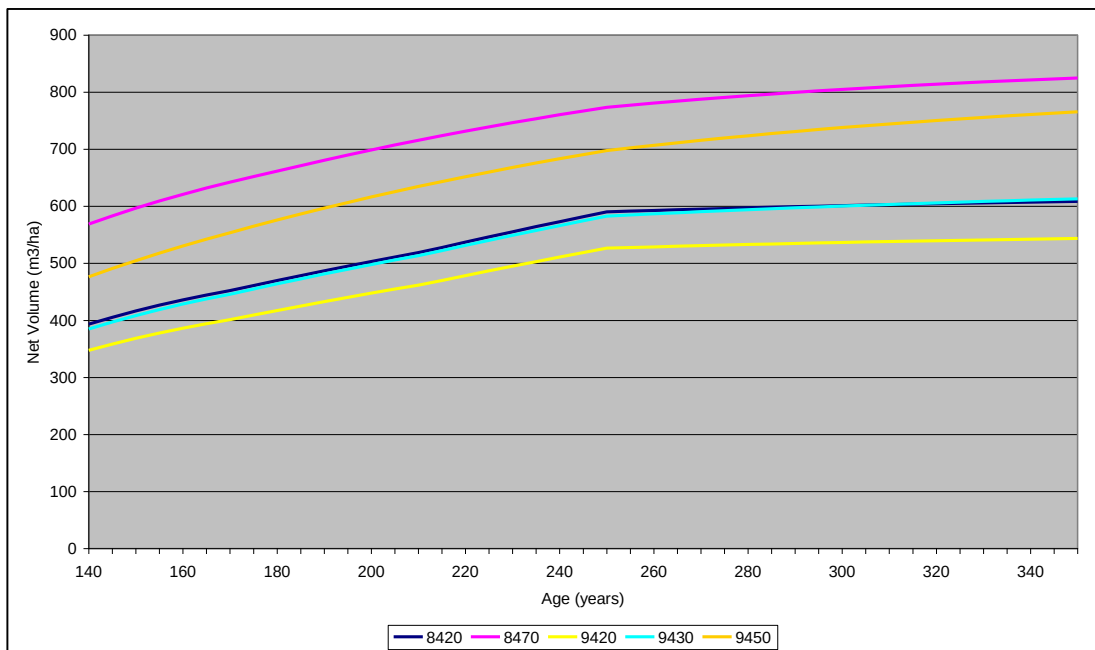


Net Merchantable Volume Yield Tables

Mature Stands

S2F, M2, M4 Ecosites

Age	Analysis Units				
	8420	8470	9420	9430	9450
140	393	569	347	385	476
145	405	583	358	397	491
150	416	597	368	409	505
155	427	609	378	419	518
160	436	621	386	429	530
165	445	632	394	438	542
170	452	642	401	446	554
175	461	652	409	455	565
180	470	662	417	464	576
185	479	671	425	473	587
190	487	681	433	481	597
195	495	690	440	490	607
200	503	699	448	498	616
205	511	707	455	506	626
210	519	716	462	514	635
215	528	724	470	522	643
220	537	732	478	531	652
225	546	739	487	540	660
230	555	746	495	549	668
235	564	754	503	558	676
240	573	760	511	566	683
245	582	767	519	575	691
250	590	774	527	583	698
255	591	777	528	585	702
260	593	781	529	587	707
265	594	784	530	589	711
270	595	788	531	590	716
275	596	791	532	592	720
280	597	794	533	594	724
285	598	797	534	596	727
290	599	800	535	597	731
295	600	802	536	599	735
300	601	805	537	600	738
305	602	807	537	602	741
310	603	810	538	603	744
315	604	812	539	605	747
320	604	814	540	606	750
325	605	816	540	607	753
330	606	818	541	608	756
335	607	820	542	610	758
340	607	822	542	611	761
345	608	823	543	612	763
350	609	825	544	613	766



Appendix D - Unmanaged Immature Yield Tables
